SUMMARY REFERENCE

Capability Matters

Case Overview

A quick-reference guide to one hundred real cases of capability — how it fails, how it has been engineered, and where the discipline is being tested next — each with a brief callout, its key sources, and the LENS courses it informs.

LDT · LENS — JOHNS HOPKINS UNIVERSITY SCHOOL OF EDUCATION

Learning Engineering for Next-Generation Systems

LENS is a specialization within the Johns Hopkins University School of Education's **Master of Education in Learning Design and Technology** (LDT). Its premise is that capability — the ability of people and institutions to perform demanding, high-consequence work reliably — can be **engineered**: specified, measured, designed for, and improved, rather than left to tradition, heroics, or chance. The program trains practitioners to treat capability as a system parameter, to build the evidence that shows whether an intervention actually worked, and to design the human and technical halves of a system together rather than in isolation.

LENS rests on five pillars: **Mission Literacy** — fluency in the operational domains where capability is contested; the **JHU Ecosystem** — the research, applied-laboratory, and school assets the program draws on; **Intersectional Expertise** — pairing learning science with engineering and domain practice; **Capability Focus** — capability itself as the unit of design; and **Flywheel Iteration** — the Practice Flywheel, the iterative loop from requirements to evidence that animates the other four. The pillars are designed to be load-bearing together, not individually.

Two hundred cases, three movements

The casebook assembles **more than two hundred** documented cases — from healthcare, defense, education, energy, finance, transportation, and government — organized in three movements, with a contemporary expansion that extends the corpus into algorithmic decision-making, learning analytics, and workforce development.

Part I — The Failure Modes groups cases under six recurring patterns (see overleaf), with an evidence-gap chapter on measurement. The taxonomy is a finding, not a theory: six categories account for almost every well-documented preventable failure in the literature.

Part II — What Works collects the successes, which share one structural feature: a **paired intervention** — a technical artifact joined to a cultural authority; a measurement instrument joined to a body that acts on it. Neither half alone moves the curve.

Part III — The Frontier concerns the cases where the discipline is being asked to specify what good looks like before the catastrophe arrives — particularly human-AI teaming. These cases are deliberately less settled.

Reading an entry

Each case is distilled to a single panel with four parts: a **header** (case number, domain tags, and year); the **title**; an ~100-word **callout** that states what happened and why it matters; one to three **key references**; and a short note on its **LENS applicability** — the courses that use it and the capability lesson it carries. The full three-page treatment (an inline-cited narrative and a Learning Engineering Lens) lives in the main casebook; this guide is the index to it.

DOMAIN TAGS

DEFENSE Defense AVIATION Aviation SPACE / AEROSPACE Space / Aerospace HEALTHCARE Healthcare ENERGY Energy

EDUCATION AT SCALE Education at Scale GOVERNMENT Government INDUSTRIAL Industrial TECHNOLOGY Technology AUTONOMOUS SYSTEMS Autonomous Systems

FAILURE-MODE CODES

In the main casebook each case carries one or more single-letter mode codes: **T** training gap · **D** capability designed out · **N** normalization of deviance · **H** human-system interface · **G** governance & trust · **K** knowledge & institutional memory. Most cases carry two or three; the primary mode is listed first.

The six failure modes

T Training gap

The operator lacked a capability the situation demanded — and the institution had the means to build it but had let it lapse or never delivered it.

D Capability designed out

The capability was removed from, or never built into, the system's design — a defect, omission, or a cost-and-packaging decision made upstream.

N Normalization of deviance

A known deviation became routine because it had not yet caused harm — until the margin ran out.

H Human-system interface

The interface misled the operator: ambiguous cues, hidden modes, or alarms that did not carry the weight the moment required.

G Governance & trust

The institution's incentives, oversight, or disclosure produced the harm — the measurement system or authority structure was the failure.

K Knowledge & institutional memory

The institution lost, or never built, the knowledge to do the work — through attrition, silos, or thin transfer between people and units.

The ten LENS courses

Each entry's "LENS applicability" names the courses it informs. Approximate case coverage is shown in parentheses.

LEN 1	Principles of learning engineering	~21
LEN 2	Human-AI teaming	~23
LEN 3	Systems integration	~18
LEN 4	Evidence and measurement	~31
LEN 5	Capability analysis	~36
LEN 6	Applied problem-solving	~13
LEN 7	Bias and governance	~49
LEN 8	Knowledge transfer	~45
LEN 9	Computational methods	~10
LEN 10	Capstone studio	~19

Cases are mapped to more than one course where they apply, so the counts sum to more than one hundred. The distribution is itself diagnostic: the literature is richest in governance, knowledge transfer, and capability analysis, and thinnest in computational instrumentation — a gap the studio sequence is built to close.

CONTENTS

All the cases

Listed in reading order; the number is the case’s catalog number in the main casebook.

CASE 1 DEFENSE 2017

USS Fitzgerald & USS John S. McCain

17 sailors killed in two avoidable destroyer collisions in the Western Pacific within nine weeks

THE FULL CASE, IN FIVE BEATS

Background	Forward-deployed Seventh Fleet at peak tempo; in-person SWOS school replaced by self-study CD-ROMs in 2003
What Happened	Fitzgerald struck by ACX Crystal off Japan; McCain collided with Alnic MC near Singapore
The Investigation	Navy and NTSB judged both avoidable, citing training shortfalls and a confusing touch-screen helm design
The Capability Gap	Waivers stacked while readiness dashboards stayed green; stated and actual readiness diverged for a decade
Aftermath & Reform	In-person pipeline rebuilt, Ready-for-Sea Assessment stood up, touch-screen helm slated for conventional replacement

CONNECTIVITY

INDUCED	1.1
LENS	D1/PT3
CLO	CLO-1, CLO-4
COURSES	LEN 1, LEN 5, LEN 8, LEN 3

KEY REFERENCES

- T. C. Miller, M. Faturechi & R. Rotella, “Years of Warnings, Then Death and Disaster,” ProPublica (2019) — the 2003 SWOS-in-a-Box shift.
- GAO, Navy Readiness, GAO-17-809T (Sept. 2017) — 37% of Japan-based warfare certifications expired by June 2017.
- NTSB, Collision between USS Fitzgerald and MV ACX Crystal, NTSB/MAR-20/02 (2020), DCA17PM018.

> LENS LESSON Fitzgerald/McCain is the canonical Training Gap case because the gap was invisible from inside the system. Operational tempo and self-study “training” combined to produce a fleet whose stated readiness and actual readiness diverged for more than a decade. The two collisions nine weeks apart converted what could have been read as an outlier into a structural diagnosis: an under-trained bridge team failing the most basic give-way rules, and an unfamiliar interface that made the same gap reach the hull. Seven sailors died on Fitzgerald; ten on McCain. The cost of the divergence was paid in lives long after it could have been measured in dollars or inspections.

CASE 2 AVIATION 2009

Air France Flight 447

228 killed; the deadliest accident in Air France's history

THE FULL CASE, IN FIVE BEATS

Background	Highly automated A330 crossed equatorial storm band carrying a known but unfitted pitot-probe vulnerability
What Happened	Pitot probes iced, autopilot disconnected, sustained nose-up input stalled the jet for four minutes
The Investigation	BEA cited airspeed loss, inappropriate inputs, and training that never produced reliable stall responses
The Capability Gap	Simulators could not reproduce high-altitude stall; reliable automation had also let manual skills atrophy
Aftermath & Reform	Pitot probes replaced; regulators mandated Upset Prevention and Recovery Training across all airlines

CONNECTIVITY

INDUCED	1.2
LENS	D2/PT4
CLO	CLO-2, CLO-3
COURSES	LEN 1, LEN 5, LEN 2

KEY REFERENCES

- Bureau d’Enquêtes et d’Analyses (BEA), Final Report on the accident on 1 June 2009 to the Airbus A330-203 registered F-GZCP operated by Air France flight AF447 (July 2012) — full report: flight, aircraft, and the known pitot-icing vulnerability with retrofit pending.
- BEA, Final Report AF447 (2012) — accident sequence: autopilot/autothrust disconnection, degraded control law, sustained nose-up input, stall, and the stall-warning logic dropping out at high angle of attack.
- BEA, Final Report AF447 (2012) — probable cause: airspeed inconsistency from pitot icing, inappropriate control inputs, and failure to recognize and recover from the stall.

> LENS LESSON AF447 is the canonical case of training that matched one envelope of operation perfectly and another not at all. Stall **prevention** training was excellent. Stall **recovery from cruise altitude** training did not exist because the simulators of the era could not faithfully produce it. Layered on top was an automation-to-human handoff that arrived in a degraded control law the crew had never flown and a sidestick architecture that hid one pilot’s inputs from the other. The pilots performed exactly as trained. The training was the wrong training, and the reform had to reach the regulator, because no single airline could close the gap unilaterally.

CASE 3 ENERGY 1979

Three Mile Island

Partial meltdown of a Babcock & Wilcox PWR; minimal off-site dose; most serious accident in U.S. commercial nuclear history; catalyst for industry-wide reform

THE FULL CASE, IN FIVE BEATS

Background	Operators trained for design-basis ruptures; a near-identical Davis-Besse precursor was never propagated to the fleet
What Happened	Relief valve stuck open while panel reported closed; operators throttled injection the core needed
The Investigation	Kemeny Commission concluded fundamental problems were people-related, reaching past operators to management and the NRC
The Capability Gap	Training mismatched the failure that arrived; capacity to learn from precursors had broken down
Aftermath & Reform	INPO formed within nine months; NRC required plant-referenced simulators and tied training to national standards

CONNECTIVITY

INDUCED	3.1
LENS	D3/PT4
CLO	CLO-4, CLO-1
COURSES	LEN 1, LEN 5

KEY REFERENCES

- Kemeny Commission, Report of the President’s Commission on the Accident at Three Mile Island (Oct. 1979) — operator training oriented to large design-basis accidents.
- M. Rogovin & G. T. Frampton, Three Mile Island: A Report to the Commissioners and to the Public, NUREG/CR-1250 (U.S. NRC, 1980) — the September 1977 Davis-Besse stuck-PORV precursor and the failure to disseminate its lessons.
- U.S. Nuclear Regulatory Commission, Backgrounder on the Three Mile Island Accident — accident sequence, misleading PORV indication, throttled high-pressure injection, 50% core damage, minimal off-site dose.

> LENS LESSON TMI is the textbook moment when an industry discovered that its capability assumptions did not survive contact with operational reality. Training to design-basis events left the operators blind to the genuinely ambiguous failures that complex systems actually produce; a control room that reported commands rather than states made correct diagnosis structurally hard; and the channel that should have carried Davis-Besse out to every plant did not exist. The reform — INPO, plant-referenced simulators, resident inspectors, RG 1.97 instrumentation, NUREG-0700 human-factors standards — built the missing infrastructure together rather than one piece at a time, which is the case’s load-bearing teaching.

CASE 4 DEFENSE ONGOING

Marine Corps Training in the INDOPACOM AOR

Structural readiness gap in DoD's stated top-priority theater

THE FULL CASE, IN FIVE BEATS

Background	Strategy names the Indo-Pacific top priority while its live-training infrastructure remains among the least mature
What Happened	Marine Corps papered over unmet range requirements with rotations, virtual substitutes, and multinational exercises
The Investigation	GAO documented decade-long unmet requirements in 2024; the Department concurred, echoing diagnoses pressed for years
The Capability Gap	Declared priority is cheap; engineered priority follows where construction and dollars actually flow
Aftermath & Reform	GAO recommendations remain open; closure depends on programmed ranges, dollars, and schedule, not another document

CONNECTIVITY

COURSES LEN 5, LEN 8

KEY REFERENCES

- 2022 National Defense Strategy of the United States of America (U.S. Department of Defense, 2022) — the Indo-Pacific as priority theater and China as the “pacing challenge.”
- U.S. Government Accountability Office, Military Readiness: Actions Needed for DOD to Address Challenges, GAO-24-107463 (May 2024) — the least-mature training infrastructure in the priority theater.
- GAO-24-107463 (2024) — the Marine Corps unable to meet INDOPACOM training requirements for nearly a decade, and the CONUS-rotation, virtual, and multinational-exercise workarounds.

> LENS LESSON INDOPACOM illustrates the difference between declared priority and engineered priority. A theater designated as DoD’s pacing challenge while the training infrastructure to operate in it remains structurally short is not a resourcing oversight — it is a capability architecture failure. Capability follows where resources actually flow, not where briefings name as critical.

CASE 5 DEFENSE ONGOING

F-35 Sustainment & Maintainer Shortage

Fleet mission-capable rate about 55% (March 2023), far short of program goals; lifecycle cost exceeds $1.7T, with ~$1.3T in operating and support; maintainer, technical-data, and depot shortfalls are the binding readiness constraint (GAO-23-105341)

THE FULL CASE, IN FIVE BEATS

Background	Most expensive program in history; most of its lifecycle cost is decades of sustainment work
What Happened	Fleet ran at half goal with over 10,000 components queued and depots still behind schedule
The Investigation	GAO traced shortfall to maintainer shortages, lack of access to technical data, and contractor dependency
The Capability Gap	Aircraft fielded faster than maintainers and data; contractor controls knowledge needed to keep jets flying
Aftermath & Reform	GAO urged full sustainment reassessment; costs still rising and fleet readiness still below program goals

CONNECTIVITY

COURSES LEN 5, LEN 8, LEN 3

KEY REFERENCES

- U.S. Government Accountability Office, F-35 Aircraft: DOD and the Military Services Need to Reassess the Future Sustainment Strategy, GAO-23-105341 (Sept. 2023) — 2,500 planned aircraft and a lifecycle cost exceeding $1.7 trillion, $1.3 trillion of it in operating and support.
- GAO-23-105341 (2023) — 55% fleet mission-capable rate (March 2023); over 10,000 components awaiting repair; 72-day average depot turnaround; depot stand-up behind schedule.
- GAO-23-105341 (2023) — sustainment shortfall traced to maintainer shortages, lack of military access to technical data, and contractor dependency.

> LENS LESSON F-35 is the live, current example of fielding a platform faster than its capability infrastructure can be built. The aircraft is the easy part; the maintainers are the hard part. A decade of program schedules treated maintainer training as a follow-on cost, not a fielding gate. The fleet is now operating at half of its design readiness, and the program of record is more than a trillion dollars over its 2018 estimate.

CASE 6 AVIATION 1989

Kegworth / British Midland 92

47 killed and 74 seriously injured when the crew shut down the wrong engine

THE FULL CASE, IN FIVE BEATS

Background	New 737-400 variant with redesigned cockpit; conversion was brief and largely self-directed, communicated as reading
What Happened	Fan blade fractured left engine; crew shut down the right reasoning from older 737 model
The Investigation	AAIB cited misapplied mental model, an unreadable vibration indicator, and the silent cabin-to-flight-deck channel
The Capability Gap	Variant treated as incremental change; manual notes never overwrote the prior reflex under emergency pressure
Aftermath & Reform	AAIB issued 31 recommendations spanning conversion training, instrument design, CRM, and crashworthiness

CONNECTIVITY

COURSES LEN 5, LEN 8

KEY REFERENCES

- Air Accidents Investigation Branch, Report on the accident to Boeing 737-400 G-OBME near Kegworth, Leicestershire, on 8 January 1989, AAIB Aircraft Accident Report 4/90 (1990) — aircraft, route, and the limited conversion training; see also U.S. FAA, Lessons Learned: G-OBME.
- AAIB Report 4/90 (1990) — accident sequence: left-engine fan-blade failure, shutdown of the serviceable right engine, and total power loss on approach; 47 killed, 74 seriously injured.
- AAIB Report 4/90 (1990) — the older-737 bleed-air mental model misapplied to the 737-400's changed configuration.

> LENS LESSON Kegworth is the textbook example of Capability Degradation Under System Change. The pilots were excellent. Their mental model was excellent — for the airframe they had flown the previous week. The difference training had not been engineered to disturb the prior model with enough force to keep it from being misapplied. The new airframe was treated as an incremental change. The crew's response revealed it was a categorical one.

CASE 7 DEFENSE 1991 – 2014

Military Fratricide — Desert Storm to Afghanistan

24% of U.S. KIA in Desert Storm from friendly fire (35 of 146) — an order of magnitude above the historical baseline

THE FULL CASE, IN FIVE BEATS

Background	Shrader's contested under-2% baseline became the yardstick that made any later rate look catastrophic
What Happened	A quarter of U.S. KIA died to friendly fire; A-10s struck Marines and British Warriors
The Investigation	Reviews cited combat chaos, weak awareness, fire-control lapses, and the absence of a shared incident database
The Capability Gap	Fratricide is integration across many systems; no single office owns what keeps the rate down
Aftermath & Reform	Better IFF and blue-force tracking followed, yet rates never settled at a confidently low level

CONNECTIVITY

COURSES LEN 5, LEN 2, LEN 8

KEY REFERENCES

- C. R. Shrader, Amicicide: The Problem of Friendly Fire in Modern War (U.S. Army Combat Studies Institute, 1982) — the under-2% estimate from 269 incidents, a baseline later challenged as too low (with estimates nearer 10–15%).
- "Friendly Fire: Facts, Myths and Misperceptions," U.S. Naval Institute Proceedings (June 1994) — Desert Storm: 35 of 146 KIA (24%) and 72 of 467 wounded (15%) by friendly fire; critique of the 2% "historical norm."
- Khafji and Warrior fratricide incidents (Feb. 1991) — see USNI Proceedings (1994) and Time, "They Didn't Have to Die". (Per-incident casualty figures vary across sources; see AUDIT.)

> LENS LESSON Fratricide is a multi-decade capability problem that resists single- intervention solutions. Each of the contributing causes — situational awareness, fire-control discipline, combat identification, joint coordination — has been the subject of dedicated programs. The rate persists because the integration across the contributors is itself the capability, and integration is the hardest property to engineer by program.

CASE 8 HEALTHCARE 2003–2017

ACGME 80-Hour Resident Duty-Hour Reform

ACGME capped U.S. resident physician work hours at 80/week to reduce fatigue-related errors; subsequent RCTs (FIRST, iCOMPARE) showed mixed effects on patient outcomes and increased hand-off-related errors

THE FULL CASE, IN FIVE BEATS

Background	Libby Zion's 1984 death made resident fatigue a decades-long argument; Bell Commission set early limits
The Intervention	ACGME capped resident work at 80 hours weekly and limited first-year shifts to 16 hours
How It Worked	Cutting hours multiplied hand-offs and cost continuity; trainees often felt less prepared, not rested
The Evidence	FIRST and iCOMPARE found flexible schedules non-inferior; ACGME relaxed the intern cap in 2017
What Transferred	Keystone, CRM, and the surgical checklist engineered supervision and hand-offs alongside the behavioral change

CONNECTIVITY

COURSES LEN 5, LEN 4, LEN 10, LEN 8

KEY REFERENCES

- B. H. Lerner, The Libby Zion Case and the Reform of Medical Education (2006); and the 1989 New York State (Bell Commission) duty-hour regulations — the origin of the reform.
- Accreditation Council for Graduate Medical Education, Common Program Requirements (2003 and 2011 revisions) — the 80-hour weekly cap and the 16-hour first-year shift limit.
- D. A. Asch et al., "Resident Duty Hours and Medical Education Policy," NEJM 370: 1671–1673 (2014); Institute of Medicine (Ulmer, Wolman & Johns, eds.), Resident Duty Hours: Enhancing Sleep, Supervision, and Safety (2008) — hand-offs and continuity trade-offs.

> LENS LESSON The clearest healthcare case in the dataset of a single-variable intervention into a multi-variable capability system. Pairs with Case 7 (fratricide) and Case 26 (V-22). The success cases — Keystone (14), CRM (12), Korean Air (23) — engineered supervisory, hand-off, and measurement architecture **together with** the behavioral change.

CASE 9 AVIATION 2009

Colgan Air Flight 3407

50 killed near Buffalo; precipitated the FAA's 1,500-hour rule and the Pilot Records Database

THE FULL CASE, IN FIVE BEATS

Background	Captain's documented training failures known to the carrier but not to the regulator
What Happened	On approach to Buffalo the captain pulled back at the stick shaker and stalled
The Investigation	NTSB cited fatigue, weak training, and a hiring system blind to the captain's history
The Capability Gap	Pilot training history existed in files but never reached the hiring or licensing decision
Aftermath & Reform	Families drove the 2010 law raising hours to 1,500 and creating the Pilot Records Database

CONNECTIVITY

COURSES LEN 4, LEN 5, LEN 8

KEY REFERENCES

- NTSB, Aircraft Accident Report: Colgan Air Flight 3407, NTSB/AAR-10/01 (2010) — probable cause (quoted).
- NTSB/AAR-10/01 (2010) — the captain’s training history, and the first officer’s fatigue and pay.
- NTSB/AAR-10/01 (2010) — the information-flow gap between known pilot history and the hiring decision.

> LENS LESSON Colgan is the canonical case for the gap between the information an institution holds about its operators and the information that reaches the decisions about whether those operators should be in that seat. The data was already in the system — multiple failed training events. The data flow that would have made that data actionable at the hiring decision did not exist.

CASE 10 AVIATION 2013

Asiana Airlines Flight 214

3 killed, 187 injured; Boeing 777 struck the seawall short of SFO runway 28L

THE FULL CASE, IN FIVE BEATS

Background	Captain new to the 777 hand flew a visual approach with the glideslope out of service
What Happened	Autothrottle sat in HOLD; airspeed decayed silently to 103 knots and the 777 struck the seawall
The Investigation	NTSB cited mismanaged approach, weak airspeed monitoring, and a faulty mental model of the automation
The Capability Gap	The mode the autothrottle was in and the mode the crew believed diverged silently without annunciation
Aftermath & Reform	Recommendations targeted autoflight training, energy state monitoring, and mode annunciation design

CONNECTIVITY

COURSES LEN 5, LEN 2

KEY REFERENCES

- NTSB, Aircraft Accident Report: Asiana Airlines Flight 214, NTSB/AAR-14/01 (2014) — probable cause and contributing factors (quoted).
- NTSB/AAR-14/01 (2014) — the autothrottle HOLD reversion and airspeed decay to 103 kt.
- NTSB/AAR-14/01 (2014) — the captain’s mental model of the autothrottle.

> LENS LESSON Asiana 214 is the aviation case for the LENS Human-AI Teaming proposition that automation transparency is a capability deliverable. The mode the autothrottle was actually in and the mode the crew believed it was in diverged silently — a textbook automation surprise. The training, the documentation, and the interface together did not surface the divergence.

CASE 11 DEFENSE 1941 – 1943

Mark 14 Torpedo Failures

Persistent torpedo failures across the first ~20 months of the Pacific War; resolved only after fleet-level testing forced acknowledgment of the defects

THE FULL CASE, IN FIVE BEATS

Background	The Bureau of Ordnance had effectively forbidden live trials, so the Mark 14 went to war unproven
What Happened	Submarine crews reported deep runs, premature detonations, and crushed contact pins from early 1942
The Investigation	Lockwood ordered fleet tests; Tinosa's eleven duds on a stopped target finally separated the three defects
The Capability Gap	The binding gap was a channel by which the bureau could be made to hear what the boats already knew
Aftermath & Reform	By late 1943 the three defects were corrected; the episode became the canonical insulated bureau case

CONNECTIVITY

COURSES LEN 10, LEN 7, LEN 8

KEY REFERENCES

- Blair, C. (1975), Silent Victory: The U.S. Submarine War Against Japan — operator reports and the Bureau's resistance (paraphrased).
- Rowland, B. & Boyd, W. (1953), U.S. Navy Bureau of Ordnance in World War II — testing policy and the three defects.
- Blair (1975) — the USS Tinosa's July 1943 attack (eleven consecutive duds on a stopped target) and Lockwood's fleet-level testing.

> LENS LESSON The Mark 14 is the canonical Navy case for the institutional refusal to believe operator feedback. The capability that was missing was not at the front. It was at the bureau that owned the weapon. The cost of the refusal was paid by the submariners forced to use it until the bureau yielded. The eventual fixes — re-setting depth calibration, replacing the magnetic exploder, redesigning the contact pin — were technical and small. The reform was institutional and slow. The capability that had to be built was the channel by which the bureau could be made to hear what the boats already knew.

CASE 12 DEFENSE 1980

Operation Eagle Claw

8 servicemembers killed in Iran; mission to rescue 52 American hostages aborted; catalyst for the founding of U.S. Special Operations Command

THE FULL CASE, IN FIVE BEATS

Background	No standing joint command existed; the rescue force was drawn ad hoc from four services
What Happened	Three RH-53s failed at Desert One; on withdrawal a helicopter struck a C-130 killing eight
The Investigation	Holloway named ad hoc assembly, untrained aircrews, and a cover story driven aircraft choice
The Capability Gap	Each service was competent in its lane; cross-service integration as a deliverable did not exist
Aftermath & Reform	Reform built JSOC, Goldwater Nichols in 1986, and USSOCOM in 1987 as standing joint capability

CONNECTIVITY

COURSES LEN 5, LEN 8

KEY REFERENCES

- Holloway Special Operations Review Group, Rescue Mission Report (1980) — the ad-hoc assembly and equipment choices (paraphrased).
- Holloway Commission (1980) — the Desert One sequence and the helicopter–C-130 collision.
- Goldwater-Nichols Department of Defense Reorganization Act of 1986, Pub. L. 99-433.

> LENS LESSON Eagle Claw is the canonical case in U.S. defense for the absence of institutionalized cross-service capability. Each service was competent inside its own boundary; the integration across them did not exist as a deliverable — until the reform that followed built the institution the mission had needed and not had.

CASE 13 AVIATION 2005

Helios Airways Flight 522

121 killed; cabin failed to pressurize after a maintenance setting was left on "manual" and the crew misread the warning

THE FULL CASE, IN FIVE BEATS

Background	A maintenance leak check left the pressurization selector on manual and pre flight checks missed it
What Happened	Cabin failed to pressurize; the crew treated the warning horn as a configuration fault and went hypoxic
The Investigation	Hellenic investigators called the warning misidentification a critical irrecoverable error
The Capability Gap	One horn carried two meanings without differentiation while training drilled only the ground meaning
Aftermath & Reform	Reforms changed Boeing warning logic, pressurization checklists, and pre flight verification training

CONNECTIVITY

COURSES LEN 5, LEN 2

KEY REFERENCES

- Hellenic Air Accident Investigation & Aviation Safety Board, Aircraft Accident Report 11/2006 — probable cause and the warning misinterpretation (paraphrased).
- AAIASB Report 11/2006 — the manual pressurization selector and failure to pressurize.
- AAIASB Report 11/2006 — the dual-meaning warning horn and the training emphasis.

> LENS LESSON Helios 522 is the textbook example of a single cue carrying two meanings without differentiation. The interface lied by ambiguity. The training filled in only one meaning. The combination produced a twenty-minute window in which the crew was solving the wrong problem.

CASE 14 AVIATION 1996

AeroPerú Flight 603

70 killed; Boeing 757 crashed into the Pacific after maintenance tape was left over static ports

THE FULL CASE, IN FIVE BEATS

Background	Ground crew taped over the static ports during cleaning and the tape was never removed
What Happened	All air data instruments fed false contradictory readings; believing they were higher the crew struck the Pacific
The Investigation	Peruvian investigators found no independent reference since every instrument fed off the blocked source
The Capability Gap	Apparent cockpit redundancy was illusory at the source and the training had assumed the case away
Aftermath & Reform	Industry tightened conspicuous covering and removal verification for static ports and pitot tubes

CONNECTIVITY

COURSES LEN 5, LEN 2

KEY REFERENCES

- Peru Dirección General de Aeronáutica Civil, accident investigation board, final report on AeroPerú 603 (December 1996; with NTSB/FAA/Boeing participation) — primary cause and the instrument cascade (paraphrased).
- Peru CIAA report (1996) — the static-port tape and the contradictory warnings.
- Peru CIAA report (1996) — the absence of an independent cockpit reference.

> LENS LESSON AeroPerú 603 is the case for redundancy that is not redundancy. Every cockpit indicator depended on the same physical sensor. The apparent redundancy in the cockpit was an illusion at the source. The training did not include the failure case because the failure case had been assumed not to occur.

CASE 15 AVIATION 2019

Atlas Air Flight 3591

3 killed; Boeing 767 cargo flight crashed in Texas after the first officer mistakenly activated go-around mode

THE FULL CASE, IN FIVE BEATS

Background	The first officer's prior training failures were undisclosed and PRIA's window was too short to surface them
What Happened	After an inadvertent go around activation a somatogravic illusion drove a hard push and a dive into Trinity Bay
The Investigation	NTSB cited startle, spatial disorientation, the training pattern, and Atlas's inability to see the full record
The Capability Gap	PRD authorization without full coverage left the same Colgan era information flow gap partly open in 2019
Aftermath & Reform	The PRD final rule took effect for Part 121 in 2022 with full historical coverage phasing in through 2024

CONNECTIVITY

COURSES LEN 4, LEN 8

KEY REFERENCES

- NTSB, Aircraft Accident Report: Atlas Air Flight 3591, NTSB/AAR-20/02 (2020) — probable cause and contributing factors (paraphrased).
- NTSB/AAR-20/02 (2020) — the inadvertent go-around activation and the dive into Trinity Bay.
- NTSB/AAR-20/02 (2020) — the first officer’s training history and Atlas’s record-access limits.

> LENS LESSON Atlas Air 3591 is the iteration test of the Colgan reform. The reform partially worked: the PRD exists. The reform was incomplete: not all carriers were covered. The case is a cautionary worked example of why partial implementation of a regulatory remedy leaves the original capability gap partially open.

CASE 16 AVIATION 2015

TransAsia Airways Flight 235

43 killed in Taiwan; crew shut down the only working engine after the other failed

THE FULL CASE, IN FIVE BEATS

Background	Captain had failed recent proficiency checks and was flagged for weak abnormal procedure handling
What Happened	Forty three seconds after takeoff the right engine auto feathered and the crew shut down the working left engine
The Investigation	Taiwan ASC found the crew shut down the wrong engine in about fifteen seconds from memory not the checklist
The Capability Gap	Under acute time pressure crews default to memory; incomplete memory drives action away from the checklist
Aftermath & Reform	TransAsia ceased operations in 2016; the case argues the wrong engine pattern is an un engineered intervention

CONNECTIVITY

COURSES LEN 5, LEN 2

KEY REFERENCES

- Aviation Safety Council (Taiwan), Aviation Occurrence Report: TransAsia Airways Flight GE235, final report (2016) — findings and the wrong-engine shutdown (paraphrased).
- ASC final report (2016) — the autofeather event and the crash sequence.
- ASC final report (2016) — non-use of the checklist and the captain’s proficiency record.

> LENS LESSON TransAsia 235 is the modern recurrence of the Kegworth pattern under stress. Crews under acute time pressure default to memory; if the memory is incomplete or wrong, the action follows the memory. The capability gap was the same as in 1989. The intervention pattern needed is also the same.

CASE 17 AVIATION 2018 – 2019

Boeing 737 MAX / MCAS

346 killed across two crashes — Lion Air 610 and Ethiopian Airlines 302; 20-month worldwide grounding; ~\$20B direct cost; FAA delegated-authority reform under the Aircraft Certification, Safety, and Accountability Act (2020)

THE FULL CASE, IN FIVE BEATS

Background	Re-engined 737 sold on no new pilot training; MCAS hid the handling change
What Happened	Single-sensor MCAS forced two jets down, killing 346 across both crashes
The Investigation	Internal records show training omission was deliberate; certification was largely self-delegated
The Capability Gap	Training absence was a contract deliverable, not an oversight; pilots engineered out
Aftermath & Reform	Twenty-month grounding; MCAS redesigned, simulator training required, FAA oversight tightened

CONNECTIVITY

INDUCED	1.1
LENS	D1+D4/PT4
CLO	CLO-1, CLO-4
COURSES	LEN 1, LEN 5, LEN 2

KEY REFERENCES

- U.S. House Committee on Transportation and Infrastructure, The Boeing 737 MAX Aircraft: Costs, Consequences, and Lessons from Its Design, Development, and Certification (Final Committee Report, Sept. 2020) — the MCAS omission, the 2013 minutes, the 2016 survey, and the “diminished safety” conclusion.
- U.S. House report (2020) and MCAS design summary — MCAS firing on a single angle-of-attack sensor; the Lion Air 610 (189 killed) and Ethiopian 302 (157 killed) accident sequences.
- U.S. House Transportation Committee report (2020) — Boeing internal communications and the certification-and-training-impact reasoning (quoted).

> LENS LESSON The 737 MAX is the documentary record of a design choice to remove human capability to avoid the regulatory and commercial cost of sustaining it. The pilots were not undertrained by accident — they were undertrained as a **requirement**. The omission of training was a contract deliverable. Once that is understood, the accident becomes legible as an engineering decision rather than a training failure.

CASE 18 HEALTHCARE TECHNOLOGY 1985 – 1987

Therac-25

Six known massive radiation overdoses; at least three deaths; canonical case in software safety engineering

THE FULL CASE, IN FIVE BEATS

Background	Hardware interlocks removed to save cost; safety case migrated silently into software
What Happened	Race condition let fast operators fire full beam with no target
The Investigation	Manufacturer denied harm; Leveson and Turner found systemic, not single-bug, failure
The Capability Gap	Interlock was the safety case; nothing took its load when removed
Aftermath & Reform	Founded software-safety engineering; deleted safeguards relocate hazard to whatever replaces them

CONNECTIVITY

COURSES LEN 5, LEN 7, LEN 2

KEY REFERENCES

- N. G. Leveson & C. S. Turner, “An Investigation of the Therac-25 Accidents,” IEEE Computer 26(7): 18–41 (1993). doi:10.1109/MC.1993.274940 — the removed hardware interlocks and the adapted software.
- Leveson & Turner (1993) — the race condition, the uninformative “Malfunction 54”, overdoses up to 100×, six accidents, and at least three deaths.
- Leveson & Turner (1993); N. G. Leveson, Safeware: System Safety and Computers (Addison-Wesley, 1995) — the manufacturer’s denial and the systemic findings.

> LENS LESSON Therac-25 is the moment when removing the human safety margin became visible as a design choice. The hardware interlock was not redundant — it was the safety case. When it was removed, no equivalent capability was put in its place. The operator was retained in the system but stripped of the ability to detect what it was doing wrong. Capability engineering would have asked, before removing the interlock, **what function takes its load?**

CASE 19 DEFENSE 1991

Patriot Missile / Dhahran

28 U.S. soldiers killed in their barracks; roughly 100 wounded

THE FULL CASE, IN FIVE BEATS

Background	Built for short European engagements; Gulf War demanded continuous fixed-site operation
What Happened	Fixed-point time error grew with uptime; Scud passed unengaged, 28 killed
The Investigation	Israeli warning preceded strike by weeks; patch arrived one day late
The Capability Gap	Original concept did not require sustained accuracy; assumption never traveled with redeployment
Aftermath & Reform	Patch distributed, reboot intervals defined; canonical lesson about assumptions outliving their context

CONNECTIVITY

COURSES LEN 5, LEN 8, LEN 9, LEN 6

KEY REFERENCES

- U.S. General Accounting Office, Patriot Missile Defense: Software Problem Led to System Failure at Dhahran, Saudi Arabia, GAO/IMTEC-92-26 (1992) — the design-context mismatch and continuous-operation use.
- R. Skeel, “Roundoff Error and the Patriot Missile,” SIAM News 25(4): 11 (1992) — the 24-bit fixed-point time truncation and the 0.34-second range-gate drift after 100 hours.
- GAO/IMTEC-92-26 (1992) — the prior Israeli warning, the software patch arriving in Dhahran on 26 February (the day after the strike), and the 28 killed.

> LENS LESSON Patriot is the canonical Designed-Out case from defense. The capability to maintain accuracy under sustained operation was omitted because the original concept of operations did not anticipate it. When the concept changed, the assumption did not travel. The system did not fail; the transition from one operational context to another did, and there was no capability infrastructure to catch it.

CASE 20 INDUSTRIAL 2008 – 2023

Takata Airbag Inflators

More than 30 deaths and hundreds of injuries linked to inflator ruptures; largest automotive recall in history

THE FULL CASE, IN FIVE BEATS

Background	Inflators built around cheap, unstable ammonium nitrate without the desiccant competitors added.
What Happened	Heat and humidity degraded the propellant; ruptures sprayed shrapnel and killed drivers.
The Investigation	Takata reported ruptures as isolated anomalies, manipulated test data, and pleaded guilty to fraud.
The Capability Gap	Designed-out capabilities were the desiccant and the regulator's independent verification pipeline.
Aftermath & Reform	Recall outlived the bankrupt firm and pushed regulators toward coordinated replacement management.

CONNECTIVITY

INDUCED	2.4
LENS	D3/PT3
CLO	CLO-3
COURSES	LEN 4, LEN 7

KEY REFERENCES

- U.S. National Highway Traffic Safety Administration, Takata air-bag inflator recall coordination materials — the ammonium-nitrate-without-desiccant design and the propellant-degradation rupture mechanism.
- NHTSA recall record and investigative reporting (Reuters, New York Times) — 100M+ inflators across 19 automakers; the largest automotive recall in history; deaths and injuries from ruptures.
- U.S. Department of Justice, settlement and guilty plea with Takata Corporation (2017) — wire fraud, $1B in fine and restitution, and Takata’s subsequent bankruptcy.

> LENS LESSON Takata is the modern Pinto: an engineering choice to omit a stabilizing component, internal data showing the consequence, and a regulatory architecture that treated the manufacturer’s representation as authoritative rather than as one input. The capability gap was at the manufacturer’s evidence pipeline as much as at the manufacturer’s bench.

CASE 21 INDUSTRIAL 2002 – 2014

GM Ignition Switch

124 deaths attributed; ~2.6M vehicles recalled for the defective switch; $900M federal forfeiture (DOJ, 2015); the fix existed for eight years before the recall

THE FULL CASE, IN FIVE BEATS

Background	Compact-car switches lacked detent torque; jostles cut power and disarmed airbags.
What Happened	Engineer approved a redesigned switch in 2006 without changing its part number.
The Investigation	Valukas found the GM nod, the GM salute, and failure to use escalation processes.
The Capability Gap	Designed out was the pathway from a known fix to the recall decision.
Aftermath & Reform	Barra restructured safety governance and the case became a teaching example in psychological safety.

CONNECTIVITY

INDUCED	5.4
LENS	D4/PT5
CLO	CLO-3
COURSES	LEN 4, LEN 7, LEN 8, LEN 6

KEY REFERENCES

- A. R. Valukas, Report to the Board of Directors of General Motors Company Regarding Ignition Switch Recalls (Jenner & Block, 2014) — the 2002 identification of the defect and the switch-torque mechanism.
- U.S. NHTSA investigation of the GM ignition switch (2014) and the GM recall record — 2.6 million vehicles; 124 deaths via the GM compensation fund.
- Valukas Report (2014) — the unchanged part number, the "GM nod" and "GM salute," and the failure to use established escalation processes (quoted).

> LENS LESSON The GM ignition switch case is the canonical example of a corporate organizational structure that suppressed the upward flow of safety information by procedural design. The fix existed for eight years before the recall. The institutional path between the fix and the recall did not.

CASE 22 SPACE / AEROSPACE 1999

Mars Climate Orbiter — Unit Mismatch

~$125M spacecraft lost on approach to Mars; ground software produced thrust output in pound-force while navigation expected newtons

THE FULL CASE, IN FIVE BEATS

Background	Faster, better, cheaper mission split between Lockheed building and JPL navigating the orbiter.
What Happened	Ground software reported pound-force seconds while navigation expected newton-seconds; orbiter burned up at Mars.
The Investigation	Board found no individual blunder; the unverified boundary between two correct halves failed.
The Capability Gap	Missing capability was an owned, specified, verified interface between the two organizations.
Aftermath & Reform	Loss tightened interface management and became the canonical software case of interface-as-requirement.

CONNECTIVITY

COURSES LEN 5, LEN 8

KEY REFERENCES

- NASA, Mars Climate Orbiter Mishap Investigation Board: Phase I Report (Nov. 1999) — mission, the Lockheed/JPL split, and the program context.
- NASA MCO MIB Report (1999) — the pound-force-second vs newton-second mismatch (factor 4.45), the accumulated navigation error, and atmospheric destruction on 23 Sept. 1999 (root-cause statement quoted).
- NASA MCO MIB Report (1999) — the missing verified interface specification, the absent end-to-end validation, and the unresolved cruise-trajectory anomalies.

> LENS LESSON Mars Climate Orbiter is the textbook case for interface boundaries as engineering deliverables. The contractors did their work. The boundary between them did not have an owner. The capability that was missing was the interface-specification verification step.

CASE 23 TECHNOLOGY 2012

Knight Capital Trading Loss

$440M loss in 45 minutes; firm forced to seek emergency capital and was effectively acquired within months

THE FULL CASE, IN FIVE BEATS

Background	Major market maker prepared routine update for the NYSE Retail Liquidity Program launch.
What Happened	Eighth server missed the update; a reused flag woke dormant Power Peg code.
The Investigation	SEC found no deployment verification, no consistency check, and no controls to halt orders.
The Capability Gap	Designed-out capability was deployment verification; dead code was a standing option on failure.
Aftermath & Reform	Case became canonical for deployment as engineering deliverable and sharpened market-access controls.

CONNECTIVITY

INDUCED	7.1
LENS	D1/PT3
CLO	CLO-1, CLO-3
COURSES	LEN 5, LEN 9

KEY REFERENCES

- U.S. Securities and Exchange Commission, Order Instituting Administrative and Cease-and-Desist Proceedings, In re Knight Capital Americas LLC (2013) — the firm, the Retail Liquidity Program launch, and the deployment.
- SEC Order (2013) and Knight Capital 8-K filing (2012) — the missed eighth server, the reactivation of the dormant “Power Peg” code, the $440 million loss in 45 minutes, and the near-collapse.
- SEC Order (2013) — the absence of a second-technician deployment verification, the lack of an automated code-consistency check, inadequate order controls, and the $12 million penalty (quoted).

> LENS LESSON Knight Capital is the financial-industry version of Mars Climate Orbiter (Case 22): a small, unspecified boundary inside a large system that took the institution down. The capability that was missing was deployment verification. The dead code was the proximate trigger; the absent procedure was the cause.

CASE 24 ENERGY 2010

Deepwater Horizon

11 killed; largest marine oil spill in U.S. history; $65B+ in damages

THE FULL CASE, IN FIVE BEATS

Background	Macondo blowout killed eleven and unleashed the largest U.S. marine oil spill
What Happened	Crew misread the negative-pressure test, accepted a wrong explanation, and displaced the mud
The Investigation	Commission found human error so systematic it indicted the whole industry's safety culture
The Capability Gap	Procedure, training, supervision, contractors, and equipment had drifted independently until none caught the others
Aftermath & Reform	Regulator was split, well-integrity rules tightened, and the multi-layer drift named as the deeper lesson

CONNECTIVITY

INDUCED	3.1
LENS	D3/PT5
CLO	CLO-3, CLO-4
COURSES	LEN 5, LEN 8, LEN 3

KEY REFERENCES

- National Commission on the BP Deepwater Horizon Oil Spill, Deep Water: The Gulf Oil Disaster and the Future of Offshore Drilling (Report to the President, 2011) — human error as root cause; the misread negative-pressure test.
- National Commission (2011) — the well-control sequence and mud-displacement decision.
- National Commission (2011) — “systematic failures in risk management... place in doubt the safety culture of the entire industry” (quoted); U.S. Chemical Safety Board, Drilling Rig Explosion and Fire at the Macondo Well (final volumes, 2014–2016) — process-safety vs. personal-safety distinction and BP’s industry-leading lost-time-injury rate as a misleading indicator.

> LENS LESSON Deepwater Horizon is the canonical multi-layer failure: training, procedure, supervision, contractor-coordination, and equipment all drifted independently until none caught the others. The negative-pressure test was the proximate trigger, but the system as a whole had been operating with accumulated procedural debt for years. The capability to recognize an unsafe well-state simply was not present at the moment it was needed. The CSB’s process-safety / personal-safety distinction is load-bearing: BP’s industry-leading lost-time-injury record was a measurement of the wrong construct, and reading it as safety was itself a normalization.

CASE 25 SPACE / AEROSPACE 1986 / 2003

Challenger & Columbia

14 astronauts killed (7 per accident); 17 years between identical organizational pathologies

THE FULL CASE, IN FIVE BEATS

Background	Shuttle flew with known O-ring and foam flaws reclassified flight by flight as routine
What Happened	Challenger broke apart on a cold morning; Columbia disintegrated on reentry seventeen years later
The Investigation	Both boards found the same suppressed safety concerns and the same culture intact
The Capability Gap	Vaughan named normalization of deviance; the diagnosis sat on record without engineered remediation
Aftermath & Reform	Reforms followed each loss, yet the cultural mechanism stayed unrepaired until the Shuttle retired

CONNECTIVITY

INDUCED	7.4
LENS	D4/PT4
CLO	CLO-1, CLO-4
COURSES	LEN 1, LEN 7, LEN 8, LEN 3

KEY REFERENCES

- Rogers Commission, Report of the Presidential Commission on the Space Shuttle Challenger Accident (1986) — the O-ring failure, the Thiokol teleconference, and the cold-weather launch decision.
- D. Vaughan, The Challenger Launch Decision: Risky Technology, Culture, and Deviance at NASA (Univ. of Chicago Press, 1996) — normalization of deviance; the working group's drift of decision rules.
- Columbia Accident Investigation Board, Report Vol. I (2003) — the foam strike, the sixteen prior missions of foam-shedding filed as "in-family," the recurrence of the cultural pattern, and the call for an independent technical authority.

> LENS LESSON Challenger/Columbia is the strongest documentary evidence that organizational culture is an engineerable property of a system, and that diagnoses without engineered remediations decay. The same pathology, twice, seventeen years apart, in the same institution, with the same human cost — and with the diagnosis already on the record. Vaughan's "normalization of deviance" names the mechanism and is the load-bearing analytic claim across both accidents. Capability engineering treats culture as a deliverable.

CASE 26 DEFENSE 1991 – PRESENT

V-22 Osprey

~65 killed across ~17 hull-loss accidents since 1991; serious-mishap rate above comparable fleets (GAO-26-108905, 2025); some fixes stretch to the 2030s

THE FULL CASE, IN FIVE BEATS

Background	Tiltrotor shared awkwardly across three services has logged seventeen hull losses since 1991
What Happened	Yakushima crash killed eight after gear cracks and a pilot pressing through warnings
The Investigation	GAO and NAVAIR found years of unaddressed risk, elevated mishap rates, eighteen-year gearbox lag
The Capability Gap	Documented shortfall persists because parallel service adjustments never converge on resolution
Aftermath & Reform	Groundings and redesigns continue while full gearbox fixes stretch toward 2034

CONNECTIVITY

COURSES LEN 5, LEN 8, LEN 3

KEY REFERENCES

- Compiled V-22 accident record — 17 hull losses and 65 fatalities since 1991, including the 2000 Marana test crash (19 Marines).
- U.S. Air Force Accident Investigation Board findings, via USNI News (Aug. 2024) — the 29 Nov. 2023 Yakushima CV-22B crash: transmission-gear cracks (X-53 inclusions) and continued flight despite warnings.
- U.S. GAO, Osprey Aircraft: Additional Oversight and Information Sharing Would Improve Safety Efforts, GAO-26-108905 (Dec. 2025) — the joint program office's failure to assess and address risks.

> LENS LESSON V-22 demonstrates the steady-state version of normalization of deviance: a platform whose shortfall has been documented, reviewed, and acted on across multiple administrations without producing sustained improvement. The capability gap has itself been normalized. Each incident is treated as an event rather than as a sample from a distribution the program continues to draw from.

CASE 27 ENERGY INDUSTRIAL 2005

Texas City BP Refinery Explosion

15 killed, 180 injured at BP's Texas City refinery; CSB found systemic safety-culture decay

THE FULL CASE, IN FIVE BEATS

Background	Deferred maintenance and degraded instruments tolerated; personal-injury rates among the industry's best
What Happened	Startup overfilled tower past safe level; vented vapor ignited, killing fifteen workers in trailers
The Investigation	CSB drew the distinction reshaping the field; personal-safety indicators are not process-safety indicators
The Capability Gap	Excellent injury record made process-safety drift harder to see; wrong measurement worse than none
Aftermath & Reform	Baker Panel followed; process-safety distinction entered mainstream regulation; count what can kill you

CONNECTIVITY

INDUCED	5.4
LENS	D3/PT5
CLO	CLO-3, CLO-4
COURSES	LEN 4, LEN 7, LEN 3

KEY REFERENCES

- U.S. Chemical Safety and Hazard Investigation Board, Refinery Explosion and Fire, Investigation Report 2005-04-I-TX (2007) — the startup, malfunctioning instruments, and 15 killed / 180 injured.
- CSB (2007) — accumulated safety-culture decay, deferred maintenance, and the siting of occupied trailers beside a hazardous unit.
- CSB (2007) — “indicators of personal safety are not indicators of process safety” (quoted).

> LENS LESSON Texas City is the foundational evidence that organizations can be measuring the wrong thing and reporting excellent performance while their actual capability gap widens. Personal-safety metrics had no information about process-safety state. The signal regime was blind in the dimension that killed people.

CASE 28 ENERGY 2002

Davis-Besse Reactor Head Corrosion

Football-sized cavity discovered in the reactor pressure-vessel head; near-miss; ~$600M recovery and extended outage

THE FULL CASE, IN FIVE BEATS

Background	Post-TMI U.S. nuclear regime knew the boric-acid corrosion hazard across the reactor fleet
What Happened	Refueling outage revealed football-sized cavity through the head; only thin cladding held back coolant
The Investigation	FirstEnergy won an NRC inspection deferral; OIG found economics weighted over safety
The Capability Gap	Plant engineering was sound; independence of the oversight layer above operations had quietly eroded
Aftermath & Reform	INPO and NRC tightened head-inspection requirements; institutional capability erodes if not re-engineered

CONNECTIVITY

COURSES LEN 7, LEN 8

KEY REFERENCES

- U.S. NRC Office of Inspector General, NRC’s Regulation of Davis-Besse Regarding Damage to the Reactor Vessel Head (Case No. 02-03S, December 2002) — the post-TMI regulatory regime and oversight failures.
- NRC event records and OIG (2002) — the 6-inch corrosion cavity, the remaining cladding, and the 2,200-psi coolant margin.
- NRC Bulletin 2001-01 and the FirstEnergy inspection-deferral decision aligned to the February 2002 outage.

> LENS LESSON Davis-Besse is the case for how regulator-operator dynamics can erode the capability of an industry that had previously demonstrated, through INPO, that it knew how to engineer safety. Regulatory capture is a capability failure at the institutional layer above operations.

CASE 29 HEALTHCARE 2005 – 2009

Mid Staffordshire NHS Foundation Trust

Excess deaths at Stafford Hospital documented across years; the Francis Inquiry produced 290 recommendations and restructured UK patient-safety governance

THE FULL CASE, IN FIVE BEATS

Background	Pursuing Foundation Trust status, the board cut ward staffing while reported targets stayed met
What Happened	Patients neglected for years in appalling conditions; mortality ran substantially above expected
The Investigation	Francis Inquiry produced 2,000 pages and 290 recommendations; system as a whole failed
The Capability Gap	Like the VA case, every governance layer acted on metrics; no one checked against patients
Aftermath & Reform	Berwick review pressed for learning over targets; reporting can run clean while reality fails

CONNECTIVITY

COURSES LEN 4, LEN 7, LEN 3

KEY REFERENCES

- R. Francis QC, Report of the Mid Staffordshire NHS Foundation Trust Public Inquiry (2013) — the staffing cuts and Foundation Trust targets.
- Healthcare Commission, Investigation into Mid Staffordshire NHS Foundation Trust (2009) — ward conditions and excess mortality.
- Francis QC (2013) — the 2,000-page report and 290 recommendations.

> LENS LESSON Mid Staffordshire is the British analog of the VA wait-time scandal (Case 84). Measurement and reality diverged over years; every layer of governance above the operating environment was acting on the measurement; patients paid the cost of the divergence. The Francis Inquiry recommendations changed how the NHS thinks about patient safety as an institutional capability.

CASE 30 ENERGY INDUSTRIAL 2006

Sago Mine Disaster

12 killed in a West Virginia coal-mine explosion; emergency-response failures compounded the initial event; MINER Act of 2006

THE FULL CASE, IN FIVE BEATS

Background	Seals, emergency plan, and self-rescue training were each marginally adequate, never tested at edge
What Happened	Lightning ignited methane; seals failed; twelve miners died of carbon-monoxide poisoning over hours
The Investigation	MSHA found inadequate seal design, plan, and lapsed training; combined shortcomings proved lethal together
The Capability Gap	No single failure was decisive; marginal-everywhere is itself a system-level hazard
Aftermath & Reform	MINER Act strengthened rescue requirements; mandated refuge chambers; reform addressed the combination, not one cause

CONNECTIVITY

COURSES LEN 5, LEN 8

KEY REFERENCES

- U.S. Mine Safety and Health Administration, Report of Investigation: Sago Mine (2007) — the seal design, emergency plan, and self-rescue training.
- MSHA (2007) — the explosion sequence: lightning ignition in the sealed area, seal failure, twelve dead and one survivor.
- MSHA (2007) — the inadequate seal design, emergency plan, and lapsed self-rescue training.

> LENS LESSON Sago is the case for the cumulative inadequacy pattern in industrial accidents. No single failure caused the disaster. Each failure on its own would have been recoverable. The combination was not, and the combination had been the operating condition of the mine for years.

CASE 31 ENERGY INDUSTRIAL 2010

Upper Big Branch Mine Explosion

29 killed in West Virginia coal mine; MSHA found systematic falsification of safety records; first U.S. mining-industry CEO convicted of a federal mine-safety charge (misdemeanor)

THE FULL CASE, IN FIVE BEATS

Background	Massey ran two sets of records; the clean log disabled the regulator's mechanism of sight
What Happened	Coal dust and methane ignited, killing twenty-nine; the sanitized records hid the conditions
The Investigation	MSHA and DOJ found suppressed readings and disabled monitors; CEO Blankenship convicted of misdemeanor conspiracy
The Capability Gap	Dual records were stable institutional practice spanning years; measurement engineered as deception
Aftermath & Reform	Conviction set accountability marker; unanswered is what independent signal could expose divergence live

CONNECTIVITY

COURSES LEN 4, LEN 7

KEY REFERENCES

- U.S. MSHA, Internal Review of MSHA's Actions at Upper Big Branch and the accident investigation (2011–2012) — the dual records and the 29 deaths.
- Governor's Independent Investigation Panel (J. McAtee, 2011) — mine conditions: methane, ventilation, and coal-dust inerting.
- MSHA and U.S. DOJ findings — suppressed methane readings, disabled monitors, and falsified pre-shift examinations.

> LENS LESSON Upper Big Branch is the case for deliberately engineered measurement deception. The dual-records practice was not error. It was capability design — by management, against the regulator. The case anchors the LENS argument that decision-grade evidence requires structural protection against the institution that produces it having a stake in what it reports.

CASE 32 ENERGY 2011

Fukushima Daiichi

Three reactor meltdowns following the Tōhoku earthquake and tsunami; ~160,000 people displaced; cleanup projected at $200B+

THE FULL CASE, IN FIVE BEATS

Background	Internal TEPCO assessments discussed larger historical waves; evidence never forced a higher seawall
What Happened	Tōhoku tsunami topped the seawall; inundated generators; three cores melted down, displacing thousands
The Investigation	Kurokawa's NAIIC called it made in Japan; Hatamura and IAEA emphasized under-estimated external hazards
The Capability Gap	Internal hazard evidence existed; Japan lacked an INPO-equivalent with independence to act on it
Aftermath & Reform	New independent Nuclear Regulation Authority created; capability institutions must be deliberately built, not assumed

CONNECTIVITY

COURSES LEN 7, LEN 8, LEN 3

KEY REFERENCES

- National Diet of Japan Fukushima Nuclear Accident Independent Investigation Commission (NAIIC; K. Kurokawa, chair), Report (2012) — the internal tsunami evidence and the regulatory-capture finding.
- NAIIC (2012) — the accident sequence: seawall overtopping, generator inundation, and three core meltdowns.
- NAIIC (2012) — the "made in Japan" cultural and regulatory-capture conclusion (quoted).

> LENS LESSON Fukushima is the post-TMI case that establishes that the INPO pattern (Case 107) is not self-executing. The U.S. industry built INPO; the Japanese industry did not. The cost of the difference, paid in 2011, is the strongest available evidence that capability institutions must be deliberately built, not assumed.

CASE 33 ENERGY 2003

Northeast Blackout

50 million people without power across eight U.S. states and Ontario; $6B+ economic loss; FERC Order 693 followed

THE FULL CASE, IN FIVE BEATS

Background	Eastern Interconnection rides through single-line loss; FirstEnergy's alarm system was silently failing for an hour
What Happened	An Ohio line sagged into a tree; silent alarms left operators blind; cascade blacked out fifty million
The Investigation	Task Force found inadequate situational awareness, vegetation lapses, weak coordinator authority; FERC Order 693 followed
The Capability Gap	Missing capability was the meta-monitor; silence is indistinguishable from a healthy, quiet grid
Aftermath & Reform	New mandatory reliability regime backed by audits and penalties; reliability is a deliverable, not best practice

CONNECTIVITY

COURSES LEN 2, LEN 8

KEY REFERENCES

- U.S.-Canada Power System Outage Task Force, Final Report on the August 14, 2003 Blackout in the United States and Canada (2004) — the tree contact, the silent alarm, and the cascade.
- Task Force (2004) — 50 million people affected across eight states and Ontario; the minute-by-minute sequence.
- Task Force (2004) — FirstEnergy “did not have adequate situational awareness,” plus the vegetation-management and reliability-coordinator findings (quoted).

> LENS LESSON The 2003 blackout is the canonical case for silent automation failure: a system that stops doing its job without telling its operators. The capability that was missing was the meta-monitor — the system that watches the monitor. The reform that followed treated grid-reliability compliance as a deliverable rather than as a best practice.

CASE 34 DEFENSE 1988

USS Vincennes & Iran Air Flight 655

290 civilians killed — the deadliest shootdown of a commercial airliner by a military force; precipitating case for Aegis CIC display and decision-aid doctrine reform, and a central reference in the human-AI teaming literature

THE FULL CASE, IN FIVE BEATS

Background	Aegis cruiser fought Iranian gunboats near a dual-use airfield in the crowded Persian Gulf
What Happened	Crew misread an ascending Airbus as a diving F-14 and fired; all 290 died
The Investigation	Aegis radar was correct; multiple operators independently saw descent under presumed-hostile framing
The Capability Gap	Interface left burden of overriding expectation to operators stripped of time by combat
Aftermath & Reform	Naval retrospective reframed loss as predictable teaming failure of unguarded decision aids

CONNECTIVITY

INDUCED 3.3
LENS D2/PT6
CLO CLO-5, CLO-2
COURSES LEN 5, LEN 2

KEY REFERENCES

- Rear Adm. W. Fogarty, Formal Investigation into the Circumstances Surrounding the Downing of Iran Air Flight 655 (US Navy, August 1988) — the engagement, the IFF mode-confusion findings, and the 290 deaths.
- Fogarty report (1988) — the Aegis SPY-1A radar functioned; the aircraft was ascending while the crew perceived a descent; “scenario fulfillment” as the psychological mechanism.
- Fogarty report (1988) — “human error under extreme stress,” confirmation bias, and “unconscious distortion of data” (quoted); shared-error finding across CIC operators.

> LENS LESSON Vincennes is the canonical case for human-AI teaming under stress. The system did not lie. The operators did not act in bad faith. The interface and the operational framing combined to make a particular misreading not just possible but likely. Correct performance was possible in principle and unsustainable in practice — and that gap is the engineering problem.

CASE 35 HEALTHCARE 2005 – PRESENT

EHR / CPOE Implementation

~$30B federal investment under HITECH; documented increase in pediatric ICU mortality post-CPOE at one institution; ongoing usability harm at scale

THE FULL CASE, IN FIVE BEATS

Background	HITECH poured thirty billion into EHRs built to billing specifications, not clinical workflow
What Happened	Disputed 2005 CPOE study tied pediatric ICU mortality to deployment; default and alert harms recurred
The Investigation	A 2023 KLAS survey across hundreds of hospitals tied usability scores to patient safety outcomes
The Capability Gap	Systems work for billing because that was the specification; clinical workflow was never engineered
Aftermath & Reform	Usability guidance has matured but no regulatory regime monitors deployed EHR safety at scale

CONNECTIVITY

COURSES LEN 7, LEN 2, LEN 9

KEY REFERENCES

- Health Information Technology for Economic and Clinical Health (HITECH) Act (2009) — the $30B EHR incentive program.
- Y. Han et al., "Unexpected Increased Mortality After Implementation of a Commercially Sold CPOE System," Pediatrics 116(6): 1506–1512 (2005) — the disputed pediatric-ICU mortality result.
- KLAS Arch Collaborative, EHR usability and safety surveys (2023) — usability as the top clinician complaint, correlated with safety outcomes.

> LENS LESSON EHR/CPOE is the canonical ongoing case of an interface designed for one set of requirements and deployed against another. The systems work for billing and administration. They fail for clinical workflow because clinical workflow was not the design constraint. Forty billion dollars later, usability remains the single largest contributor to in-system harm in U.S. healthcare.

CASE 36 AUTONOMOUS SYSTEMS 2018

Uber ATG / Tempe Fatality

One pedestrian killed — the first fatality involving a self-driving vehicle striking a pedestrian

THE FULL CASE, IN FIVE BEATS

Background	Uber ATG tested self-driving cars using safety operators as passive surveillance of rare failures
What Happened	A Volvo killed Elaine Herzberg in Tempe; the operator was watching a phone video
The Investigation	NTSB faulted Uber for ignoring automation complacency; emergency braking was suppressed and pedestrian classification limited
The Capability Gap	The monitor role lacked interface, training, and authority; design assumed failure would never arrive
Aftermath & Reform	Uber exited self-driving; industry shifted toward two-operator teams and driver-monitoring systems

CONNECTIVITY

COURSES LEN 2

KEY REFERENCES

- NTSB, Collision Between Vehicle Controlled by Developmental Automated Driving System and Pedestrian, Highway Accident Report HAR-19/03 (2019) — the Tempe collision and probable cause.
- NTSB HAR-19/03 (2019) — the safety operator's distraction (watching a video) before impact.
- NTSB HAR-19/03 (2019) — Uber "did not adequately recognize the risk of automation complacency"; training and policy failures (quoted in part).

> LENS LESSON Uber ATG is the defining case for the LENS Human-AI Teaming competency. A human was retained in the system not because the designers believed a human could meaningfully act, but because the regulatory architecture required a human be present. The role of "monitor" was assigned without the capability infrastructure to support it. The system that resulted was performable only on the assumption that the failure case would not arrive — until it did.

CASE 37 AVIATION 1972

Eastern Air Lines Flight 401

101 killed in the Everglades; the entire flight crew fixated on a landing-gear indicator bulb while the autopilot silently disengaged

THE FULL CASE, IN FIVE BEATS

Background	Three-crew L-1011 on night Miami approach, with no designed division of attention
What Happened	Crew fixated on a burned-out gear bulb while autopilot disengaged and chime went unheeded
The Investigation	NTSB found crew failed to monitor flight path; trivial trigger produced catastrophic outcome
The Capability Gap	No designed division of attention across crew; alert lacked priority weight to cut through
Aftermath & Reform	Formative event behind Crew Resource Management and prioritized cockpit alerting standards

CONNECTIVITY

COURSES LEN 5, LEN 2

KEY REFERENCES

- NTSB, Aircraft Accident Report: Eastern Air Lines Flight 401, NTSB-AAR-73-14 (1973) — the night approach and full flight deck.
- NTSB-AAR-73-14 (1973) — the landing-gear-bulb fixation, the inadvertent autopilot disengagement, and 101 deaths of 176 aboard.
- NTSB-AAR-73-14 (1973) — the crew's preoccupation with the landing-gear indication that distracted them from maintaining altitude (paraphrased).

> LENS LESSON Eastern 401 is the canonical attention-failure case. The capability that was missing was a designed division of attention across the crew. CRM exists because Eastern 401 happened, and the discipline of cockpit alert design exists because the altitude warning chime did not carry the priority weight it needed to.

CASE 38 AVIATION 1991, 1994

Boeing 737 Rudder Hardovers

157 killed across United 585 (Colorado Springs, 1991) and USAir 427 (Pittsburgh, 1994); rudder Power Control Unit malfunctions traced to a thermal-shock condition

THE FULL CASE, IN FIVE BEATS

Background	737 rudder PCU servo valve presumed fault-tolerant by certification, hiding a latent failure mode
What Happened	United 585 and USAir 427 rolled and dove unrecoverably; rudders moved opposite to commanded input
The Investigation	NTSB took years to identify thermal-shock servo-valve jam from cold fluid striking hot valve
The Capability Gap	No recoverable cockpit action existed; gap sat in certification and maintenance, not pilots
Aftermath & Reform	Boeing redesigned PCU, retrofitted fleet, and reshaped how rare unrecoverable modes are hunted

CONNECTIVITY

COURSES LEN 5, LEN 7, LEN 3

KEY REFERENCES

- NTSB, Uncommanded Flight Control Movements (USAir 427 / 737 rudder PCU), NTSB-AAR-99-01 (1999) — the thermal-shock servo-valve finding.
- NTSB, Aircraft Accident Report: United Airlines Flight 585, NTSB-AAR-92-06 (1992) — the initial (undetermined) Colorado Springs investigation, later reopened.
- NTSB-AAR-99-01 (1999) — the servo-valve reversal mechanism and the absence of any recoverable crew action.

> LENS LESSON The 737 rudder cases are the case for an unrecoverable operational state hidden inside an apparently working system. The capability gap was not in the pilots — there was no capability that would have recovered the aircraft. The gap was in the certification process that had not surfaced the failure mode and in the maintenance procedures that did not test for it.

CASE 39 TECHNOLOGY 2024

CrowdStrike Falcon Outage

8.5 million Windows machines crashed; airlines, hospitals, broadcasters, and banks affected simultaneously; largest IT outage on record

THE FULL CASE, IN FIVE BEATS

Background	Falcon sensor runs in Windows kernel; pipeline treated rapid content updates as lighter than code
What Happened	Faulty content config crashed kernel on 8.5 million machines simultaneously; no staged rollout existed
The Investigation	Post-incident review found content lacked code-grade testing; category boundary did not match operational reality
The Capability Gap	For deployment safety content is code; customers had outsourced a missing safety gate to the vendor
Aftermath & Reform	Staged rollouts, kernel-access review, and scrutiny of vendor concentration risk followed the outage

CONNECTIVITY

COURSES LEN 5, LEN 2

KEY REFERENCES

- CrowdStrike, Falcon Content Update: Preliminary Post-Incident Review (July 2024) — the content-vs-code testing and staged-rollout gap (paraphrased).
- CrowdStrike PIR (2024) — the configuration-file logic error, the kernel crash, and 8.5 million affected Windows machines.
- Microsoft resilient-engineering analyses and Windows kernel-access review (2024).

> LENS LESSON The CrowdStrike outage is the live case for what happens when a deployment safety architecture treats categories of artifact differently than the operational system actually does. Content looked operationally identical to code; it was treated as different in the deployment pipeline. The mismatch became the largest IT outage in history.

CASE 40 DEFENSE 1983

Stanislav Petrov / 1983 False Alert

Soviet early-warning system reported five incoming U.S. ICBMs; Lt. Col. Petrov correctly assessed the signal as false; retaliation averted

THE FULL CASE, IN FIVE BEATS

Background	Soviet Oko satellite system fed launch-on-warning posture; duty officer was the single verification checkpoint
What Happened	Oko reported five inbound U.S. ICBMs; Petrov judged it false because a real strike means hundreds
The Investigation	Sunlight on high-altitude clouds at particular geometry fooled infrared sensors; algorithm later modified
The Capability Gap	No automated check could catch unimagined mode; human contextual judgment plus override authority saved it
Aftermath & Reform	Decision credited with averting nuclear war; case preserved in training as argument for human-in-loop

CONNECTIVITY

COURSES LEN 2

KEY REFERENCES

- D. Hoffman, The Dead Hand: The Cold War Arms Race and Its Dangerous Legacy (2009) — the Oko system and Soviet launch-on-warning posture.
- Accounts of the 26 Sept. 1983 incident (Hoffman; Petrov interviews) — the five-missile alert and Petrov's assessment.
- Investigations of the Oko incident (incl. V. Aksenov, 2004) — sunlight-on-clouds satellite geometry as the false-positive cause.

> LENS LESSON Petrov is the canonical positive case for human-in-the-loop nuclear command-and-control. The automation failed in a mode its designers had not anticipated. The human's contextual judgment was the backstop that allowed the failure to be recovered before it became catastrophic. The case is the strongest argument in this book that the design choice to retain humans in some loops is not nostalgia but capability engineering.

CASE 41 TECHNOLOGY 2018

TSB Bank IT Migration

1.9 million UK customers locked out of accounts; £330M+ in compensation and remediation; CEO resigned

THE FULL CASE, IN FIVE BEATS

Background	TSB needed to migrate five million accounts off Lloyds onto Sabadell platform in one weekend
What Happened	Nearly every component failed at relaunch; 1.9 million customers locked out, £330M cost, CEO resigned
The Investigation	Slaughter and May found unrealistic load testing, unchallenged readiness certification, override of technical advice
The Capability Gap	Technical signal existed but had no authority; governance let executive schedule overrule a known not-ready
Aftermath & Reform	TSB rebuilt testing and migration governance; FCA penalized the override of technical objection

CONNECTIVITY

COURSES LEN 7, LEN 8

KEY REFERENCES

- Slaughter and May, Independent Review of the TSB Migration (2019) — the single-weekend cutover and the testing failures.
- Slaughter and May (2019) and FCA materials — 1.9 million customers locked out, £330M+ in costs, and the CEO's resignation.
- Slaughter and May (2019) — inadequate load testing and an unchallenged readiness certification.

> LENS LESSON TSB is the canonical case for schedule pressure overriding technical signal in a regulated industry. The technical signal existed. The decision authority was at the executive layer where the signal arrived weakened by passage through multiple intermediate layers. The institutional architecture did not allow the technical layer to halt the migration.

CASE 42 SPACE / AEROSPACE HUMAN SPACEFLIGHT SAFETY ENGINEERING 2013

NASA EVA-23 — Water Intrusion Inside a Spacesuit

During the second of two ISS spacewalks, ESA astronaut Luca Parmitano's helmet filled with up to 1.5 liters of water from the suit's cooling-and-ventilation loop, blinding him, fouling his communications, and threatening drowning in vacuum; the EVA was terminated and Parmitano recovered to the airlock guided by tether feel — the NASA Mishap Investigation Board called the outcome a near-fatality

THE FULL CASE, IN FIVE BEATS

Background	EVA-23 (July 16 2013) — Parmitano's helmet fills with up to 1.5 L of water; EVA terminated, return traverse by tether feel
What Happened	Proximate cause (NASA MIB): blocked drum-holes in the Fan/Pump/Separator allowed cooling-loop water into the ventilation loop
The Investigation	Deeper cause: EVA-22 the week prior had a smaller water event mis-attributed to a drink-bag leak; no teardown before EVA-23 was approved
The Capability Gap	Corrective actions: hardware fix, Helmet Absorption Pad and snorkel, in-suit water-detection instrumentation, anomaly-investigation discipline
Aftermath & Reform	Open lifecycle question: how decades-certified hardware developed a contamination pathway not represented in its anomaly catalog

CONNECTIVITY

INDUCED	3.4
LENS	D5/PT5
CLO	CLO-3, CLO-5
COURSES	LEN 3, LEN 5, LEN 7

KEY REFERENCES

- NASA International Space Station Program (2013), "International Space Station EVA Suit Water Intrusion High Visibility Close Call" — Mishap Investigation Board final report, December 2013.
- NASA Aerospace Safety Advisory Panel (2014), Annual Report — EVA-23 follow-up actions including helmet absorption pad and snorkel.
- ESA (2013), debrief and operational statement on the Parmitano EVA-23 event.

> LENS LESSON EVA-23 is the canonical modern human-spaceflight case of a failure mode the suit was not instrumented to detect, surfacing first as a near-fatality because the prior weak signal was rationalized away. The hardware fix and the new in-suit instrumentation are necessary; the safety-culture half — anomaly is anomaly — is what generalizes.

CASE 43 AVIATION DEFENSE HUMAN FACTORS 2008 – 2012

F-22 OBOGS Hypoxia — The Sustainment-Era Instrumentation Gap

Between 2008 and 2012 USAF F-22 Raptor pilots reported a cluster of physiological events consistent with hypoxia; one fatal accident (Capt. Jeffrey Haney, Nov 2010) was attributed in part to operator response to a bleed-air shutoff; the fleet was stood down in 2011, and the USAF Scientific Advisory Board found no single root cause but a contributing oxygen-delivery issue rooted in the On-Board Oxygen Generation System (OBOGS) and associated equipment

THE FULL CASE, IN FIVE BEATS

Background	F-22 physiological-event cluster 2008–2012; symptoms consistent with hypoxia, often with persistent post-flight effects
What Happened	Capt. Haney fatal accident Nov 2010 — bleed-air shutoff cut OBOGS supply; AIB findings contested but the burden on pilot recognition is clear
The Investigation	USAF fleet stood down May 2011; USAF SAB and NASA review
The Capability Gap	Load-bearing finding: no single root cause; contributing factors include OBOGS, upper-pressure-garment vest, aircrew equipment fit, absence of in-cockpit physiological monitoring
Aftermath & Reform	Symptom reports decreased after bundle of corrections; attribution to any single component not possible from available evidence

CONNECTIVITY

INDUCED	2.4
LENS	D3/PT5
CLO	CLO-3, CLO-5
COURSES	LEN 3, LEN 5, LEN 7

KEY REFERENCES

- USAF Scientific Advisory Board (2012), “Aircraft Oxygen Generation Study” — final report on F-22 physiological-event investigation.
- NASA Engineering and Safety Center (2012), report contributing to the F-22 hypoxia review.
- USAF Accident Investigation Board (2011), Capt. Jeffrey Haney F-22 accident report (Nov 2010).

> LENS LESSON F-22 OBOGS is the canonical recent sustainment-instrumentation- gap case. A high-performance platform was fielded without in-cockpit measurement of pilot oxygenation, the diagnostic process took years against a noisy self-report channel, and the corrective action was an irreducibly multi-cause bundle. The “no single root cause” finding is the case.

CASE 44 GENERAL AVIATION AVIONICS PILOT TRAINING 2010

Glass-Cockpit Transition in Light Aircraft — Technology Outran Training

An NTSB safety study of ~8,000 piston aircraft over 2002–2006 found that glass-cockpit aircraft had no better overall safety record — and a higher fatal accident rate — than comparable conventional-instrument aircraft over the period studied; the Board attributed this to inadequate equipment-specific training and issued recommendations A-10-36 through A-10-40

THE FULL CASE, IN FIVE BEATS

Background	Glass-cockpit avionics introduced into light piston aircraft over the 2000s as fleet-wide modernization; assumed positive safety move
What Happened	NTSB safety study NTSB/SS-10/01 examined ~8,000 small piston aircraft 2002–2006; matched comparison glass vs. conventional fleets
The Investigation	Headline against expectation: no better overall safety record, higher fatal accident rate for glass-cockpit fleet over the period studied
The Capability Gap	NTSB attribution: inadequate equipment-specific training, not the technology; recommendations A-10-36 through A-10-40 to the FAA
Aftermath & Reform	Open verdict preserved: advanced avionics 'can increase the safety potential' but 'was not yet realized in the period studied'

CONNECTIVITY

INDUCED	7.1
LENS	D1+D5/PT1
CLO	CLO-1, CLO-2
COURSES	LEN 5, LEN 7, LEN 9

KEY REFERENCES

- National Transportation Safety Board (2010). Introduction of Glass Cockpit Avionics into Light Aircraft, Safety Study NTSB/SS-10/01. https://www.ntsb.gov/safety/safety-studies/Documents/SS1001.pdf — the case’s primary investigation document.
- NTSB Safety Recommendations A-10-36 through A-10-40 (2010), issued to the FAA — the engineering response the open verdict points to.
- Wiener, E. L., & Curry, R. E. (1980). Flight-deck automation: Promises and problems. Ergonomics, 23(10):995–1011 — the foundational literature on automation-induced failure modes that the glass-cockpit transition re-introduced at the GA scale.

> LENS LESSON The NTSB’s glass-cockpit GA study is the canonical capability-under-system-change failure at the consumer-aviation scale: a positive technology transition that nevertheless outran the inherited certification of operator proficiency. The Board’s attribution is to inadequate equipment-specific training, and the verdict is open — the safety potential is there, and the transition has not yet realized it.

CASE 45 EDUCATION AT SCALE 2014

inBloom

$100M initiative collapsed in ~14 months; 9 states withdrew; data infrastructure for education set back years

THE FULL CASE, IN FIVE BEATS

Background	Gates-funded \$100M shared student-data store; technically sound, built by enterprise engineers on commercial cloud
What Happened	Launched without consent, engagement, or parent voice; nine states withdrew within fourteen months
The Investigation	Data and Society read it as technocratic reform assuming engineering quality confers legitimacy
The Capability Gap	Stakeholder trust treated as add-on rather than precondition; no patch retrofits trust under fire
Aftermath & Reform	Set shared education infrastructure back years; drove a wave of state student-privacy laws

CONNECTIVITY

COURSES LEN 1, LEN 10, LEN 7, LEN 6

KEY REFERENCES

- M. Bulger, P. McCormick & M. Pitcan, The Legacy of inBloom, Data & Society Research Institute (2017) — inBloom as a failure of technocratic reform.
- Education Week and Hechinger Report coverage of the state withdrawals (2013–2014) — nine states exiting.
- Bulger et al. (2017) — governance, not technology, as the cause; "the technology was not the problem."

> LENS LESSON inBloom is the purest governance failure in this dataset. Nothing technical was wrong. Everything sociotechnical was. The case is the strongest argument in the book for treating ethics-as-design- constraint not as ideology but as engineering: a $100M empirical test of what happens when you do not.

CASE 46 GOVERNMENT 2013

Healthcare.gov Launch

~27,000 federal-marketplace enrollments in the first month vs. a 7M first-year target; hundreds of millions in remediation

THE FULL CASE, IN FIVE BEATS

Background	Federal ACA marketplace built under fixed political deadline by teams lacking consumer-launch expertise
What Happened	No clear integrator; no end-to-end test; site collapsed serving 27,000 against seven-million target
The Investigation	GAO and HHS-IG found no one understood project status; governance chain surfaced no readiness signal
The Capability Gap	Capability failure wearing a technology costume; missing match of human capability to system requirement
Aftermath & Reform	Rescue team became U.S. Digital Service; rare case producing durable institutional reform

CONNECTIVITY

COURSES LEN 1, LEN 5, LEN 7, LEN 6

KEY REFERENCES

- U.S. GAO, Healthcare.gov reports (2014–2016) — the launch, the capability gaps, and the absent end-to-end testing.
- HHS Office of Inspector General, Case Study of CMS Management of the Federal Marketplace, OEI-06-14-00350 (2016) — unclear ownership and the CMS/CGI integration confusion.
- HHS OIG (2016) — "no single person had a clear understanding of the project's status" (quoted).

> LENS LESSON Healthcare.gov is a capability case wearing a technology costume. The technology was salvageable in weeks once the right people arrived. The original failure was that the wrong people had been assembled in the first place, and the governance chain that should have noticed the mismatch had no signal to surface it. The aftermath — USDS — is the rare case in this book of a governance failure that produced permanent institutional reform.

CASE 47 INDUSTRIAL 1984

Bhopal

≈ 15,000–20,000 killed; ≈ 500,000 injured; worst industrial disaster in history

THE FULL CASE, IN FIVE BEATS

Background	Union Carbide plant stored bulk MIC under cost pressure; understaffed, safety systems off-line, training thin
What Happened	Water entered an MIC tank; non-operational defenses let forty tons vent over the sleeping city
The Investigation	Investigators cited operating errors, design flaws, maintenance failures, training deficiencies, economy measures endangering safety
The Capability Gap	Capability hollowed across training, maintenance, design, staffing, oversight; no layer owned the integrated whole
Aftermath & Reform	Reshaped industrial safety worldwide; catalyzed the U.S. Chemical Safety Board and process-safety regime

CONNECTIVITY

INDUCED	7.4
LENS	D4/PT4
CLO	CLO-1, CLO-4
COURSES	LEN 5, LEN 7, LEN 3

KEY REFERENCES

- Union Carbide and government investigation reports (1985) — MIC storage, the disabled safety systems, and plant understaffing.
- Accounts of the 2–3 Dec. 1984 release — the contested toll (thousands of immediate deaths; 15,000–20,000 total estimates; 500,000 exposed). (Figures vary widely across sources; see AUDIT.)
- New York Times investigation (1985) — “operating errors, design flaws, maintenance failures, training deficiencies and economy measures that endangered safety” (quoted).

> LENS LESSON Bhopal is the largest-magnitude capability-and-governance failure on record. It is also the catalyst for the creation of the U.S. Chemical Safety Board, an INPO-equivalent for industrial chemistry. The pattern that follows nearly every case in this book — diagnose, then engineer remediation at scale — runs through Bhopal in canonical form.

CASE 48 INDUSTRIAL 2017

Grenfell Tower

72 killed in a residential tower fire in London; decades of regulatory failure

THE FULL CASE, IN FIVE BEATS

Background	1970s tower refurbished with combustible aluminium-composite cladding despite expert cautions against high-rise use
What Happened	Kitchen fire climbed the exterior cladding; stay-put advice held residents inside; seventy-two died
The Investigation	Inquiry found systematic dishonesty by cladding firms; sixteen inspections missed effectively banned products
The Capability Gap	Distributed capability failure; fraud, capture, incompetence, lost memory converged with no integrated owner
Aftermath & Reform	Phase 2 report and government response drove building-safety, cladding, and fire-service reforms

CONNECTIVITY

INDUCED	7.4
LENS	D4/PT4
CLO	CLO-2, CLO-4
COURSES	LEN 10, LEN 7, LEN 8, LEN 3

KEY REFERENCES

- Grenfell Tower Inquiry, Phase 1 Report (2019) — the fire’s spread up the cladding and the failure of “stay put.”
- Grenfell Tower Inquiry, Phase 2 Report (2024) — decades of failure and the combustible-cladding decision.
- Phase 2 Report (2024) — cladding firms’ “systematic dishonesty” and the inspection failures across sixteen visits.

> LENS LESSON Grenfell is the strongest evidence in the dataset that capability failure can be distributed across many actors, each of whom contributes a small piece, none of whom is accountable for the whole. The inquiry’s “grey elephant” framing — known but ignored — describes a pattern that LENS treats as a primary governance problem in any high-consequence domain.

CASE 49 EDUCATION AT SCALE 2014–2019

Summit Learning / Personalized Learning Rollout

Personalized-learning platform deployed across ~380 U.S. schools; parent revolts in Brooklyn, Cheshire, McPherson, Kennebunk; multiple districts withdrew within two years

THE FULL CASE, IN FIVE BEATS

Background	CZI-backed personalized-learning platform offered free from 2015 on defensible competency-based pedagogy
What Happened	Reached 380 schools, 80,000 students; by 2019 parent revolts and withdrawals over screen time and privacy
The Investigation	Analysts located failure in deployment governance; no evaluation framework, data agreement, or exit path
The Capability Gap	Working pedagogy with no accountability contract collapsed; same pattern as inBloom recurring at scale
Aftermath & Reform	Districts withdrew, CZI revised outreach; episode became standard caution in ed-tech adoption

CONNECTIVITY

COURSES LEN 7, LEN 10, LEN 8

KEY REFERENCES

- N. Bowles, “Silicon Valley Came to Kansas Schools. That Started a Rebellion,” New York Times (2019) — the parent revolt.
- N. Singer, “The Silicon Valley Billionaires Remaking America’s Schools,” New York Times (2017) — the CZI/Summit rollout.
- B. Herold, Education Week coverage of Summit / CZI implementation (2019) — district adoptions and withdrawals.

> LENS LESSON Summit Learning is a clean test of the book’s central claim: technology that worked at the pedagogical level still failed because the **governance** architecture (consent, evidence, measurement, exit) had not been engineered alongside it. The pattern — well-intentioned tool, well-funded rollout, no institutional contract with the families and teachers operating inside it — recurs across the educational-technology dataset (Cases 45, 85, 113) and is the educator’s-side analog of the governance failures in Cases 78 and 77.

CASE 50 EDUCATION AT SCALE 2009–2018

Tennessee Voluntary Pre-K Study

Vanderbilt longitudinal RCT of a state-funded universal pre-K program: early gains faded by third grade; sixth-grade outcomes were worse than the control group on several measures

THE FULL CASE, IN FIVE BEATS

Background	Tennessee Voluntary Pre-K served 18,000 four-year-olds; oversubscription enabled a rare lottery-based RCT
What Happened	Kindergarten gains faded by third grade; sixth-grade pre-K children did somewhat worse than controls
The Investigation	Field contested the methods and largely declined to internalize findings, defending policy rather than interrogating
The Capability Gap	Measurement instrument worked; institutional architecture for absorbing inconvenient evidence did not exist
Aftermath & Reform	Episode became methodological touchstone in early-childhood research, argued over more than acted upon

CONNECTIVITY

COURSES LEN 4, LEN 7, LEN 10

KEY REFERENCES

- M. Lipsey, D. Farran & K. Durkin, “Effects of the Tennessee Pre-Kindergarten Program… Through Third Grade,” Early Childhood Research Quarterly 45: 155–176 (2018) — the RCT and fade-out.
- K. Durkin, M. Lipsey, D. Farran & E. Wiesen, “Effects of a Statewide Pre-Kindergarten Program… Through Sixth Grade,” Developmental Psychology 58(3): 470–484 (2022) — the sixth-grade reversal (quoted).
- Responses and counter-analyses to the TN-VPK findings (2018–2022) — the contested reception.

> LENS LESSON Tennessee Pre-K is the cleanest case in the dataset for what happens when a discipline has not engineered its own capacity to update on contrary evidence. The measurement instrument worked. The institutional architecture for acting on what it found did not. Compare to Case 18 (Makary methodology debate) and Case 84 (VA Wait-Time): in each, a measurement that returned an unwelcome answer was contested rather than absorbed.

CASE 51 EDUCATION AT SCALE ONGOING

Algorithmic Bias in Educational Predictive Analytics

Predictive "at-risk" models show racial calibration bias that can misdirect support away from Black and Latinx students; a large majority of U.S. public colleges now use predictive analytics

THE FULL CASE, IN FIVE BEATS

Background	Most U.S. public colleges now use predictive analytics to flag at-risk students for early advising support
What Happened	Research finds racial calibration bias misclassifying Black and Latinx students; magnitude depends on construct
The Investigation	Researchers traced calibration gaps to training data encoding historical discrimination; deficit framing compounds harm
The Capability Gap	Bias lives in the construct definition of at-risk; capability-engineering decision made implicitly through labels
Aftermath & Reform	Ongoing and quiet across hundreds of dashboards; no single collapse forces a reckoning

CONNECTIVITY

COURSES LEN 4, LEN 7, LEN 9

KEY REFERENCES

- Surveys of predictive-analytics adoption in U.S. higher education — a large majority of public colleges now use some form.
- K. Bird et al., “Are Algorithms Biased in Education? Exploring Racial Bias in Predicting Community College Student Success,” Journal of Policy Analysis and Management 44 (2025), 379–402 — racial calibration bias, 5× higher at the bottom decile depending on the “at-risk” construct.
- D. Gándara, H. Anahideh, M. Ison & L. Picchiarini, “Inside the Black Box: Detecting and Mitigating Algorithmic Bias across Racialized Groups in College Student-Success Prediction,” AERA Open (2024) — bias traced to training data encoding historical discrimination.

> LENS LESSON Educational predictive analytics is the ongoing live case for algorithmic bias at the construct level. The bias is not in the sigmoid; it is in the definition of “at-risk.” Defining at-risk in one way allocates support to one population; defining it in another way allocates support to another. The choice of definition is a capability-engineering decision with measurable equity consequences.

CASE 52 TECHNOLOGY GOVERNMENT 1999 – 2015

UK Post Office Horizon Scandal

~900 sub-postmasters wrongfully prosecuted; many imprisoned; documented suicides; described as the most wide-spread miscarriage of justice in UK history

THE FULL CASE, IN FIVE BEATS

Background	Sub-postmasters bore personal liability for shortfalls reported by Fujitsu's Horizon accounting system from 1999
What Happened	Phantom shortfalls drove around 900 prosecutions for theft; imprisonment, bankruptcy, and suicides followed
The Investigation	Released documents showed engineers knew Horizon could err; convictions began being quashed in 2021
The Capability Gap	No regulator, prosecutor, or court had standing to interrogate Fujitsu's claim of reliability
Aftermath & Reform	Parliament exonerated convictions and opened compensation schemes; the public inquiry continues its work

CONNECTIVITY

COURSES LEN 7, LEN 2

KEY REFERENCES

- Post Office Horizon IT Inquiry hearings and exhibits (2020–) — the system, the prosecutions, and the human toll.
- Hamilton & Others v. Post Office Limited (Court of Appeal, 2021) — quashed convictions.
- Internal Fujitsu and Post Office documents released through litigation — engineers’ knowledge of Horizon bugs.

> LENS LESSON Horizon is the canonical case for institutional deference to automated systems whose internal evidence was already known to be flawed. The capability gap was at every layer that took the software’s output as authoritative — including the courts. “The computer said so” became, for two decades, an institutionally sufficient basis for criminal prosecution.

CASE 53 HEALTHCARE TECHNOLOGY 2003 – 2018

Theranos

$9B valuation collapsed; thousands of patients given unreliable results; founder convicted on multiple counts of wire fraud

THE FULL CASE, IN FIVE BEATS

Background	Theranos claimed an Edison device running hundreds of tests from one finger-stick, reaching a nine-billion-dollar valuation
What Happened	The device did not work; Theranos secretly ran samples on commercial analyzers and reported them as its own
The Investigation	Carreyrou's reporting and CMS inspectors surfaced the fraud after the FDA-CLIA seam left validation unowned
The Capability Gap	Neither regulator validated novel clinical claims end-to-end; a celebrity board offered prestige instead of scrutiny
Aftermath & Reform	CLIA certificate revoked, the company collapsed, and Holmes was convicted of multiple wire-fraud counts

CONNECTIVITY

COURSES LEN 4, LEN 7

KEY REFERENCES

- United States v. Elizabeth Holmes (N.D. Cal., 2018–2022) — the indictment and conviction.
- J. Carreyrou, Bad Blood (2018) — the device's failure and the commercial-analyzer substitution.
- CMS inspection reports and the revocation of Theranos's CLIA certificate (2015–2016).

> LENS LESSON Theranos is the canonical case for fraud exploiting the seam between two regulatory regimes. The FDA had not approved the device; CLIA accepted the laboratory operation. Neither layer had the capability to validate the claims independently. The capability gap was at the regulatory architecture.

CASE 54 TECHNOLOGY GOVERNMENT 2011 – 2016

Wells Fargo Fake Accounts

~3.5 million unauthorized accounts opened; ~$3B in penalties; CEO resigned; the Federal Reserve capped the bank's assets

THE FULL CASE, IN FIVE BEATS

Background	Wells Fargo drove cross-selling with aggressive branch quotas that pay and jobs depended on hitting
What Happened	Employees opened about 3.5 million unauthorized accounts; the bank fired front-line staff and kept the incentives
The Investigation	CFPB and OCC actions and a 2017 independent-directors report tied misconduct to the sales-target architecture
The Capability Gap	The governance layer that designed the incentives mistook a designed outcome for individual moral failure
Aftermath & Reform	The Federal Reserve capped Wells Fargo's assets; the case anchored teaching on measurement-gaming and incentive design

CONNECTIVITY

INDUCED	2.1
LENS	D3/PT5
CLO	CLO-3, CLO-4
COURSES	LEN 4, LEN 7

KEY REFERENCES

- Consumer Financial Protection Bureau, Consent Order against Wells Fargo (2016) — the unauthorized accounts and penalties.
- Office of the Comptroller of the Currency, Consent Order AA-EC-2016-66 (2016) — unsafe or unsound sales practices tied to the incentive-compensation structure (paraphrased).
- Independent Directors of Wells Fargo, Sales Practices Investigation Report (2017) — how the sales-target architecture drove the conduct.

> LENS LESSON Wells Fargo is the strongest available evidence that incentive architecture is a capability-engineering deliverable. The measurement system used to manage employees produced the behavior the institution then prosecuted. The gap was not at the front line. It was at the governance layer that had designed the incentives.

CASE 55 INDUSTRIAL GOVERNMENT 2015

Volkswagen Dieselgate

~11 million vehicles equipped with defeat-device software; $33B+ in penalties and remediation; multiple criminal convictions

THE FULL CASE, IN FIVE BEATS

Background	VW promised clean diesel meeting nitrogen-oxide limits its engineers did not believe they could deliver
What Happened	Defeat-device software detected the test bench and enabled controls only there across about eleven million vehicles
The Investigation	A West Virginia team comparing road to lab emissions exposed the cheat; documents showed institutional authorization
The Capability Gap	The regulator's test ran in a regime the vehicle could detect, inviting the gaming it was meant to catch
Aftermath & Reform	VW paid more than thirty-three billion in penalties; the EU introduced real-world driving emissions testing

CONNECTIVITY

INDUCED	2.2
LENS	D3/PT3
CLO	CLO-3
COURSES	LEN 4, LEN 7

KEY REFERENCES

- U.S. EPA, Notice of Violation to Volkswagen (2015) — the defeat device and emissions exceedances.
- West Virginia University / ICCT real-world diesel-emissions study (2014) — the discovery comparing road to lab.
- U.S. Department of Justice Plea Agreement with Volkswagen AG (2017) — the institutional decision, $33B+ in penalties, and convictions (quoted).

> LENS LESSON Dieselgate is the canonical case for institutionally engineered evasion of a measurement system. The capability gap was at the regulator's test protocol: it operated in a regime the vehicle could detect. The reform — real-world emissions testing — was a capability deliverable upgrade at the regulator's instrument boundary.

CASE 56 TECHNOLOGY 2014 – 2018

Cambridge Analytica / Facebook

~87 million Facebook profiles harvested without informed consent; FTC $5B penalty; foundational data-governance reform

THE FULL CASE, IN FIVE BEATS

Background	Facebook's Graph API let apps collect quiz-takers' data and their friends' data without friends consenting
What Happened	A personality-quiz app taken by 270,000 people harvested about 87 million profiles for political micro-targeting
The Investigation	UK ICO and U.S. FTC investigations, prompted by Guardian reporting, found Facebook responsible for what its design permitted
The Capability Gap	The architecture worked as designed; a load-bearing assumption of developer benevolence was never red-teamed
Aftermath & Reform	GDPR and CCPA accelerated; the FTC imposed a five-billion-dollar penalty and a continuing consent decree

CONNECTIVITY

INDUCED	5.3
LENS	D4/PT5
CLO	CLO-4
COURSES	LEN 1, LEN 7, LEN 6

KEY REFERENCES

- U.S. FTC, In the Matter of Facebook, Inc., Consent Order (2019) — the $5B penalty; Facebook gave developers "far more user data than was necessary" (quoted).
- UK Information Commissioner's Office, report on Cambridge Analytica / data analytics in political campaigns (2018) — the scope of the harvesting.
- C. Cadwalladr & E. Graham-Harrison, The Guardian investigation (2018) — the disclosure.

> LENS LESSON Cambridge Analytica is the canonical case for platform-governance failure: an API contract that assumed benevolent intent and was not engineered against systematic abuse. The capability gap was not technical. The architecture worked exactly as designed; the design assumption was wrong.

CASE 57 TECHNOLOGY GOVERNMENT 2017

Equifax Data Breach

147 million Americans' personal data exposed; CEO resigned; ~$700M settlement; foundational U.S. data-breach case

THE FULL CASE, IN FIVE BEATS

Background	Equifax held identity data on most US adults; security flagged an Apache Struts patch to IT
What Happened	Unapplied patch let attackers exfiltrate 147M Americans' data over 2.5 months
The Investigation	Senate subcommittee found systematically inadequate patching and no comprehensive IT asset inventory
The Capability Gap	Routine work — patching, inventory, monitoring, response — each below standard, starved as a cost center
Aftermath & Reform	$700M settlement funded compensation; patching, inventory, and disclosure timelines rose on the agenda

CONNECTIVITY

INDUCED	6.2
LENS	D4/PT3
CLO	CLO-2
COURSES	LEN 5, LEN 7

KEY REFERENCES

- U.S. Senate Permanent Subcommittee on Investigations, How Equifax Neglected Cybersecurity and Suffered a Devastating Data Breach (2019) — the unpatched Apache Struts vulnerability.
- The breach record — 147 million affected, exploitation from May 2017, public disclosure in September 2017.
- Senate PSI (2019) — “Equifax lacked a comprehensive IT asset inventory” (quoted).

> LENS LESSON Equifax is the canonical institutional-cybersecurity case for cumulative inadequacy in routine work: patching, inventory, monitoring, response. Each function was below industry standard. None alone produced the breach. The combination was the breach.

CASE 58 INDUSTRIAL 1981

Hyatt Regency Walkway Collapse

114 killed and 216 injured in Kansas City when suspended walkways collapsed; foundational U.S. engineering-ethics case

THE FULL CASE, IN FIVE BEATS

Background	The Kansas City Hyatt's atrium walkways were originally designed to hang from single continuous steel rods
What Happened	A two-rod construction change doubled the upper connection's load; on 17 July 1981 the walkways fell, killing 114
The Investigation	The National Bureau of Standards found the as-built connection had carried roughly twice its intended load
The Capability Gap	Nothing required the engineer of record to re-analyze a load-bearing change before it was built
Aftermath & Reform	Missouri revoked engineers' licenses; shop-drawing review and engineer-of-record responsibility hardened into rule

CONNECTIVITY

COURSES LEN 5, LEN 7

KEY REFERENCES

- National Bureau of Standards, Investigation of the Kansas City Hyatt Regency Walkways Collapse, NBSIR 82-2465 (1982) — the doubled-load connection (quoted).
- The 17 July 1981 collapse — 114 killed and 216 injured.
- NBSIR 82-2465 (1982) — the as-built connection inadequate even under the building code.

> LENS LESSON The Hyatt Regency case is the canonical engineering-ethics case for institutional review of construction changes. The change looked small; the load implication was catastrophic. The capability gap was at the review process that should have caught the doubled load.

CASE 59 ENERGY 2018

Camp Fire / PG&E

85 killed in Paradise, California; deadliest U.S. wildfire in a century; PG&E pleaded guilty to 84 counts of involuntary manslaughter

THE FULL CASE, IN FIVE BEATS

Background	PG&E ran aging transmission lines across drying foothills, with hardware approaching a century old
What Happened	A worn C-hook failed in high winds, igniting a fire that overran Paradise; 85 died, the deadliest in a century
The Investigation	CalFire and the Butte County DA found PG&E had known about deteriorating infrastructure and deferred maintenance
The Capability Gap	Neither utility nor regulator owned the capability to update operations to the actual, hotter, drier risk
Aftermath & Reform	PG&E pleaded guilty to 84 manslaughter counts; California mandated wildfire-mitigation plans and power shutoffs

CONNECTIVITY

INDUCED	7.3
LENS	D4/PT1
CLO	CLO-1, CLO-4
COURSES	LEN 7, LEN 8

KEY REFERENCES

- CalFire, Camp Fire Investigation Report (2019) — the worn transmission-line hardware as ignition source.
- The Camp Fire record — 85 killed; PG&E’s guilty plea to 84 counts of involuntary manslaughter (2020).
- Butte County District Attorney, The Camp Fire Public Report (2020) — PG&E’s knowledge and deferred maintenance (quoted).

> LENS LESSON The Camp Fire is the canonical climate-era case for utility- capability failure under changing risk conditions. The infrastructure was designed for one set of conditions and operated in another. The capability to update operations to match actual conditions did not exist as an institutional deliverable.

CASE 60 HEALTHCARE CLINICAL AI GOVERNANCE 2017 – 2021

Epic Sepsis Model

External validation across 38,455 hospitalizations found AUROC 0.63 versus reported 0.76–0.83, missing ~67% of sepsis at the operational threshold; deployed in hundreds of hospitals without independent validation or FDA clearance

THE FULL CASE, IN FIVE BEATS

Background	Most-deployed clinical AI in US medicine — embedded as a default Epic EHR feature; no independent validation
What Happened	Wong et al. external validation: AUROC 0.63, missed ~67% of sepsis at threshold, 12% PPV, alert burden
The Investigation	Governance seam: EHR-embedded proprietary models fell outside FDA device oversight by classification, not by design
The Capability Gap	Disconfirmation came as a published external validation, not from a standing post-deployment surveillance regime
Aftermath & Reform	Negative pair to TREWS; part of the AI-delegation typology (Epic / SyRI / Watson / TREWS)

CONNECTIVITY

INDUCED	2.4
LENS	D3+D5/PT6
CLO	CLO-3, CLO-4, CLO-5
COURSES	LEN 4, LEN 7, LEN 2

KEY REFERENCES

- Wong et al. (2021), “External Validation of a Widely Implemented Proprietary Sepsis Prediction Model in Hospitalized Patients,” JAMA Internal Medicine 181(8):1065–1070, doi:10.1001/jamainternmed.2021.2626.
- Habib et al. (2021), commentary on Wong et al., JAMA Internal Medicine — on the implications for proprietary clinical AI.
- FDA, Clinical Decision Support Software: Final Guidance (2022) — the post-Wong reframing of the EHR-embedded oversight question.

> LENS LESSON The Epic Sepsis Model is the canonical case of consequential clinical-AI delegation at scale without independent validation. The structural lesson is not the model’s poor performance; it is the governance seam that let the deployment proceed without the validation and surveillance machinery that the medical-device pathway would have required, surfaced only by post-deployment external work.

CASE 61 FINANCE ALGO FAIRNESS 2018 – 2022

Algorithmic Mortgage Lending — Omitting the Variable Did Not Fix the Disparity

Algorithmic mortgage underwriting reduced face-to-face discrimination but preserved a measured pricing disparity even when race was excluded from the inputs — the variable's omission did not fix the harm

THE FULL CASE, IN FIVE BEATS

Background	Algorithmic underwriting reaches the mortgage market at scale; the question is whether automation reduces or preserves discrimination
What Happened	Bartlett et al. decompose discrimination along acceptance and pricing margins across millions of applications
The Investigation	Algorithmic acceptance is more equitable than face-to-face; algorithmic pricing preserves a measured disparity even with race excluded
The Capability Gap	Excluded variable returns through correlated features (geography, credit history); omission shifts but does not close the channel
Aftermath & Reform	Fairness through unawareness is not fairness; the deliverable is output-level disparate-impact measurement, not input-level assurance

CONNECTIVITY

INDUCED	8.2
LENS	D3+D5/PT6
CLO	CLO-3, CLO-4, CLO-5
COURSES	LEN 4, LEN 7, LEN 9

KEY REFERENCES

- Bartlett, Morse, Stanton, & Wallace (2022), "Consumer-lending discrimination in the FinTech era," Journal of Financial Economics 143(1):30–56, doi:10.1016/j.jfineco.2021.05.047.
- Consumer Financial Protection Bureau, Mortgage Market Activity and Trends (annual HMDA reports), supporting the population-level disparities backdrop.
- Dwork et al. (2012), "Fairness Through Awareness," ITCS 2012 — the foundational technical statement of the limits of unawareness.

> LENS LESSON Bartlett et al. is the canonical mortgage-finance instance of why "fairness through unawareness" is not fairness. Algorithmic underwriting reduced the acceptance-margin disparity but the pricing-margin disparity persisted even with race excluded — because correlated admissible features carry the same signal. The capability deliverable is output-level disparate-impact measurement, not input-level assurance.

CASE 62 FINANCE ALGO FAIRNESS FINTECH 2025

Fintech Lending Fairness Audit — When Including Race Reduces Disparity

A fintech-lending fairness audit reports that incorporating protected attributes under controlled mitigation reduces measured disparity below the unawareness baseline — surfacing the next teaching point: competing fairness definitions disagree, and the choice is a judgment

THE FULL CASE, IN FIVE BEATS

The Shift	Past omission: when controlled inclusion of a protected attribute reduces measured disparity below the unawareness baseline
What Is Emerging	Coots audit examines a deployed fintech-lending model under unawareness vs. controlled mitigation
The Capability Question	Frontier point: competing fairness definitions disagree about the same model; impossibility results force a choice
Early Evidence	Preprint-tier evidence; metric specifics and magnitudes may move in peer review; future validation will continue
Open Problems	The capability deliverable is to pre-specify the fairness definition, audit on outputs, and decide under irreducible disagreement

CONNECTIVITY

INDUCED	8.1
LENS	D3/PT6
CLO	CLO-3, CLO-5
COURSES	LEN 4, LEN 7, LEN 9

KEY REFERENCES

- Coots, Bartlett, Nyarko, & Goel (2025), "Algorithmic Bias in Lending: Evidence from a Fintech Audit," arXiv:2512.20753 — audit of 80,000 personal loans showing model miscalibration disparities and that controlled inclusion of protected attributes could correct them; the evidence-tier flag is binding until peer review completes.
- Bartlett, Morse, Stanton, & Wallace (2022), "Consumer-lending discrimination in the FinTech era," Journal of Financial Economics 143(1):30–56 — the paired big-tier case (103).
- Chouldechova (2017), "Fair Prediction with Disparate Impact," Big Data 5(2):153–163 — the impossibility result for calibration and error-rate parity.

> LENS LESSON The Coots audit is the small-tier frontier instance of the finding that controlled inclusion of a protected attribute can produce lower measured disparity than omission. With Bartlett (Case 61) it forms the canonical lending pair: omission does not fix the harm; competing fairness definitions disagree about what fix is. Evidence is preprint-tier; future validation will continue.

CASE 63 CLINICAL MEDICINE MEDICAL EDUCATION HEALTH EQUITY 2016

Racial Bias in Pain Assessment — The False-Belief Mechanism

About half of surveyed medical students and residents endorsed at least one false belief about biological differences between Black and White people; those who held more false beliefs rated Black patients' pain as less severe and recommended less accurate treatment

THE FULL CASE, IN FIVE BEATS

Background	Documented pain-undertreatment disparity for Black patients in US clinical settings; mechanism less precisely named
What Happened	Hoffman et al. survey medical trainees on a battery of false biological-difference beliefs; ~half endorse at least one
The Investigation	Experimental layer: respondents who endorse more false beliefs rate Black mock patients' pain as less severe and treat less accurately
The Capability Gap	Mechanism is specific and nameable: a small set of false beliefs, not diffuse implicit bias — curriculum and instrumentation can target it
Aftermath & Reform	Trio with eGFR (construct) and pulse oximetry (validation): same surface harm at three distinct layers — gap attribution is the deliverable

CONNECTIVITY

INDUCED	8.1
LENS	D3/PT5
CLO	CLO-3
COURSES	LEN 1, LEN 4, LEN 7

KEY REFERENCES

- Hoffman, Trawalter, Axt, & Oliver (2016), “Racial bias in pain assessment and treatment recommendations, and false beliefs about biological differences between blacks and whites,” PNAS 113(16):4296–4301, doi:10.1073/pnas.1516047113.
- Anderson, Green, & Payne (2009), “Racial and ethnic disparities in pain: causes and consequences of unequal care,” Journal of Pain 10(12):1187–1204 — the population-level disparity.
- Sabin & Greenwald (2012), “The influence of implicit bias on treatment recommendations for 4 common pediatric conditions,” American Journal of Public Health — the diffuse-mechanism backdrop the Hoffman precision improves on.

> LENS LESSON Hoffman et al. is the human-development case in the race-construct trio. The pain-assessment disparity in medical settings is mediated, in measurable part, by a nameable set of false biological-difference beliefs held by clinicians in training. The capability deliverable is curriculum that specifically disconfirms the beliefs and instrumentation that surfaces the bedside rating gap when it persists.

CASE 64 GOVERNMENT/WELFARE (NETHERLANDS) ALGORITHMIC DECISION-MAKING PUBLIC-SECTOR AI 2014 – 2020

Dutch SyRI — Welfare-Fraud Risk Scoring Halted on Rights Grounds

An opaque risk-scoring system that combined up to 17 categories of previously siloed government data on citizens — disproportionately deployed in low-income and minority neighborhoods — halted by the District Court of The Hague in 2020 as a violation of Article 8 ECHR; also reported as operationally ineffective

THE FULL CASE, IN FIVE BEATS

Background	SyRI combines up to 17 categories of siloed government data; deployed 2014; targeted at low-income and minority neighborhoods
What Happened	Civil-society and citizen coalition sue; District Court of The Hague halts program 5 Feb 2020 as Article 8 ECHR violation
The Investigation	Investigative reporting: on its own terms, SyRI did not work — no new fraud detected in six years of algorithmic investigations
The Capability Gap	Court leaves open that a more transparent and narrower system could pass; the ruling is system-specific, not categorical
Aftermath & Reform	Pair with OU (Case 122): governance-objection diagnostic — when design dissolves the objection vs. when the objection is correct

CONNECTIVITY

INDUCED	5.1
LENS	D4+D5/PT6
CLO	CLO-4, CLO-5
COURSES	LEN 4, LEN 7, LEN 9

KEY REFERENCES

- District Court of The Hague (2020), judgment of 5 February 2020 (NJCM et al. v. State of the Netherlands), ECLI:NL:RBDHA:2020:865.
- Rachovitsa & Johann (2022), “The Human Rights Implications of the Use of AI in the Digital Welfare State,” Human Rights Law Review 22(2), doi:10.1093/hrlr/ngac010.
- Library of Congress Global Legal Monitor (2020), report on the SyRI ruling.

> LENS LESSON SyRI is the canonical case in the corpus of a governance objection that was correct. The system was both rights-violating (Article 8 ECHR) and operationally ineffective (no new fraud detected). The court ruling is system-specific — it does not foreclose all public-sector AI — which is what makes the case a governance-objection diagnostic teaching point rather than a categorical verdict.

CASE 65 PHARMA SAFETY REGULATORY PROGRAMS CLINICAL CARE 2006 – 2011

iPLEDGE Isotretinoin REMS — When the Authorization Mechanism Underperforms

An FDA-mandated risk-evaluation and mitigation program with pregnancy testing, two-method contraception requirements, and lockout authorization before dispensing — Kaiser Permanente cohort (n=8,344; 9,912 treatment courses) reported 29 fetal exposures and 'no evidence that iPLEDGE significantly decreased the risk of fetal exposure' relative to the prior SMART program

THE FULL CASE, IN FIVE BEATS

The Shift	iPLEDGE 2006 — first REMS-class authorization for isotretinoin; mandatory pregnancy test, two-method contraception, pharmacy lockout per cycle
What Is Emerging	Brinker et al. 2011 Kaiser cohort (n=8,344; 9,912 courses) — 29 fetal exposures; 'no evidence iPLEDGE significantly decreased risk' vs. prior SMART
The Capability Question	Approximately 150 isotretinoin-exposed pregnancies continue annually in the US despite the program; non-adherence is the documented driver
Early Evidence	Structural form same as SUBSAFE and WHO checklist; the mechanism alone does not deliver capability without adherence support
Open Problems	Most analytically useful 'mixed' case in v2 — the form has demonstrated successes; form-without-adherence-support is the gap

CONNECTIVITY

INDUCED 4.4
LENS D4/PT5
CLO CLO-3, CLO-5
COURSES LEN 5, LEN 7, LEN 9

KEY REFERENCES

- Brinker, Kornegay, Nourjah, Smith, & Reichman (2011), “The impact of the iPLEDGE program on isotretinoin fetal exposure in an integrated health care system,” Journal of the American Academy of Dermatology, PMID:21565419.
- FDA, iPLEDGE program documentation (2006 – present) — REMS architecture and enrollment requirements.
- Pinheiro et al. (2013), “Isotretinoin and pregnancy in the era of iPLEDGE,” Journal of the American Academy of Dermatology — broader outcome literature documenting the 150 annual exposures figure.

> LENS LESSON iPLEDGE is the productive counterpoint to SUBSAFE and the WHO Surgical Checklist. The structural form — mandatory pre-authorization gating a consequential action — is the same; the measured outcome is very different because the capability the program exists to enforce depends on patient adherence the program does not instrument. The “no significant decrease” finding is load-bearing and survives into the case verbatim.

CASE 66 ED-TECH ALGO FAIRNESS ASSESSMENT 2022

Algorithmic Bias in Automated Exam Proctoring

The first quantitative study of facial-detection bias in automated exam proctoring software found that students with darker skin tones and Black students were significantly more likely to be flagged for instructor review for potential cheating; at the race–sex intersection, women with the darkest skin tones were far more likely to be flagged than other groups

THE FULL CASE, IN FIVE BEATS

Background	COVID-era expansion of remote-learning automated exam proctoring; webcam-based face detection flagging suspicious behavior for instructor review
What Happened	Yoder-Himes et al. _Frontiers in Education_ 2022 — first published quantitative study of facial-detection bias in this class of software
The Investigation	Students with darker skin tones and Black students significantly more likely to be flagged for instructor review for potential cheating
The Capability Gap	Intersectional finding: women with the darkest skin tones far more likely to be flagged than other groups
Aftermath & Reform	Failure / diagnosis case: documents the disparity, not a remediation; trio with Cases 119 (eGFR), 106 (pulse oximetry), 107 (Hoffman pain bias)

CONNECTIVITY

INDUCED 8.2
LENS D3/PT5
CLO CLO-3, CLO-4
COURSES LEN 5, LEN 8, LEN 9

KEY REFERENCES

- Yoder-Himes, D. R., Asif, A., Kinney, K., Brandt, T. J., Cecil, R. E., Himes, P. R., Cashon, C., Hopp, R. M. P., & Ross, E. (2022). Racial, skin tone, and sex disparities in automated proctoring software. Frontiers in Education, 7:881449. doi:10.3389/feduc.2022.881449 — the case’s primary study.
- Buolamwini, J., & Gebru, T. (2018). Gender shades: Intersectional accuracy disparities in commercial gender classification. Proceedings of Machine Learning Research, 81:77–91 — the foundational intersectional-bias finding in face recognition.
- Raji, I. D., & Buolamwini, J. (2019). Actionable auditing: Investigating the impact of publicly naming biased performance results of commercial AI products. AAAI/ACM Conference on AI, Ethics, and Society — the audit-and-disclosure mechanism the case calls for.

> LENS LESSON Automated exam proctoring is the assessment-domain counterpart to pulse oximetry’s medical-device bias. Yoder-Himes et al. 2022 is the first quantitative study of the disparity in this class of software, with the intersectional finding the broader face-recognition literature predicts. The case documents the disparity in one product; the remediation is not yet documented.

CASE 67 HIGHER EDUCATION (AFRICA) DATA PRIVACY LAW LEARNING ANALYTICS 2022

Data Privacy and Learning Analytics on the African Continent

Mapped the legal and regulatory privacy landscape across African jurisdictions and surfaced the structural governance seam — African higher education frequently uses learning platforms and analytics hosted by external providers, creating cross-regime gaps where student data crosses jurisdictions with inconsistent protection — and recommended common cross-border data-sharing frameworks

THE FULL CASE, IN FIVE BEATS

The Shift	African higher education frequently uses LA platforms hosted by external providers — cross-regime seam between operating, regulating, and hosting jurisdictions
What Is Emerging	Prinsloo, Slade, & Khalil 2022 (BJET) map the African privacy landscape: growing trend, few frameworks enacted, cross-border policy uneven
The Capability Question	Recommendation: common cross-border data-sharing frameworks — the architectural response to the inter-regime seam
Early Evidence	Frontier — outcome open; the architecture is not yet built and no continent-scale resolution exists
Open Problems	Names the extraterritorial-platform-governance pattern: increasingly universal as everyone builds on services they do not control

CONNECTIVITY

INDUCED	5.3
LENS	D4/PT6
CLO	CLO-4, CLO-5
COURSES	LEN 4, LEN 7, LEN 9

KEY REFERENCES

- Prinsloo, Slade, & Khalil (2022), "Data privacy on the African continent: Opportunities, challenges and implications for learning analytics," British Journal of Educational Technology 53(4):894–913, doi:10.1111/bjet.13226.
- African Union Convention on Cyber Security and Personal Data Protection (Malabo Convention, 2014) — continental policy backdrop, partial ratification.
- Slade & Prinsloo (2013), American Behavioral Scientist — earlier framing on which the 2022 paper builds.

LENS LESSON The African data-privacy case names a governance seam the existing curriculum underweights: extraterritorial-platform governance, where the operating, regulating, and hosting regimes diverge. The case is a frontier — peer-reviewed mapping, architectural recommendation, no continent-scale resolution yet built. The pattern is increasingly universal, and the case-grounded basis is the case for naming a sub-competency the framework do

CASE 68 HEALTHCARE/ONCOLOGY COMMERCIAL AI PROCUREMENT 2013 – 2018

Watson for Oncology — Delegation Marketed Ahead of Capability

IBM marketed Watson for Oncology with tumor-board concordance rates as high as 96%; STAT News investigation (Ross & Swetlitz, 2017–2018) and internal IBM documents documented unsafe and incorrect cancer-treatment recommendations and that the system was trained on a small number of synthetic cases rather than real patient outcomes; hospitals worldwide had purchased the tool on the marketed concordance claim

THE FULL CASE, IN FIVE BEATS

Background	Watson for Oncology marketed as cancer-treatment recommendation engine with tumor-board concordance rates as high as 96%
What Happened	STAT News investigation (Ross & Swetlitz, 2017–2018) drawing on leaked IBM documents: trained on small synthetic-case set curated by MSK, not real patient outcomes
The Investigation	Documented unsafe and incorrect recommendations; internal IBM engineers raised concerns about gap between marketed and operating capability
The Capability Gap	Hospitals worldwide purchased on marketed claim; major deployments (MD Anderson and others) later wound down
Aftermath & Reform	Evidence tier: journalism-grade (STAT primary, academic secondary); no independent peer-reviewed evaluation of operating accuracy; future validation ongoing

CONNECTIVITY

INDUCED	2.4
LENS	D5/PT6
CLO	CLO-1, CLO-3
COURSES	LEN 5, LEN 7, LEN 9

KEY REFERENCES

- Ross & Swetlitz (2017–2018), "IBM Watson recommended unsafe and incorrect cancer treatments" and adjacent investigations, STAT News — the load-bearing primary source; investigative journalism drawing on leaked IBM internal documents.
- Strickland (2019), "How IBM Watson Overpromised and Underdelivered on AI Health Care," IEEE Spectrum — independent retrospective synthesis of the public record.
- Topol (2019), Deep Medicine, Basic Books — secondary academic situating of Watson within the broader clinical-AI delegation discourse.

LENS LESSON Watson for Oncology is the canonical instance of clinical-AI delegation marketed ahead of validated capability. The operating capability was substantially smaller than the marketed concordance claims; the procurement mechanism evaluated the marketing rather than the validation; the deployments were wound down as the gap surfaced. The evidence-tier flag is binding: journalism-grade reporting is the load-bearing source, and future validation ongoing remains the honest qualifier.

CASE 69 K-12 EDUCATION SCHOOL SAFETY INFRASTRUCTURE RACIAL DISPARITIES 2010S – 2022

School Surveillance and Black Student Outcomes — Infrastructure as the Mechanism

Peer-reviewed study (Journal of Criminal Justice, 2022) examining school-surveillance infrastructure and Black student outcomes; the mechanism is the surveillance infrastructure (cameras, metal detectors, school resource officers), not the students, and the infrastructure drove a measured share of the outcome gap — one of the motivating cases for the CLO Gap attribution

THE FULL CASE, IN FIVE BEATS

Background	US school-surveillance infrastructure (cameras, metal detectors, SROs, ID checks) distributed unevenly — schools with predominantly Black students carry more
What Happened	Johnson et al. 2022 (Journal of Criminal Justice): separates student-level from school-level variables; measured share of outcome gap attributable to the infrastructure, not the students
The Investigation	Mechanism is the infrastructure, not the students — attributing the gap to the population mis-locates the mechanism in a measurable sense
The Capability Gap	Extends race-construct trio (Cases 119 eGFR, 106 pulse oximetry, 107 Hoffman) into K-12 education at the institutional-infrastructure layer
Aftermath & Reform	One of the motivating cases for the CLO Gap attribution — discipline of asking which share is the infrastructure vs. the population

CONNECTIVITY

INDUCED	8.1
LENS	D3/PT5
CLO	CLO-3, new Gap-attribution CLO
COURSES	LEN 4, LEN 7, LEN 8

KEY REFERENCES

- Johnson and colleagues (2022), school-surveillance infrastructure and Black student outcomes, Journal of Criminal Justice — the load-bearing peer-reviewed source for the case.
- Hoffman, Trawalter, Axt, & Oliver (2016), “Racial bias in pain assessment and treatment recommendations, and false beliefs about biological differences between blacks and whites,” PNAS 113(16):4296–4301 — race-construct trio at the cognitive-baseline layer (Case 63).
- Inker, Eneanya, Coresh, et al. (2021), “New Creatinine- and Cystatin C–Based Equations to Estimate GFR without Race,” NEJM — race-construct trio at the formula layer (Case 119).

> LENS LESSON The Johnson school-surveillance study locates the mechanism of an outcome disparity in the institutional infrastructure, not in the population th[illegible] to K-12 education at the infrastructure layer and is one of the motivating cases for the CLO Gap attribution. COI under the title — shared institution — is binding and rendered openly.

CASE 70 AUTONOMOUS VEHICLES REGULATORY GOVERNANCE INCIDENT DISCLOSURE 2023

Cruise's Partial Disclosure — How Disclosure Posture Decides Deployment

On 24 October 2023 the California DMV suspended Cruise's driverless deployment and testing permits, citing the company's misrepresentation of safety-relevant information after a robotaxi dragged a pedestrian roughly 20 feet at ~7 mph following an initial stop — disclosure posture, not the underlying collision sequence, was the load-bearing failure

THE FULL CASE, IN FIVE BEATS

Background	2 October 2023: pedestrian struck by human-driven vehicle, propelled into path of Cruise robotaxi; Cruise vehicle stopped, then pullover maneuver dragged pedestrian ~20 ft at ~7 mph
What Happened	Cruise initial disclosure to regulators and reporters showed only the initial stop; full pullover sequence not disclosed
The Investigation	DMV learned of pullover from another agency; obtained fuller video weeks after incident
The Capability Gap	24 October 2023: DMV Order of Suspension revokes driverless deployment and testing permits, citing misrepresentation of safety-relevant information
Aftermath & Reform	Journalism-tier flag: Order is investigation-grade; partial-disclosure mechanism reconstructed from TechCrunch, NBC News, SF Standard, Mission Local — future validation ongoing

CONNECTIVITY

INDUCED	5.4
LENS	D4/PT6
CLO	CLO-4, NEW CLO Delegation-with-revocation
COURSES	LEN 4, LEN 8, LEN 9

KEY REFERENCES

- California DMV (24 October 2023), Order of Suspension, Cruise LLC — driverless deployment and testing permits.
- NBC News (October 2023), reporting on the Cruise pedestrian incident and DMV suspension.
- TechCrunch (October 2023), incident-disclosure reconstruction.

> LENS LESSON Cruise is the foil to Waymo: same regulatory regime, same delegation problem, opposite governance choice. The DMV’s Order of Suspension is investigation-grade; the partial-disclosure mechanism is reconstructed from journalism. The evidence-tier flag is load-bearing — the internal timeline is journalism-tier and future validation continues as more of the company’s own post-mortem becomes public.

CASE 71 HIGHER EDUCATION PREDICTIVE ANALYTICS ACCESS PRICING 2010S – PRESENT (BROOKINGS SYNTHESIS 2021)

Enrollment-Algorithm Yield Optimization Across U.S. Higher Education

Vendor case studies report 23% yield increases (Washington), 33% net tuition increases with a 6-point cut to discount rate (EAB), and 173 additional freshmen without aid-budget increases (Othot); algorithms across at least seven major vendors price aid offers against each accepted applicant's modeled willingness to pay

THE FULL CASE, IN FIVE BEATS

Background	Two-stage architecture: predict enrollment probability per accepted applicant, then optimize aid offer for net tuition or yield
What Happened	Seven major vendors named: Ruffalo Noel Levitz, EAB, Rapid Insight, Capture Higher Ed, Othot, Whiteboard, Civitas Learning
The Investigation	Vendor-reported case studies: 23% yield gain (Washington), 33% net tuition gain with 6-point discount cut (EAB), 173 added freshmen (Othot)
The Capability Gap	Inversion of Case 112 (Georgia State support-trigger) and pair with Cases 61 (Bartlett lending) and 138 (Gándara community college)
Aftermath & Reform	Engler hedges binding: vendor obscurity, algorithmic vs. manual leveraging, no audit of specific protected-class impact; future validation ongoing

CONNECTIVITY

INDUCED	8.3
LENS	D4/PT5
CLO	CLO-3, CLO-4
COURSES	LEN 3, LEN 5, LEN 8

KEY REFERENCES

- Engler, A. (2021), “Enrollment algorithms are contributing to the crises of higher education,” Brookings Institution, 14 Sept 2021.
- Bettinger, E. et al. (2019), “Merit aid and college completion,” literature reviewed in Engler — graduation-odds effect sizes for $1,000 of additional aid.
- Goldrick-Rab, S. (2016), Paying the Price — broader synthesis on net-price, unsubsidized loans, and low-income completion.

> LENS LESSON Enrollment-algorithm yield optimization is the construct-choice case at the pricing layer of higher-education access: prediction is used to reduce aid per applicant, not to trigger support, and the operational target is “willingness to pay.” The evidence- tier flag is binding — vendor case studies are not auditable, the algorithmic-vs-manual distinction is bounded, and the protected-class fairness question requires audit access the public synthesis cannot supply. Future validation ongoing.

CASE 72 HIGHER EDUCATION FINANCIAL AID POLICY SOCIAL MOBILITY 2001 – 2017 (BURD IPEDS ANALYSIS); 2024 (LIFTING THE VEIL)

Crisis Point — Merit-Aid Leveraging at Public Flagships

Burd's IPEDS analysis: nearly $32 billion of public four-year institutional aid 2001–2017 went to students the federal government deemed able to pay without aid — about $2 of every $5 of institutional aid; financially needy students at high-merit-aid publics face larger unmet-need gaps

THE FULL CASE, IN FIVE BEATS

Background	Burd IPEDS analysis 2001–2017: nearly $32B of public institutional aid to students federally deemed able to pay — $2 of every $5
What Happened	Mechanism: state disinvestment + ranking-driven adoption of private-sector enrollment-management practices at public flagships
The Investigation	Consequence: low-income students at high-merit-aid publics face larger unmet-need gaps; need-based dollars redirected to yield
The Capability Gap	2024 Lifting the Veil (Harvard Ed Press, Burd ed.): multi-author synthesis — researchers, journalists, industry insiders
Aftermath & Reform	Construct substitution: 'students served' → 'net tuition revenue per matriculant'; never deliberated as policy; pair with Case 71

CONNECTIVITY

INDUCED	8.1
LENS	D4/PT5
CLO	CLO-3, CLO-4
COURSES	LEN 3, LEN 5, LEN 8

KEY REFERENCES

- Burd, S. J. (2020), “Crisis Point: How Enrollment Management and the Merit-Aid Arms Race Are Derailing Public Higher Education,” New America, ERIC ED604970.
- Burd, S. J. (ed., 2024), Lifting the Veil on Enrollment Management: How a Powerful Industry Is Limiting Social Mobility in American Higher Education, Harvard Education Press, ISBN 9781682538920.
- U.S. Department of Education IPEDS — public four-year institutional aid distribution 2001–2017, the data backbone of Burd’s analysis.

> LENS LESSON Burd’s IPEDS analysis is the construct-substitution case at public-flagship scale. Nearly $32 billion of institutional aid over seventeen years was redirected from need-based to merit- based distribution without a deliberative public-policy decision. The 2024 Harvard Education Press volume extends the documentation across multiple author types. Evidence-tier flag binding — policy-tier analysis, multi-author synthesis, causal-share decomposition bounded; future validation ongoing.

CASE 73 HIGHER EDUCATION REGULATORY OVERSIGHT ONLINE EDUCATION 2022

GAO Online Program Manager Oversight Gap (GAO-22-104463)

GAO found at least 550 colleges contracted with OPMs to support at least 2,900 academic programs as of July 2021; revenue-share arrangements typically gave the OPM 40–60% of tuition revenue per recruit, some up to 80%; ED lacked instructions for auditors to detect incentive-compensation violations and was not collecting the information it needed to oversee the arrangements

THE FULL CASE, IN FIVE BEATS

Background	1992 HEA banned incentive compensation for recruiters; 2011 ED guidance exempted OPMs under bundled-services construct
What Happened	GAO 2022: at least 550 colleges, 2,900 programs, OPM revenue-share typically 40–60% of tuition (some 80%) per recruit
The Investigation	Oversight gap: colleges/auditors lacked instructions to detect violations; ED not collecting information needed to oversee arrangements
The Capability Gap	2024 partial rescission + 2U Chapter 11 closed one boundary; successor OPMs and underlying delegation problem persist
Aftermath & Reform	Investigation-grade delegation-with-revocation case at population scale; pair with Case 74 (USC × 2U Luna) and Case 71 (Engler)

CONNECTIVITY

INDUCED	5.3
LENS	D4/PT6
CLO	CLO-4, CLO-5
COURSES	LEN 3, LEN 5, LEN 8

KEY REFERENCES

- U.S. Government Accountability Office (2022), “Higher Education: Education Needs to Strengthen Its Approach to Monitoring Colleges’ Arrangements with Online Program Managers,” GAO-22-104463.
- U.S. Department of Education (2011), “Dear Colleague” guidance on incentive-compensation bundled-services exemption — the regulatory artifact the GAO audits.
- Higher Education Act of 1992, incentive-compensation prohibition — the statutory basis the 2011 guidance interpreted.

> LENS LESSON GAO-22-104463 is the investigation-grade delegation-with- revocation case at population scale. A regulatory exemption permitted a contracting structure across 550+ colleges and 2,900+ programs; the monitoring architecture required to police the exemption’s boundaries was never built. The 2024 partial rescission and 2U Chapter 11 closed one boundary; the underlying delegation problem persists for successor OPMs.

CASE 74 HIGHER EDUCATION ONLINE PROGRAM MANAGEMENT PROFESSIONAL LICENSURE 2010S – 2024

USC × 2U Online MSW — When the Delegation Becomes the Product (Luna v. USC)

USC's online MSW grew from ~300 students per cohort pre-2010 to >3,000 per cohort through the 2U partnership; 2023 class action alleges USC marketed the online program as the 'same' as the residential program while outsourcing recruiting, advising, and clinical placement to 2U employees; partnership terminated 2024

THE FULL CASE, IN FIVE BEATS

Background	USC online MSW grew ~300 to >3,000 students per cohort via 2U; tuition tracked residential pricing (>$100K)
What Happened	Luna 2023 complaint: USC marketed online program as 'same' as residential while outsourcing recruiting, advising, clinical placement to 2U
The Investigation	Complaint alleges aggressive targeting of students of color and veterans; usc.edu email cover on OPM-employee operations
The Capability Gap	Licensure half: clinical-placement quality independently unstudied; downstream what-can-graduates-do question carried as gap
Aftermath & Reform	Partnership terminated 2024; pair with Case 73 (GAO regulator-side) and Case 71 (Engler pricing); journalism-tier flag binding

CONNECTIVITY

INDUCED	5.4
LENS	D4/PT6
CLO	CLO-4, CLO-5
COURSES	LEN 3, LEN 5, LEN 8

KEY REFERENCES

- Stephanie Luna v. University of Southern California, class action complaint, Los Angeles County Superior Court, May 2023.
- Higher Ed Dive reporting on the Luna complaint, May 2023; classaction.org and topclassactions.com summaries.
- Project on Predatory Student Lending statement on USC-2U partnership termination, 2024.

> LENS LESSON Luna v. USC is the consumer-side journalism-tier counterpart to GAO-22-104463 (Case 73). The complaint alleges USC marketed the online MSW as the “same” as the residential program while outsourcing the student-facing operations to 2U; the licensure-board half of the consequence chain is independently unstudied. Journalism-tier flag is binding — the predatory-targeting reconstruction is allegation-tier; future validation ongoing on litigation outcome and licensure consequences.

CASE 75 HIGHER EDUCATION ALGORITHMIC DECISION-MAKING HUMAN-COMPUTER INTERACTION 2025

Algorithmic College Admissions — Vendors' Claims vs. Applicants' Perceptions

Eighteen semi-structured interviews with recent U.S. university applicants, using speculative-design probes, surfaced systematic distances between vendor marketing claims (efficiency, fairness, enhanced fit) and applicants' own perceptions (opacity, distrust, anticipated discrimination)

THE FULL CASE, IN FIVE BEATS

The Shift	Pyle & Andalibi CSCW 2025: 18 semi-structured interviews with U.S. university applicants, speculative-design probes
What Is Emerging	Systematic distance: vendor marketing (efficiency, fairness, fit) vs. applicant perceptions (opacity, distrust, anticipated discrimination)
The Capability Question	Consent-side companion to Case 71 (Engler deployment) and Case 73 (GAO regulator-side); applicants as structurally absent voice
Early Evidence	Authors' hedge: 18 interviews is right for speculative-design depth, not for prevalence claims; future validation ongoing
Open Problems	Anchors the applicant-perception strand alongside Cases 61 (Bartlett) and 138 (Gándara) in the equity-in-prediction thread

CONNECTIVITY

INDUCED	8.4
LENS	D4/PT6
CLO	CLO-4, CLO-5
COURSES	LEN 3, LEN 5, LEN 8

KEY REFERENCES

- Pyle, C., & Andalibi, N. (2025), “Algorithmic College Admissions in the U.S.: Distances Between Vendors’ Claims and Applicants’ Perceptions,” Proceedings of the ACM on Human-Computer Interaction 9(7), CSCW369, doi:10.1145/3757550.
- Engler (2021), Brookings — paired deployment-side mapping (Case 71).
- GAO-22-104463 (2022) — paired regulator-side audit (Case 73).

> LENS LESSON Pyle and Andalibi is the peer-reviewed consent-side case at the applicant end of the enrollment-management deployment. Eighteen interviews surface systematic distance between vendor marketing and applicant perception across three axes; the authors’ methodological hedge is binding on prevalence claims. The case completes the partial-information triangle with Cases 71 (deployment-side) and 180 (regulator-side).

CASE 76 HIGHER EDUCATION SECURITIES LAW ENROLLMENT MANAGEMENT 2019 – 2022

In re 2U, Inc. Securities Class Action — When Yield Management Crashes Into Disclosure

Consolidated federal securities class action in the District of Maryland (TDC-19-3455, consolidated with TDC-20-1006); §10(b) and §20(a) allegations that 2U executives misled investors about declining enrollment projections during the Feb 26 2018 – Jul 30 2019 class period; $37M settlement July 2022; final approval Dec 9 2022 by Hon. Theodore D. Chuang

THE FULL CASE, IN FIVE BEATS

Background	Consolidated federal securities class action TDC-19-3455 + TDC-20-1006, D. Md., Hon. Theodore D. Chuang; filed Dec 2019
What Happened	Class period Feb 26 2018 – Jul 30 2019; §10(b) and §20(a) allegations on enrollment-projection disclosure to investors
The Investigation	Lead plaintiff Fiyyaz Pirani; co-lead Oklahoma City Employees Retirement System
The Capability Gap	$37M settlement July 2022; final approval Dec 9 2022; not an admission of liability — case teaches disclosure pattern, not fault
Aftermath & Reform	Investor-side complement to Case 73 (regulator audit) and Case 74 (consumer litigation); pair with Cases 71 and 72

CONNECTIVITY

INDUCED	5.4
LENS	D4/PT5
CLO	CLO-3, CLO-4
COURSES	LEN 3, LEN 5, LEN 8

KEY REFERENCES

- In re 2U, Inc. Securities Class Action, Civil Action No. TDC-19-3455 (consolidated with TDC-20-1006), D. Md., before Hon. Theodore D. Chuang; consolidated complaint filed December 2019.
- Stipulation of settlement and motion for final approval, In re 2U, Inc. Securities Class Action, July 2022; final approval order, December 9, 2022.
- Securities Exchange Act of 1934, §10(b) and §20(a); SEC Rule 10b-5 — the statutory and regulatory basis the complaint sounded in.

> LENS LESSON In re 2U is the investor-side disclosure-architecture case at OPM-delegation scale. A federal securities class action over enrollment-projection communications during a weakening- enrollment window settled for $37 million without admission of liability. The case teaches the disclosure pattern, not adjudicated fault, and completes the regulator-side and consumer-side oversight triangle with Cases 73 and 74 across the same calendar window.

CASE 77 GOVERNMENT SOCIAL WELFARE ALGORITHMIC ADMINISTRATION 2016 – 2023

Australia Robodebt — Algorithmic Debt-Recovery and the Royal Commission Verdict

Income-averaging algorithm used by the Australian Department of Human Services and Centrelink raised approximately 470,000 wrongful debts against welfare recipients between 2016 and 2019; Royal Commission led by Catherine Holmes AC SC delivered its final report July 7 2023 finding the scheme unlawful and circumstantially attributing multiple deaths to its operation

THE FULL CASE, IN FIVE BEATS

Background	Income-averaging algorithm: ATO annual income ÷ 26 compared to Centrelink fortnightly reports; arithmetic cannot establish overpayment
What Happened	~470,000 wrongful debts raised 2016–2019; burden of proof reversed onto recipients; agency legal advice flagged the seam and was set aside
The Investigation	Prygodicz 2019 Federal Court judgment found the method unlawful; Royal Commission final report July 7 2023 adjudicated the governance question
The Capability Gap	Commission attribution on deaths is circumstantial — not individual legal findings of causation; the careful language is part of the record
Aftermath & Reform	Pair with Case 64 (SyRI precedent), Case 69 (Johnson algorithmic public administration), Case 60 (Epic Sepsis delegation without validation)

CONNECTIVITY

INDUCED	5.2
LENS	D4+D5/PT6
CLO	CLO-4, CLO-5
COURSES	LEN 5, LEN 8, LEN 9

KEY REFERENCES

- Royal Commission into the Robodebt Scheme, Final Report, Commissioner Catherine Holmes AC SC, July 7, 2023 (Commonwealth of Australia).
- Prygodicz v Commonwealth of Australia (No 2) [2021] FCA 634 — Federal Court class-action settlement following the 2019 unlawfulness finding.
- Australian National Audit Office, Centrelink's Compliance Activities — Income Compliance Program, performance audit reports across 2017–2020.

> LENS LESSON Robodebt is the load-bearing reference for algorithmic public administration deployed at population scale without the burden of proof being honored. The Royal Commission's final report adjudicated approximately 470,000 wrongful debts and circumstantially attributed multiple deaths to the [illegible] ther, and neither is smoothed in the casebook's rendering.

CASE 78 GOVERNMENT EDUCATION AT SCALE HIGH-STAKES ASSESSMENT 2020

UK Ofqual A-Level Algorithm — National-Scale Grading Replaced by Algorithm, Withdrawn in Days

Ofqual standardisation algorithm applied to summer 2020 A-level grades following examination cancellation downgraded approximately 39.1% of teacher-estimated grades; results released August 13 2020; algorithm withdrawn August 17 2020 after four days of public protest; Centre Assessment Grades (teacher estimates) substituted; UK House of Commons Education Committee report (2021) documented disproportionate downgrade rates for state-school students

THE FULL CASE, IN FIVE BEATS

Background	Summer 2020 examinations cancelled; Ofqual deployed standardisation algorithm combining Centre Assessment Grades and Centre historical performance
What Happened	~39.1% of teacher-estimated grades downgraded; state-school students in large cohorts downgraded at higher rates than independent-school students in small cohorts
The Investigation	Results released Aug 13 2020; withdrawn Aug 17 2020 after four days of public protest; Centre Assessment Grades substituted
The Capability Gap	Cohort-size dependence of model is structural; technical report acknowledges incompatibility of standardisation with individual-level fairness
Aftermath & Reform	Pair with Case 186 (Gándara community-college equity), Case 189 (LiveHint bias), Case 69 (Johnson school surveillance)

CONNECTIVITY

INDUCED	5.4
LENS	D3+D4/PT5
CLO	CLO-3, CLO-4
COURSES	LEN 3, LEN 5, LEN 8

KEY REFERENCES

- Ofqual, Awarding GCSE, AS, A level, advanced extension awards and extended project qualifications in summer 2020: interim report, August 2020.
- UK House of Commons Education Committee, Getting the grades they've earned: COVID-19: the cancellation of exams and "calculated" grades, HC 254, July 2021.
- Royal Statistical Society, Submission to the Office for Statistics Regulation on the summer 2020 grading process, 2020 — independent statistical review of the standardisation methodology.

> LENS LESSON The Ofqual A-level case is the rare national-scale algorithmic deployment withdrawn within days under public pressure. The technical report was internally honest about the cohort-size dependence and the incompatibility of population-level standardisation with individual-level fairness; the four-day withdrawal is the evidence that the internal honesty did not travel out of the document to the affected population in advance of deployment.

CASE 79 CRIMINAL JUSTICE PREDICTIVE ANALYTICS ALGO FAIRNESS 2014 – 2018

COMPAS Recidivism Prediction — Calibration vs. Equal Error Rate

Northpointe COMPAS (Correctional Offender Management Profiling for Alternative Sanctions) risk-assessment instrument used in pretrial, parole, and sentencing decisions across multiple U.S. jurisdictions; ProPublica's May 2016 investigation reported a 2× higher false-positive rate for Black defendants; Chouldechova (2017) and Kleinberg, Mullainathan & Raghavan (2017) independently formalized the impossibility of simultaneously satisfying calibration and equal false-positive/false-negative rates across groups with unequal base rates

THE FULL CASE, IN FIVE BEATS

Background	Northpointe COMPAS deployed across U.S. pretrial, parole, sentencing decisions; ProPublica May 2016 audit on ~7,000 Broward County defendants
What Happened	ProPublica finding: ~2× false-positive rate for Black defendants among non-reoffenders; Northpointe response: predictive parity within risk scores
The Investigation	Both findings correct by their respective definitions; Chouldechova 2017 and Kleinberg/Mullainathan/Raghavan 2017 formalize the impossibility result
The Capability Gap	Calibration and equal FPR/FNR cannot be simultaneously satisfied when base rates differ across groups except in degenerate cases — binding mathematics
Aftermath & Reform	Pair with Case 61 (Bartlett), Case 62 (Coots), Case 64 (SyRI); central reference for the algorithmic-fairness literature

CONNECTIVITY

INDUCED	8.4
LENS	D3+D4/PT6
CLO	CLO-3, CLO-4
COURSES	LEN 3, LEN 5, LEN 8

KEY REFERENCES

- Angwin, J., Larson, J., Mattu, S., & Kirchner, L. (2016), “Machine Bias,” ProPublica, May 23, 2016 — the audit investigation of COMPAS scores in Broward County, Florida.
- Dieterich, W., Mendoza, C., & Brennan, T. (2016), COMPAS Risk Scales: Demonstrating Accuracy Equity and Predictive Parity, Northpointe Inc. response document.
- Chouldechova, A. (2017), “Fair Prediction with Disparate Impact: A Study of Bias in Recidivism Prediction Instruments,” Big Data 5(2):153–163, doi:10.1089/

> LENS LESSON COMPAS is the central reference in the contemporary algorithmic-fairness literature because the impossibility result was formalized against its au [illegible] dictive parity within risk scores) are correct by their respective definitions; the Chouldechova and Kleinberg/Mullainathan/Raghavan results show that the two cannot be simultaneously satisfied when base rates differ. The choice between fairness criteria is governance work, not technique.

CASE 80 FINANCIAL SERVICES CONSUMER CREDIT ALGORITHMIC DECISION-MAKING 2019 – 2021

Apple Card / Goldman Sachs — When the Lender Cannot Explain Its Own Model

New York State Department of Financial Services investigation March 2021 found no violation of New York anti-discrimination law in Apple Card credit-limit decisions following David Heinemeier Hansson's November 2019 viral allegation that his wife received a credit limit approximately 20 times lower despite shared assets; DFS documented "lack of transparency" as the structural problem and required Goldman Sachs to overhaul its customer-service process

THE FULL CASE, IN FIVE BEATS

Background	Nov 7 2019: Heinemeier Hansson Twitter thread on ~20× Apple Card credit-limit disparity; Wozniak names similar pattern; viral within days
What Happened	Goldman Sachs (issuing bank) cannot explain individual credit decisions to challenging applicants; explainability gap is operational, not narrowly algorithmic
The Investigation	NY DFS investigation opened late Nov 2019, findings released March 2021; reviewed ~400,000 New York credit-line decisions
The Capability Gap	DFS: no violation of NY anti-discrimination law under applicable statutory standard; structural finding is "lack of transparency"
Aftermath & Reform	Goldman required to overhaul customer-service process, build appeal mechanisms, document individual-applicant credit-decisioning explanations

CONNECTIVITY

INDUCED	5.2
LENS	D4/PT6
CLO	CLO-4, CLO-5
COURSES	LEN 3, LEN 5, LEN 8

KEY REFERENCES

- New York State Department of Financial Services, Report on Apple Card Investigation, March 23, 2021.
- Heinemeier Hansson, D. (2019), Twitter thread of November 7, 2019, archived in DFS investigation record and contemporaneous press coverage (Bloomberg, New York Times, Wall Street Journal, November 2019).
- Vigdor, N. (2019), “Apple Card Investigated After Gender Discrimination Complaints,” The New York Times, November 10, 2019.

> LENS LESSON Apple Card is the consumer-credit explainability case at deployment scale. DFS found no violation of New York anti- discrimination law under the applicable statutory standard, and DFS also found lack of transparency as the structural problem; Goldman Sachs was required to overhaul its customer-service process and build individual-applicant appeal mechanisms. The load-bearing hedge is the precision of the DFS finding — neither “fair” nor “unfair,” but “no violation under this standard, transparency gap as the structural problem.”

CASE 81 TECHNOLOGY HIRING ALGORITHMS MACHINE LEARNING 2014 – 2018

Amazon Hiring AI — Trained Bias, Deprecated 2018

Amazon internal recruiting-algorithm project initiated 2014, deprecated 2017 – 2018 after engineers determined the model could not be debiased; trained on ten years of historical resume data in which men predominated in technical roles; the model downgraded resumes containing the word "women's" and resumes from all-women's colleges; Reuters single-source investigation Oct 10 2018

THE FULL CASE, IN FIVE BEATS

Background	Amazon internal recruiting-algorithm project 2014 – 2018; goal: automate resume screening for technical roles; trained on 10 years of historical resume data
What Happened	Training-data composition encoded gender imbalance; model downgraded resumes containing "women's" and resumes from all-women's colleges
The Investigation	Feature-engineering attempts to debias failed: remaining features carried correlated signal reproducing the same downgrade pattern
The Capability Gap	Amazon deprecated the project 2017 – 2018; did not deploy at production scale; deprecation is the load-bearing decision
Aftermath & Reform	Evidence is Reuters single-source reporting (Oct 10 2018); Amazon never published technical detail; journalism-tier flag binding

CONNECTIVITY

INDUCED	8.1
LENS	D2+D3/PT6
CLO	CLO-3, CLO-4
COURSES	LEN 3, LEN 5, LEN 8

KEY REFERENCES

- Dastin, J. (2018), “Amazon scraps secret AI recruiting tool that showed bias against women,” Reuters, October 10, 2018 — the primary investigation.
- Subsequent commentary: Kim, P. T. (2017), “Data-Driven Discrimination at Work,” William & Mary Law Review 58(3):857–936 — academic frame for the structural pattern the Amazon case instantiates.
- Raghavan, M., Barocas, S., Kleinberg, J., & Levy, K. (2020), “Mitigating Bias in Algorithmic Hiring: Evaluating Claims and Practices,” Proceedings of FAT* 2020, pp. 469–481 — survey of the algorithmic-hiring landscape into which the Amazon case is positioned.

> LENS LESSON Amazon Hiring AI is the trained-bias-cannot-be-debiased case at major-technology-company scale. The engineering team determined that the model could not be made fair through feature engineering and the organization deprecated the project rather than deploying it. The journalis[illegible]; the structural teaching point — choose the construct, then ask whether the historical record supports learning to predict it — is the case's curricular anchor.

CASE 82 AVIATION CUSTOMER SERVICE AI AGENTS 2022 – 2024

Air Canada Chatbot Liability — Delegation Without Revocation

British Columbia Civil Resolution Tribunal ruled February 14 2024 in Moffatt v. Air Canada, 2024 BCCRT 149, that Air Canada was liable for bereavement-fare-policy misinformation provided to passenger Jake Moffatt by the airline's website chatbot; tribunal rejected Air Canada's argument that the chatbot was a "separate legal entity" responsible for its own outputs; small-claims-tribunal ruling with limited precedential weight outside BC but cited widely as articulating the principle that organizations are liable for representations made by their AI agents

THE FULL CASE, IN FIVE BEATS

Background	Nov 2022: Air Canada chatbot represents bereavement fare claimable retroactively; passenger Jake Moffatt books in reliance; Air Canada refuses claim
What Happened	BC Civil Resolution Tribunal small-claims forum; ruling Feb 14 2024 by Tribunal Member Christopher Rivers; \$650.88 in damages
The Investigation	Air Canada argued chatbot was "separate legal entity" responsible for its own outputs; tribunal rejected, finding no support in law
The Capability Gap	Principle: organizations are responsible for representations made by their AI agents; agents are not separate legal persons
Aftermath & Reform	Small-claims ruling with limited precedential weight outside BC; case teaches the form more than it establishes settled law

CONNECTIVITY

INDUCED	5.2
LENS	D5/PT6
CLO	CLO-5, CLO-4
COURSES	LEN 5, LEN 8, LEN 9

KEY REFERENCES

- Moffatt v. Air Canada, 2024 BCCRT 149 (British Columbia Civil Resolution Tribunal, February 14, 2024), Tribunal Member Christopher Rivers presiding.
- Cecco, L. (2024), “Air Canada ordered to pay customer who was misled by airline’s chatbot,” The Guardian, February 16, 2024 — contemporaneous press coverage of the ruling.
- Air Canada bereavement-fare policy text (as in effect November 2022 and through the period covered by the ruling) — referenced in the tribunal decision as the divergence the chatbot’s representation produced.

> LENS LESSON Air Canada chatbot is the delegation-without-revocation case at customer-interaction-AI-agent scale. The BC Civil Resolution Tribunal’s ruling holds that organizations are liable for representations made by their AI agents and that the agents are not separate legal persons; the small-claims venue limits the precedential weight, but the principle has been cited widely as the first clear judicial articulation of the form. The case teaches the form more than it establishes settled law.

CASE 83 EDUCATION AT SCALE PREDICTIVE ANALYTICS ALGORITHMIC EQUITY 2012 – 2024

Wisconsin DEWS — A Decade of Algorithmic Dropout Prediction

Wisconsin Department of Public Instruction's Dropout Early Warning System (DEWS) deployed since 2012, producing risk scores for approximately 200,000 sixth- through ninth-grade students per year; Perdomo, Britton, Hardt, & Abebe FAccT 2025 regression-discontinuity analysis across ~10 years of data found the analysis cannot rule out zero treatment effect on graduation; DPI's own 2021 internal equity audit and The Markup's 2023 investigation found the system was less accurate for Black, Hispanic, and English-learner students

THE FULL CASE, IN FIVE BEATS

Background	Wisconsin DPI Dropout Early Warning System deployed since 2012; ~200,000 students per year in grades 6 – 9 receive risk scores
What Happened	Perdomo, Britton, Hardt, Abebe FAccT 2025 RDD on ~10 years of data: cannot rule out zero treatment effect of being above DEWS threshold on graduation
The Investigation	Wisconsin DPI 2021 internal equity audit "Is DEWS Fair?": less accurate for Black, Hispanic, English-learner students; agency continued unchanged
The Capability Gap	The Markup 2023 investigation (Feathers) documented disparate-impact finding and agency response
Aftermath & Reform	Both streams load-bearing; pair with Case 69 (Johnson), Case 186 (Gándara), Case 101 (Purdue Course Signals reverse causality)

CONNECTIVITY

INDUCED	8.3
LENS	D3+D4/PT6
CLO	CLO-3, CLO-4
COURSES	LEN 3, LEN 5, LEN 8

KEY REFERENCES

- Perdomo, J. C., Britton, T., Hardt, M., & Abebe, R. (2025), “Difficult Lessons on Social Prediction from Wisconsin Public Schools,” Proceedings of FAccT 2025, doi:10.1145/3715275.3732175 (also arXiv:2304.06205).
- Wisconsin Department of Public Instruction (2021), Is DEWS Fair? — internal equity audit of the Dropout Early Warning System.
- Feathers, T. (2023), “False Alarm: How Wisconsin Uses Race and Income to Label Students ‘High Risk,’” The Markup, April 27, 2023 — investigation documenting the disparate-impact finding and the agency response.

> LENS LESSON Wisconsin DEWS is the load-bearing case for a prediction system operating at population scale for more than a decade without evidence tha[illegible]ers outcome- changing intervention and with evidence that the prediction’s accuracy varies across protected-attribute subgroups. Both the peer-reviewed null and the agency-audit disparate-impact finding travel together; the case carries both evidence streams in one entry rather than parallel ones.

CASE 84 GOVERNMENT 2014

VA Wait-Time Scandal

Veterans died waiting for care; 300,000+ on waiting lists or waiting 6+ months; staff falsified records

THE FULL CASE, IN FIVE BEATS

Background	A 14-day access target pressed hardest on schedulers, the highest-turnover front-line role
What Happened	Phoenix and nationwide staff hid waits on secret lists while official metrics reported success
The Investigation	Fifteen years of GAO and IG warnings landed on a continually relearning scheduler workforce
The Capability Gap	Measurement gaming hardened into routine practice once turnover erased memory of deviance
Aftermath & Reform	Resignations and the 2014 Choice Act followed; trustworthy measurement proved harder to rebuild

CONNECTIVITY

COURSES	LEN 4, LEN 7, LEN 8

KEY REFERENCES

- VA Office of Inspector General, Report 14-02603-267 (2014) — secret waiting lists and falsified appointment data.
- GAO Veterans Health Administration reports (2000–2019) — fifteen years of data-reliability warnings.
- VA OIG reports (2005, 2007, 2008) — prior, unactioned findings.

> LENS LESSON The VA case is the canonical example of measurement failure as a capability failure. The system that should have surfaced the gap instead generated reports that hid it. The turnover among schedulers — the human capability operating the measurement instrument — meant the institution lost the knowledge to even notice it was lying to itself. The case stands as the strongest argument in this book for treating decision-grade evidence as a design requirement, not a reporting requirement.

CASE 85 DEFENSE EDUCATION AT SCALE 2012 – PRESENT

GIFT and the Adoption Gap

Active open-source ITS framework with demonstrated learning effectiveness; ubiquitous fielded adoption in routine military training has not been achieved

THE FULL CASE, IN FIVE BEATS

The Shift	Decades of research settled efficacy; the open question became why adaptive tutoring isn't everywhere
What Is Emerging	ARL and DEVCOM sustain GIFT as open-source authoring infrastructure with releases and symposium
The Capability Question	Ubiquitous fielded adoption across routine military training remains limited despite the working framework
Early Evidence	Tutoring-effectiveness literature is robust and GIFT-based studies continue to show learning gains
Open Problems	Procurement, pipeline integration, instructor workflows, and authority structure remain unbuilt adoption artifacts

CONNECTIVITY

COURSES LEN 1, LEN 10, LEN 8, LEN 6

KEY REFERENCES

- K. VanLehn, "The Relative Effectiveness of Human Tutoring, Intelligent Tutoring Systems, and Other Tutoring Systems," Educational Psychologist 46(4) (2011) — tutoring effectiveness.
- GIFT Project, gifttutoring.org — the framework, releases, and development under ARL / DEVCOM.
- Editors' synthesis of the GIFT adoption record — active development but limited ubiquitous fielding (quoted).

> LENS LESSON GIFT is the most directly relevant case in this book to the learning-engineering discipline itself. The technology works, the learning science works, the framework is active and supported. What has not been built is the institutional adoption pathway — procurement, training-pipeline integration, instructor workflow redesign, the authority structure — that would make adaptive tutoring a default rather than an experiment.

CASE 86 EDUCATION AT SCALE DEFENSE ONGOING

xAPI / Total Learning Architecture — Interoperability Gap

Despite xAPI adoption, most implementations remain siloed; cross-organizational learning-data interoperability largely unrealized

THE FULL CASE, IN FIVE BEATS

The Shift	Learning crosses many systems, so the field needed common records of experiences
What Is Emerging	xAPI and ADL's Total Learning Architecture envisioned records and credentials flowing across organizations
The Capability Question	Implementations remain siloed inside individual LMS platforms; cross-organizational data sharing has not scaled
Early Evidence	Data model and reference implementations work; ownership, consent, and cross-organization quality assurance lag
Open Problems	Like inBloom, a sound standard arrived ahead of the governance needed for sharing

CONNECTIVITY

COURSES LEN 4, LEN 8, LEN 6

KEY REFERENCES

- Advanced Distributed Learning Initiative, Total Learning Architecture documentation — the cross-boundary vision.
- xAPI specification (xapi.com) — the technical standard and reference implementations.
- IEEE ICICLE proceedings on TLA adoption challenges — the persistence of siloed implementations.

> LENS LESSON xAPI/TLA is the technical-standard analog of the implementation gap — the discipline has the data model, the spec, and the reference implementations. What it does not have is the governance and institutional architecture to make cross-organizational learning data flow. This is the canonical case in this book for treating data governance as a capability-engineering deliverable.

CASE 87 HEALTHCARE ONGOING

Implementation Science in Healthcare — The 17-Year Gap

Average time from research finding to clinical practice: 17 years; only ~14% of research findings ever reach practice

THE FULL CASE, IN FIVE BEATS

The Shift	Implementation science arose because knowing what works and practicing it are different problems
What Is Emerging	Translation averages seventeen years, and only about fourteen percent of findings ever reach practice
The Capability Question	Effective interventions exist; the system to adopt, sustain, adapt, and measure them does not
Early Evidence	Active Implementation Frameworks and EPIS show implementation can be engineered rather than left to chance
Open Problems	Building, funding, and owning the adoption-and-measurement pathway is the general unsolved problem

CONNECTIVITY

COURSES LEN 1, LEN 10, LEN 8, LEN 6

KEY REFERENCES

- E. A. Balas & S. A. Boren (2000), Yearbook of Medical Informatics — the 17-year / 14% translation figures.
- Z. Morris, S. Wooding & J. Grant, “The answer is 17 years, what is the question,” J. Royal Society of Medicine (2011) (quoted).
- D. Fixsen et al., Implementation Research: A Synthesis of the Literature (2005) — the Active Implementation Frameworks.

> LENS LESSON The 17-year gap is the structural problem that LENS exists to address. It is the difference between knowing what works and having it deployed. Every case in this book is a sample from a distribution governed by that gap. The success cases shorten it; the failure cases let it run.

CASE 88 AVIATION 1972 – PRESENT

NASA Saturn V Documentation

Foundational U.S. aerospace case for the cost of losing the institutional capability to build a system

THE FULL CASE, IN FIVE BEATS

Background	Saturn V was one of history's best-documented engineering programs with exhaustive drawings and records
What Happened	By the 2000s NASA could redesign Saturn V but no longer reproduce its making-knowledge
The Investigation	Documentation captured the what; the tacit how walked out with the retired workforce
The Capability Gap	Institutional capability lives in people and practices, not in the artifacts they leave behind
Aftermath & Reform	Serious programs now use apprenticeship and team continuity to keep tacit capability alive

CONNECTIVITY

INDUCED	6.3
LENS	D4/PT2
CLO	CLO-2
COURSES	LEN 1, LEN 8

KEY REFERENCES

- NASA Constellation Program documentation and reviews (2005–2010) — the difficulty of reproducing Saturn V capability.
- R. E. Bilstein, Stages to Saturn: A Technological History of the Apollo/Saturn Launch Vehicles (NASA SP-4206, 1980) — the program and its workforce.
- The documentation-vs-capability distinction (editors’ synthesis of the Saturn V record).

> LENS LESSON The Saturn V case is the canonical evidence that institutional knowledge is not equivalent to documentation. The documents persist; the capability does not. Capability engineering must treat the people who hold tacit knowledge as a primary institutional asset.

CASE 89 SPACE / AEROSPACE 2019 – 2024

Boeing Starliner

Multiple delays; the 2024 crewed flight left two NASA astronauts at the ISS for months; contemporary case for capability erosion at a legacy contractor

THE FULL CASE, IN FIVE BEATS

Background	Boeing won Commercial Crew on a human-spaceflight heritage once definitive in American aerospace
What Happened	Software errors in 2019, valve corrosion in 2021, and 2024 propulsion trouble stranded astronauts
The Investigation	GAO and NASA found generational retirement plus cost pressure eroding both contractor and buyer
The Capability Gap	Reputation outran reality because no instrument measured the legacy contractor's current capability
Aftermath & Reform	NASA leaned on independent reviews and SpaceX as the verified alternative absorbing the failure

CONNECTIVITY

INDUCED	6.3
LENS	D4/PT2
CLO	CLO-1
COURSES	LEN 5, LEN 8, LEN 6

KEY REFERENCES

- U.S. GAO, NASA Commercial Crew Program, GAO-20-121 (Jan. 2020) and GAO-19-504 (2019) — schedule slips and technical risks across the program.
- The Starliner test history — 2019 software errors, 2021 valve scrub, and the 2024 crewed flight and ISS stay.
- NASA reviews and reporting on the Starliner test program — issues across software, valves, propulsion, and integration testing (editors' synthesis).

> LENS LESSON Starliner is the case for sustained capability erosion at a legacy contractor whose track record had previously been definitive. The erosion happened over decades and was visible in retrospect; the institutional architecture for catching it did not exist.

CASE 90 SPACE / AEROSPACE 1996

Ariane 5 Flight 501

Maiden flight destroyed itself 37 seconds after launch; ~$500M payloads lost; reused Ariane 4 code never re-verified for Ariane 5

THE FULL CASE, IN FIVE BEATS

Background	Ariane 5 reused inertial reference software flown reliably on Ariane 4 to reduce risk
What Happened	The 1996 maiden flight veered off course and broke up 37 seconds after launch
The Investigation	A horizontal-velocity overflow shut down both redundant reference systems simultaneously through identical inherited code
The Capability Gap	Code is fit only for the envelope it was verified against; reuse demands re-verification
Aftermath & Reform	Safety-critical reuse became a verification event with every inherited assumption explicitly re-checked

CONNECTIVITY

COURSES LEN 5, LEN 8

KEY REFERENCES

- J. L. Lions (chair), Ariane 5 Flight 501 Failure Inquiry Board Report (1996) — the data-conversion overflow (quoted).
- Lions Report (1996) — the 37-second breakup and the lost payloads.
- Lions Report (1996) — the 64-bit-to-16-bit conversion overflow and the simultaneous shutdown of both inertial reference systems.

> LENS LESSON Ariane 5 is the canonical software-engineering case for the fallacy of risk-free code reuse. Code is fit for its envelope of operation. Reusing it in a different envelope is a new design decision and must be re-verified as one. The institutional capability that was missing was the requirement to re-verify.

CASE 91 NEUROSCIENCE SCIENCE POLICY (EU) 2013 – 2023

EU Human Brain Project — Top-Down Vision That Unraveled

A €1B EU-flagship initiative to simulate a living brain governed top-down under a single PI; the project unraveled as the field disputed both feasibility and approach, restructured under protest, and concluded in 2023 with the founding framing abandoned

THE FULL CASE, IN FIVE BEATS

Background	EU FET Flagship 2013 with ~€1B over a decade and a single-PI top-down vision: simulate a living brain
What Happened	Community contestation surfaces quickly: feasibility, leadership breadth, no scope-revision process
The Investigation	EU commissions mediation and restructuring; scope re-shaped around infrastructure platforms, not simulation
The Capability Gap	Project runs to 2023 conclusion; founding framing is not what was delivered
Aftermath & Reform	Pair with BRAIN (Case 121): same era and ambition, opposite governance models, opposite trajectories — governance is the variable

CONNECTIVITY

INDUCED	5.1
LENS	D1+D4/PT4
CLO	CLO-1, CLO-4
COURSES	LEN 1, LEN 7, LEN 8

KEY REFERENCES

- MIT Technology Review (2021), retrospective on big-science brain projects — the principal critical assessment of HBP alongside BRAIN.
- In Silico (2020), documentary by Noah Hutton — long-form follow of Markram and the HBP through the contestation period.
- Science and Nature contemporaneous coverage of the open letters and the mediation/restructuring (2014–2016).

> LENS LESSON The EU Human Brain Project is the paired contrast to the BRAIN Initiative (Case 121): same era, same field-scale ambition, opposite governance models, opposite trajectories. Top-down single-PI governance with a unified framing did not survive community contestation; the program was mediated away from its founding goal and ran to its scheduled conclusion in 2023. Evidence base is journalism-tier and the flag is rendered under the title; future validation will continue.

CASE 92 DEFENSE SUSTAINMENT AUDIT 2022 (GAO-22-104533)

GAO Weapon-System Sustainment Reviews — Operating Costs without Decision-Grade Data

GAO-22-104533 reviewed sustainment of selected DoD weapon systems and found that operating-and-support (O&S) costs — the dominant share of a system's lifecycle expense — are not reported at a level of completeness, consistency, or comparability that supports portfolio-level sustainment decisions; the finding recurs across GAO's weapon-system portfolio reporting and is a frontier evidence-architecture problem, not a failure attributable to any single program

THE FULL CASE, IN FIVE BEATS

The Shift	O&S cost typically dominates weapon-system lifecycle expense; sustainment decisions ride on O&S evidence
What Is Emerging	GAO-22-104533 and predecessor reports: O&S data not reported with completeness, consistency, comparability portfolio decisions require
The Capability Question	Structural problem across services and categories — not attributable to any single program manager
Early Evidence	GAO recommends standardized categories, time-series completeness, comparability; DoD implementation ongoing, gap narrowing in places
Open Problems	Frontier evidence-architecture case: how to make portfolio-scale decisions when the available evidence is acknowledged as insufficient

CONNECTIVITY

INDUCED	1.4
LENS	D1/PT4
CLO	CLO-1, CLO-3
COURSES	LEN 1, LEN 5, LEN 7

KEY REFERENCES

- Government Accountability Office (2022), “Weapon System Sustainment: Selected Aircraft and Combat Vehicles Sustainment Costs,” GAO-22-104533.
- Government Accountability Office (recurring), annual “Weapon Systems Annual Assessment” portfolio reports — context for the structural finding.
- Department of Defense (2022), DoD response to GAO-22-104533 — acceptance and implementation status of recommendations.

> LENS LESSON GAO-22-104533 is the frontier evidence-architecture case at portfolio scale: the O&S data that dominate weapon-system lifecycle decisions are not reported in a form the decisions require, and the gap is structural across the enterprise. The case is the worked example of the CLO **Judgment under inadequate evidence** at the budget- authority scale.

CASE 93 HEALTHCARE 1999 – PRESENT

Medical Errors as Systemic Failure

IOM 1999 estimate of 44,000–98,000 US deaths/year from medical error; Makary & Daniel (2016) estimate of ~250,000 deaths/year — substantively contested on methodological grounds; 2023 NEJM inpatient-harm study confirms persistence; 25-year reform arc with bounded successes and an unmoved population count

THE FULL CASE, IN FIVE BEATS

Background	Death certificates record proximate physiology with no field for medical error
What Happened	Makary and Daniel estimated 250,000 annual U.S. error deaths from systemic care failings
The Investigation	Critics challenged the extrapolation, arguing counterfactual attribution of error-caused death resists clean tallying
The Capability Gap	Without a tracked instrument and robust attribution, safety programs cannot prove worth against invisible harm
Aftermath & Reform	Targeted reforms cut bounded harms while system-wide mortality kept escaping measurement and persisted at scale

CONNECTIVITY

INDUCED	2.1
LENS	D3/PT5
CLO	CLO-3
COURSES	LEN 1, LEN 4, LEN 10, LEN 6

KEY REFERENCES

- Institute of Medicine, To Err Is Human: Building a Safer Health System (1999); Crossing the Quality Chasm (2001); Improving Diagnosis in Health Care (2015) — the field-defining trilogy and the 44,000–98,000 estimate; the systems framing.
- Makary, M. & Daniel, M. (2016), “Medical error — the third leading cause of death in the US,” BMJ 353:i2139 — the 250,000 estimate, the quoted framing, the extrapolation from four prior studies (IOM 1999; OIG 2010; Landrigan 2010; Classen 2011).
- Shojania, K. & Dixon-Woods, M. (2017), “Estimating deaths due to medical error: the ongoing controversy and why it matters,” BMJ Quality & Safety 26(5):423–428; with companion commentaries including Carr in Health Affairs — methodological contestation of the Makary extrapolation and CDC-ranking comparison.

> LENS LESSON The Makary data is the anchor evidence for the LENS argument because of its scale, its provenance (Johns Hopkins), and its framing — system failure rather than individual error. The seventeen years between **To Err Is Human** and Makary’s reassessment is also the implementation gap of Case 87 in another guise — and the cost of leaving it open is measured in lives at population scale.

CASE 94 GOVERNMENT TECHNOLOGY 2003 – 2012

LIBOR Manipulation

Major banks fined more than $9B; the benchmark underlying ~$350T of contracts was manipulable; replaced by SOFR

THE FULL CASE, IN FIVE BEATS

Background	Benchmark set from panel banks' rate estimates rather than observed transactions
What Happened	Traders pressured submitters to shade rates favoring their derivatives books for years
The Investigation	DOJ and UK regulators levied over nine billion in fines; Wheatley Review blamed design
The Capability Gap	Declarations rather than observations made systematic gaming the architecture's predictable output
Aftermath & Reform	LIBOR phased out; SOFR built on observed Treasury repo transactions replaced it

CONNECTIVITY

INDUCED	2.2
LENS	D3/PT3
CLO	CLO-3
COURSES	LEN 4, LEN 7

KEY REFERENCES

- U.S. Department of Justice settlements with Barclays, UBS, Deutsche Bank, RBS et al. (2012–2015) — the manipulation and fines.
- Wheatley, M., The Wheatley Review of LIBOR: Final Report (2012) — the estimate-vs-transaction design flaw (paraphrased).
- Wheatley Review (2012) — recommendation to anchor benchmarks in observed transactions.

> LENS LESSON LIBOR is the canonical financial-system case for an instrument whose design invites the gaming it then experiences. The capability that was missing was the basic architectural choice to measure observations rather than declarations. The reform — SOFR — re-engineered the measurement at the architecture level.

CASE 95 EDUCATION AT SCALE GOVERNMENT 2009 – 2015

Atlanta Public Schools Cheating Scandal

~180 educators implicated; 35 indicted; 11 convicted under RICO statute; foundational U.S. education measurement-gaming case

THE FULL CASE, IN FIVE BEATS

Background	District administered and reported the very high-stakes tests determining bonuses and recognition
What Happened	Educators across dozens of schools organized erasure parties to change students' answers
The Investigation	Newspaper analysis and state investigators documented scheme; thirty-five indicted, eleven convicted under RICO
The Capability Gap	Institution being measured operated the instrument measuring it, with no independent audit
Aftermath & Reform	Convictions made APS the prominent U.S. high-stakes-testing fraud case fuelling reassessment

CONNECTIVITY

INDUCED	2.2
LENS	D3/PT3
CLO	CLO-3
COURSES	LEN 4, LEN 7

KEY REFERENCES

- Special Investigators, Final Report on Atlanta Public Schools (2011) — the organized cheating and the quoted finding.
- Atlanta Journal-Constitution investigative series (2009–2011) — the erasure-rate analysis.
- State of Georgia v. Hall et al. (2014–2015) — indictments and RICO convictions.

> LENS LESSON The APS cheating scandal is the strongest available case for measurement-gaming under high-stakes incentives in education. The capability gap was at the measurement architecture: the institution being measured operated the instrument that measured it. The reform pattern is direct — independent measurement audit — but rarely implemented at scale.

CASE 96 GOVERNMENT TECHNOLOGY 1992 – 2008

Bernard Madoff / SEC Failures

~$65B Ponzi scheme — the largest in history; SEC repeatedly investigated and cleared Madoff; foundational regulator-capability case

THE FULL CASE, IN FIVE BEATS

Background	Former NASDAQ chairman reported steady returns that were entirely fictitious Ponzi fabrications
What Happened	Markopolos delivered SEC a detailed quantitative memo in 2005; investigation closed without action
The Investigation	SEC Inspector General found numerous missed opportunities and staff deference to Madoff's stature
The Capability Gap	Regulator's technical-evaluation pipeline lacked people able to check the checkable math
Aftermath & Reform	Collapse prompted SEC reforms including Office of Market Intelligence for technical triage

CONNECTIVITY

INDUCED	6.2
LENS	D4/PT3
CLO	CLO-2
COURSES	LEN 4, LEN 7

KEY REFERENCES

- SEC Office of Inspector General, Investigation of Failure of the SEC to Uncover Madoff's Ponzi Scheme, Report OIG-509 (2009) — the quoted finding.
- Markopolos, H. (2010), No One Would Listen — the 2005 memo and its dismissal.
- United States v. Madoff (2009) — guilty plea and the $65B figure.

> LENS LESSON The Madoff case is the canonical example of a regulator without the technical capability to evaluate the evidence it received. The complaint was specific. The math was checkable. The institution did not have the people to check it. The capability gap was at the regulator's expertise pipeline, not at the evidence.

CASE 97 DEFENSE GOVERNMENT 1996 – 2001

9/11 Intelligence Sharing Failures

2,977 killed; the 9/11 Commission found systemic intelligence-sharing failures across the U.S. government

THE FULL CASE, IN FIVE BEATS

Background	Multiple agencies held pieces; no institution owned integration as a required architectural function
What Happened	CIA tracked future hijackers from Kuala Lumpur; FBI flagged flight training; none aggregated
The Investigation	9/11 Commission and Joint Inquiry documented specific sharing failures across FBI, CIA, NSA
The Capability Gap	Architecture was missing; shared watch-lists, mandated handoffs, and a fusion body existed nowhere
Aftermath & Reform	2004 Intelligence Reform Act created ODNI and NCTC, building integration as institutional deliverable

CONNECTIVITY

COURSES LEN 7, LEN 8, LEN 3

KEY REFERENCES

- National Commission on Terrorist Attacks Upon the United States, The 9/11 Commission Report (2004) — the quoted “failure of imagination” and the specific sharing failures.
- 9/11 Commission Report (2004) — the Kuala Lumpur tracking and the Phoenix/Minneapolis flagging.
- Joint Inquiry into Intelligence Community Activities Before and After September 11, 2001 (2002) — cross-agency information-sharing failures.

> LENS LESSON 9/11 is the foundational U.S. case for cross-agency information- sharing as an engineering deliverable. The architecture had not been built. The reform built it. The cost of the missing architecture was paid in 2001. The discipline LENS represents is the kind of work that, applied across the U.S. intelligence community in the 1990s, would have produced the architecture earlier.

CASE 98 HEALTHCARE 1999 – 2004

Vioxx Withdrawal

Tens of thousands of excess cardiovascular events estimated (FDA's Graham, 2004; Lancet 2005); Merck withdrew Vioxx in September 2004; ~\$4.85B settlement; FDA Amendments Act of 2007 and the Sentinel Initiative (2008) institutionalized active post-market surveillance

THE FULL CASE, IN FIVE BEATS

Background	Widely prescribed COX-2 inhibitor approved 1999; post-market surveillance thin relative to exposure
What Happened	VIGOR trial showed five-fold heart attack rate; Merck read comparator as protective instead
The Investigation	Hearings and Graham testimony showed cardiovascular signals present in trial record years earlier
The Capability Gap	Disclosure architecture between manufacturer data, FDA reviewers, and prescribers failed to aggregate signal
Aftermath & Reform	Merck settled near five billion; REMS and Sentinel built active post-market surveillance

CONNECTIVITY

INDUCED	2.4
LENS	D3+D5/PT5
CLO	CLO-3, CLO-5
COURSES	LEN 4, LEN 7

KEY REFERENCES

- Bombardier, C. et al. (2000), “Comparison of upper gastrointestinal toxicity of rofecoxib and naproxen in patients with rheumatoid arthritis,” VIGOR trial, NEJM 343(21):1520–1528 — the five-fold myocardial-infarction signal and the naproxen-protective hypothesis.
- Bresalier, R. et al. (2005), “Cardiovascular events associated with rofecoxib in a colorectal adenoma chemoprevention trial,” APPROVe, NEJM 352(11):1092–1102 — placebo-controlled confirmation of cardiovascular risk; early trial termination.
- Graham, D. J. et al. (2005), “Risk of acute myocardial infarction and sudden cardiac death in patients treated with COX-2 selective and non-selective NSAIDs: nested case-control study,” Lancet 365(9458):475–481 — Kaiser Permanente population-level cardiovascular risk analysis.

> LENS LESSON Vioxx is the canonical pharmaceutical case for post-market surveillance as a capability deliverable. The signal existed; the institutional architecture to aggregate it did not. The reform pattern — Sentinel — built the architecture. The case is the drug-industry analog of the EHR case (Case 35): a measurement architecture too thin for the system that depended on it.

CASE 99 CORPORATE L&D TRAINING EVALUATION WORKFORCE DEV 1959 – PRESENT

The Kirkpatrick Chain of Evidence — Where Corporate L&D Stops

Across a US corporate-training market sized in the >\$125B/year range, the dominant evaluation framework structurally collapses: most teams stop at Levels 1–2 (reaction + learning) and never measure Level 3 (behavior change) or Level 4 (business results) — measuring the variable that flatters the program

THE FULL CASE, IN FIVE BEATS

The Shift	Kirkpatrick four levels (Reaction / Learning / Behavior / Results) — dominant framework for sixty years; framed as 'chain of evidence'
What Is Emerging	Practice-literature synthesis: most organizations stop at Levels 1–2; Levels 3 and 4 require data the training org typically can't access
The Capability Question	US corporate-training market sized >\$125B/year per ASTD; measurement concentrated on variable that flatters the program
Early Evidence	Evidence-tier flag: practice-synthesis, not single peer-reviewed study; pattern is consistent, magnitudes still consolidating
Open Problems	Capability deliverable is an evaluation architecture that crosses the Level-2/Level-3 seam; pair with Blume (Case 100) for transfer

CONNECTIVITY

INDUCED	2.1
LENS	D3/PT5
CLO	CLO-3
COURSES	LEN 4, LEN 7, LEN 8

KEY REFERENCES

- Kirkpatrick (1959–1960), original four-level evaluation series in Journal of the ASTD; updated as Kirkpatrick & Kirkpatrick, Evaluating Training Programs (3rd ed., 2006).
- Devlin Peck, “Kirkpatrick Model: A Guide to the Four Levels of Training Evaluation” — synthesis of the stop-at-L2 pattern in corporate practice.
- D2L, “Kirkpatrick’s 4 Levels of Training Evaluation,” practitioner guide documenting the same pattern.

> LENS LESSON The Kirkpatrick stop-at-L2 pattern is the corporate-scale instance of the enthusiasm-evidence gap and the direct illustration of the revised “decision-grade evidence” point: the evidence most L&D decisions ride on is structurally sub-decision-grade. Evidence base is practice-synthesis tier; the pattern is consistent across sources, the magnitudes still consolidating; future validation will continue.

CASE 100 CORPORATE L&D TRAINING TRANSFER ORGANIZATIONAL BEHAVIOR 2010

Training Transfer — The Gap Between Learning and Doing

A meta-analysis of 89 empirical studies finds training transfer is positively related to cognitive ability, conscientiousness, motivation — and decisively to a supportive work environment; the literature carries explicit hedges about inconsistent measurement and significant variability

THE FULL CASE, IN FIVE BEATS

The Shift	Blume et al. meta-analysis of 89 studies on training transfer — extent to which training produces on-job behavior change
What Is Emerging	Transfer positively related to cognitive ability, conscientiousness, motivation, and work environment
The Capability Question	Work environment (supervisor + peer support) among strongest predictors and the decisive system-layer variable
Early Evidence	Load-bearing hedge: 'inconsistent measurement of transfer and significant variability in findings' — preserved in case
Open Problems	Pair with Kirkpatrick (Case 99) — together they close the corporate-L&D gap and motivate the return-to-work interface

CONNECTIVITY

INDUCED	2.3
LENS	D2/PT4
CLO	CLO-2, CLO-3
COURSES	LEN 2, LEN 4, LEN 7

KEY REFERENCES

- Blume, Ford, Baldwin, & Huang (2010), “Transfer of Training: A Meta-Analytic Review,” Journal of Management 36(4):1065–1105, doi:10.1177/0149206309352880.
- Baldwin & Ford (1988), “Transfer of Training: A Review and Directions for Future Research,” Personnel Psychology 41(1):63–105 — the foundational synthesis the 2010 meta-analysis updates.
- Burke & Hutchins (2007), “Training Transfer: An Integrative Literature Review,” Human Resource Development Review 6(3):263–296 — the integrative-review companion synthesis.

> LENS LESSON Blume et al. is the strongest current peer-reviewed synthesis on training transfer: cognitive ability, conscientiousness, motivation, and decisively the work environment predict whether training produces on-job behavior change. The literature carries explicit load-bearing hedges about inconsistent measurement and significant variability; the case is included with the hedges intact.

CASE 101 HIGHER-ED ANALYTICS EARLY-WARNING SYSTEMS EVIDENCE ARCHITECTURE 2012 – 2013

Purdue Course Signals — The Reverse-Causality Retention Claim

Purdue's widely cited claim that students taking two or more Course Signals classes were 21% more likely to be retained was challenged by Mike Caulfield in 2013, who identified the dose–response curve as an artifact of selection: students persist and therefore take more Signals courses, not the reverse — Alfred Essa reproduced the apparent retention gain by substituting 'received a piece of chocolate' for 'took a Signals class' in a simulation

THE FULL CASE, IN FIVE BEATS

Background	Arnold & Pistilli LAK 2012 — 21% retention advantage for students taking two or more Course Signals classes; widely cited
What Happened	Caulfield 2013 (_e-Literate_, _Inside Higher Ed_) — identifies dose–response curve as reverse-causality artifact: persistence enables more Signals classes, not the reverse
The Investigation	Essa simulation — substituting 'received a piece of chocolate' for 'took a Signals class' reproduces the apparent retention gain
The Capability Gap	Original study never peer-reviewed outside conference proceedings yet became one of the most-referenced learning-analytics studies
Aftermath & Reform	Deeper-evidence-of v1 Cases 51 (predictive-analytics bias) and 39 (Georgia State); a named methodological failure distinct from the bias finding

CONNECTIVITY

INDUCED	2.1
LENS	D3/PT5
CLO	CLO-3, CLO-4
COURSES	LEN 2, LEN 5, LEN 8

KEY REFERENCES

- Arnold, K. E., & Pistilli, M. D. (2012). Course Signals at Purdue: Using learning analytics to increase student success. Proceedings of LAK 2012, 267–270. doi:10.1145/2330601.2330666 — the original study at the center of the critique.
- Caulfield, M. (2013). A discussion of the Purdue Course Signals retention numbers. e-Literate blog series and Inside Higher Ed commentary — the reverse-causality critique.
- Essa, A. (2013). Simulation reproducing the Course Signals retention curve using a placebo treatment (“received a piece of chocolate”) — methodological demonstration of the artifact.

> LENS LESSON Course Signals is the named methodological failure in the predictive-analytics conversation: the institution measured a number that felt like the failure made it cared about, using a design that could not isolate effect from selection. The original study was never peer-reviewed outside conference proceedings, the correction came from outside the original network. Both are part of the cautionary reading.

CASE 102 K-12 EDUCATION SCALABLE INTERVENTIONS EQUITY 2019

Growth-Mindset National Experiment — Heterogeneous Effects

A nationally representative RCT of US 9th-graders found a less-than-1-hour online growth-mindset intervention improved grades among lower-achieving students and increased advanced-math enrollment, but the effect was conditional on peer norms — the intervention changed grades only where peer norms aligned with the intervention's message

THE FULL CASE, IN FIVE BEATS

The Shift	Growth-mindset interventions — short scalable psychological interventions; substantial laboratory and small-school evidence base by the late 2010s
What Is Emerging	Yeager et al. _Nature_ 2019 — nationally representative RCT of US 9th-graders; less-than-1-hour online module; pre-registered moderator analysis
The Capability Question	Headline outcome: improved grades among lower-achieving students; increased advanced-math enrollment relative to active control
Early Evidence	Conditional preserved verbatim: 'effect was conditional on peer norms' — intervention changed grades only where peer norms aligned with the intervention's message
Open Problems	Methodological importance: treatment-effect heterogeneity as the finding, not as a nuisance; pair with Case 186 (Gándara) at the scalable-equity-intervention layer

CONNECTIVITY

INDUCED	8.3
LENS	D2/PT5
CLO	CLO-2, CLO-4
COURSES	LEN 2, LEN 5, LEN 8

KEY REFERENCES

- Yeager, D. S., Hanselman, P., Walton, G. M., Murray, J. S., Crosnoe, R., Muller, C., Tipton, E., Schneider, B., Hulleman, C. S., Hinojosa, C. P., Paunesku, D., Romero, C., Flint, K., Roberts, A., Trott, J., Iachan, R., Buontempo, J., Yang, S. M., Carvalho, C. M., Hahn, P. R., Gopalan, M., Mhatre, P., Ferguson, R., Duckworth, A. L., & Dweck, C. S. (2019). A national experiment reveals where a growth mindset improves achievement. Nature, 573(7774):364–369. doi:10.1038/s41586-019-1466-y — the case’s primary trial.
- National Study of Learning Mindsets, ICPSR 37353 — the trial dataset.
- Dweck, C. S. (2006). Mindset: The New Psychology of Success. Random House — the broader theoretical framework the intervention rests on.

> LENS LESSON The growth-mindset national experiment is the methodologically clean model of how a population-scale RCT can earn the heterogeneity-as-finding stance. The intervention improved grades among lower-achieving students and increased advanced-math enrollment — and the effect was conditional on peer norms. The qualifying language is the load-bearing hedge; headline-only readings in either direction miss the substance.

CASE 103 AVIATION 1981 – PRESENT

Crew Resource Management & CAST

83% reduction in U.S. commercial aviation fatality risk (1998–2008); 95% reduction in fatalities per 100M passengers over 20 years

THE FULL CASE, IN FIVE BEATS

Background	Tenerife showed crashes came from crew coordination breakdowns, not individual lack of flying skill
The Intervention	United formalized CRM in 1981 with protocols, authority gradients, and structured briefings as procedure
How It Worked	Engineered the interaction system so junior challenges had a named, authorized route to the decision
The Evidence	CAST closed-loop hazard work cut fatality risk 83 percent; Collier Trophy in 2008
What Transferred	Design logic of paired cultural change plus measurement loop exported to surgery and AI systems

CONNECTIVITY

COURSES LEN 1, LEN 4, LEN 2, LEN 3

KEY REFERENCES

- FAA Advisory Circular 120-51E, Crew Resource Management Training — CRM protocols and authority-gradient training.
- Helmreich, Merritt & Wilhelm (1999), "The Evolution of Crew Resource Management Training in Commercial Aviation," International Journal of Aviation Psychology.
- Spanish CIAIAC / ALPA reports on the 1977 Tenerife collision — the overridden crew challenge.

> LENS LESSON CRM is the canonical evidence that capability is engineerable at the system level, not just the individual. Tenerife was not solvable by hiring better pilots — only by changing the authority structure inside the cockpit. The intervention worked because it was **paired**: a procedural change plus a cultural change in how the procedure was authorized. Either alone fails.

CASE 104 HEALTHCARE 2008 – PRESENT

WHO Surgical Safety Checklist

Death rate 1.5% → 0.8% in eight-site pilot; complications down >33%; adopted by the majority of surgical providers worldwide; Ontario population study (2014) found no significant mortality benefit after a mandated rollout

THE FULL CASE, IN FIVE BEATS

Background	Surgical harm was widespread because teams in motion rarely paused to verify shared understanding
The Intervention	WHO and Harvard introduced a nineteen-item checklist piloted across eight hospitals in eight countries
How It Worked	Forced pauses at three junctures required teams to halt and speak confirmations aloud together
The Evidence	NEJM study showed death rate halved and complications dropped a third across all sites
What Transferred	Ontario mandated rollout produced no benefit; the pause works only when genuinely authorized

CONNECTIVITY

INDUCED	2.3
LENS	D3/PT5
CLO	CLO-3, CLO-4
COURSES	LEN 4, LEN 10

KEY REFERENCES

- Haynes, A. et al. (2009), "A Surgical Safety Checklist to Reduce Morbidity and Mortality," NEJM — the 1.5%→0.8% result and major-complication reduction.
- WHO Safe Surgery Saves Lives campaign documentation — the nineteen-item checklist and the three junctures.
- Gawande, A. (2009), The Checklist Manifesto — the pause as the active mechanism.

> LENS LESSON The Surgical Safety Checklist is the canonical evidence that a tiny artifact — one page, nineteen items — can produce population-scale effects when paired with the structural change of **requiring a pause.** The checklist alone is paper. The pause alone is anxiety. Together they constitute the smallest effective capability intervention in the dataset. The Ontario follow-up underscores the secondary requirement: the artifact carries the effect only when the institution actually enforces the pause. Where mandate replaced authorization, the measured effect attenuated.

CASE 105 HEALTHCARE 2004 – PRESENT

Keystone ICU / Pronovost Checklist

Central-line-associated bloodstream infections (CLABSI) reduced to near zero across 103 Michigan ICUs; ~1,500 lives saved in 18 months; ~$100M saved; sustained at ten years

THE FULL CASE, IN FIVE BEATS

Background	Central-line infections persisted because nurses lacked procedural path to intervene across the authority gradient
The Intervention	Pronovost paired a five-item sterile checklist with a required nurse stop authority
How It Worked	Requirement, not permission, plus an impersonal trigger made the stop usable against senior physicians
The Evidence	CLABSI rates fell near zero across 103 ICUs; effect sustained at ten years
What Transferred	Packaged as the AHRQ CUSP toolkit, replicated in forty states and internationally as paired design

CONNECTIVITY

COURSES LEN 4, LEN 10, LEN 5

KEY REFERENCES

- Pronovost, P. et al. (2006), "An Intervention to Decrease Catheter-Related Bloodstream Infections in the ICU," NEJM 355 — the trial and the near-zero result.
- Pronovost & Vohr (2010), Safe Patients, Smart Hospitals — the checklist-plus-nurse-authority pairing (paraphrased).
- Lipitz-Snyderman, A. et al. (2011), BMJ — sustained effect at follow-up.

> LENS LESSON Keystone is the clearest evidence in healthcare that a technical intervention without authority intervention produces no durable change, and authority intervention without a technical artifact produces no measurable change. Both are necessary. The empirical record of Keystone is the strongest available argument for designing interventions as **pairs** — and treating the cultural half as engineering, not aspiration.

CASE 106 DEFENSE 2018 – PRESENT

Navy Surface Warfare Readiness Reform

Tripled ship-driving training hours; 10 pass-or-fail career assessments; Ready-for-Sea Assessments — 3 of 18 forward-deployed ships immediately sidelined

THE FULL CASE, IN FIVE BEATS

Background	Two fatal 2017 collisions exposed seamanship training degraded to CD-ROM self-study
The Intervention	Navy restored SWOS classroom instruction, tripled training hours, and created ten career assessments
How It Worked	Reform paired simulators and restored curricula with aviation-style debriefing and explicit gate ownership
The Evidence	GAO noted the Navy still lacks systematic evaluation of whether reforms improve readiness
What Transferred	Live in-progress reform shows mature capability engineering must build measurement infrastructure from the start

CONNECTIVITY

COURSES LEN 4, LEN 10, LEN 5

KEY REFERENCES

- GAO-21-168 (and GAO-20-154), Navy readiness reform — the lack of systematic outcome evaluation (paraphrased).
- Readiness Reform Oversight Council, One-Year Report (2019) — restored training, assessments, and gates.
- Navy and NTSB reports on the Fitzgerald and McCain collisions (2017–2019) — the training-degradation antecedent.

> LENS LESSON The Navy reform is a paired intervention in progress: technical (training restored, simulators procured, assessments created) plus cultural (debriefing, gate ownership). What it lacks is the third half: evidence that the intervention has worked. LENS treats this as a teachable case for what mature capability engineering should include from the start — the measurement infrastructure that lets the institution know whether its investment is producing the capability it bought.

CASE 107 ENERGY 1979 – PRESENT

INPO and the Nuclear Academy

No INES-level event at U.S. commercial reactors post-INPO; sustained improvement in INPO/WANO performance indicators across the industry

THE FULL CASE, IN FIVE BEATS

Background	Three Mile Island exposed an industry where lessons at one plant never reached others
The Intervention	Utilities founded INPO within months on the premise one accident threatened everyone's license
How It Worked	Honest peer review by working operators gave a non-statutory body its enforcement weight
The Evidence	No significant INES-level event since founding; fleet-wide performance indicators improved broadly across the industry
What Transferred	Shared exposure, regulatory legitimacy, and peer review crossed borders to WANO after Chernobyl

CONNECTIVITY

COURSES LEN 1, LEN 8, LEN 3

KEY REFERENCES

- Rees, J. (1994), Hostages of Each Other: The Transformation of Nuclear Safety since Three Mile Island — INPO's design and the "hostages" premise (paraphrased).
- Report of the President's Commission on the Accident at Three Mile Island (Kemeny Commission, 1979) — the pre-TMI culture.
- Nuclear Energy Institute, "Lessons from the 1979 Accident at Three Mile Island"; National Academy for Nuclear Training — accreditation and peer evaluation.

> LENS LESSON INPO is the strongest evidence in any domain that capability engineering can be undertaken at the level of an **industry**, not just an organization. The conditions that made it possible — shared catastrophic exposure, regulatory legitimacy, an honest peer-review architecture — appear wherever a single failure can damage every operator.

CASE 108 AVIATION 2000 – PRESENT

Korean Air Safety Transformation

From industry pariah (16 aircraft written off, 700+ lives lost, 1970–1999) to spotless passenger safety record since 1999

THE FULL CASE, IN FIVE BEATS

Background	Korean Air's pre-2000 loss rate was seventeen times United's, rooted in cockpit authority gradients
The Intervention	Korean Air hired David Greenberg from Delta to mandate English, adapt CRM, and modernize
How It Worked	Switching to English stripped out Korean honorifics that had silenced first officers and absorbed warnings
The Evidence	No fatal passenger accident since the reforms; the 2006 Phoenix Award recognized the turnaround
What Transferred	Cultural legacy is not destiny; locate the binding feature and engineer the medium that carries it

CONNECTIVITY

COURSES LEN 2, LEN 8

KEY REFERENCES

- NTSB, Aircraft Accident Report: Korean Air Flight 801, Guam (2000) — authority gradients as a root cause.
- Air Transport World Phoenix Award documentation (2006) — recognition of the turnaround.
- Gladwell, M. (2008), Outliers — the Korean Air chapter on power distance and the English-language change.

> LENS LESSON Korean Air is the strongest aviation evidence that cultural legacy is not destiny. A specific cultural feature — high power distance in the cockpit — was the binding capability constraint. Once it was redesigned, the safety record changed categorically. The intervention is also methodologically interesting because it operated on language: English became the cockpit language because English has no honorifics to enforce. The interface for cultural redesign was a linguistic one.

CASE 109 INDUSTRIAL 1950S – PRESENT

Toyota Production System / Andon Cord

Front-line authority to stop the line resolves most defects quickly at the source; defect-propagation cost minimized; the system adopted globally

THE FULL CASE, IN FIVE BEATS

Background	The cheapest moment to fix a defect is when the seer has least standing to halt
The Intervention	Toyota installed a pull-cord letting any worker signal a problem and stop the line
How It Worked	Mechanism paired with psychological safety, rapid supervisor response, no-blame analysis, and Five Whys
The Evidence	American copies failed because workers feared pulling it; Toyota's protected authority is the variable
What Transferred	Authority interventions and technical artifacts are inseparable; the manufacturing twin of Keystone nurse-stop

CONNECTIVITY

COURSES LEN 10, LEN 2, LEN 8

KEY REFERENCES

- Liker, J. (2020), The Toyota Way (2nd ed.) — the andon cord, the cultural pairing, and the American imitation (paraphrased).
- Spear, S. & Bowen, H. (1999), “Decoding the DNA of the Toyota Production System,” HBR — the response protocol and embedded learning.
- Shingo, S. (1989), A Study of the Toyota Production System — the technical mechanism.

> LENS LESSON The Andon Cord is the foundational evidence that authority interventions and technical artifacts are inseparable. The cord means nothing without the protected authority to pull it. The protected authority means nothing without the cord to act through. The pair is irreducible — and that irreducibility is the LENS co-optimization commitment in physical form.

CASE 110 HEALTHCARE 2006 – PRESENT

TeamSTEPPS

Improved teamwork, communication, and patient-safety culture across diverse settings; OR on-time first start +21%; adopted by thousands of healthcare organizations

THE FULL CASE, IN FIVE BEATS

Background	High-reliability research showed teamwork drives safety, but clinical settings lacked a structured curriculum
The Intervention	AHRQ and DoD jointly released TeamSTEPPS in 2006, training four core team competencies
How It Worked	Funding master-trainer programs, toolkits, and a support center alongside the curriculum compressed the gap
The Evidence	Studies show improved teamwork and safety culture; on-time surgical first starts rose 21 percent
What Transferred	Cross-domain capability transfer is engineerable when the funded path to sustained practice is included

CONNECTIVITY

COURSES LEN 1, LEN 10, LEN 8, LEN 3

KEY REFERENCES

- AHRQ, TeamSTEPPS 3.0 Curriculum (2023) — the framework and four competencies.
- DoD / AHRQ partnership documentation — the joint development and implementation infrastructure.
- Salas, E., Rosen, M. et al. (2009) — cross-domain team-training evidence base.

> LENS LESSON TeamSTEPPS is the canonical evidence that cross-domain capability transfer is engineerable and that it can shorten the implementation gap dramatically when the transfer is funded as part of the intervention rather than as an afterthought. The four competencies map directly to the LENS curriculum’s argument for what capability engineering competence looks like as a deliverable.

CASE 111 DEFENSE 1954 – PRESENT

U.S. Nuclear Navy / Rickover Training Model

Zero reactor accidents in 60+ years of U.S. Naval nuclear operations; the most demanding nuclear operator training program in the world

THE FULL CASE, IN FIVE BEATS

Background	Reactors at sea allowed no accident rate, forcing the human operating system to extreme standards
The Intervention	Rickover required every nuclear sailor to pass zero-defect qualification and continuous re-examination
How It Worked	Technical mastery paired with a mandatory questioning posture that obliges challenging superiors' assumptions
The Evidence	Zero reactor accidents across sixty years and thousands of reactor-years; duration is the key evidence
What Transferred	Same Navy let surface training decay; internal contrast isolates training philosophy as the variable

CONNECTIVITY

COURSES LEN 5, LEN 8, LEN 3

KEY REFERENCES

- Polmar, N. & Allen, T. (2007), Rickover: Father of the Nuclear Navy — the program and Rickover's philosophy (paraphrased).
- Naval Nuclear Propulsion Program documentation (NRC/DOE) — qualification standards and the accident record.
- Admiral Hyman G. Rickover, "Doing a Job" (Columbia University commencement address, 1982) — "people, not organizations... get things done" (quoted).

> LENS LESSON The Nuclear Navy is the longest-running continuous capability- engineering program in any high-consequence domain. The choice to treat training as a system parameter rather than as a cost center has produced sixty-plus years of zero reactor accidents. The contrast with the Surface Navy is the cleanest available test of what happens when capability is engineered versus when it is deferred — and the price of that engineering (the qualification ladder, the zero-defect oral boards, the continuous re-qualification) is visible on the budget line.

CASE 112 EDUCATION AT SCALE 2012 – PRESENT

Georgia State University Predictive Analytics

Six-year graduation rate 32% → 54%; equity gaps in graduation eliminated; 2,000+ more graduates per year

THE FULL CASE, IN FIVE BEATS

Background	GSU graduated only a third of students in six years, with large race and income gaps
The Intervention	GSU deployed daily monitoring of 800 risk factors with equity as a primary design constraint
How It Worked	Alerts route through advisors as decision support, delivering proactive outreach rather than gatekeeping
The Evidence	Graduation rose 32 to 54 percent and the equity gap was eliminated rather than narrowed
What Transferred	Construct definition and human-loop architecture, not the model itself, determine whether prediction helps

CONNECTIVITY

COURSES LEN 1, LEN 4, LEN 7

KEY REFERENCES

- Renick, T. & Strom, A. (2020) on GSU's advising transformation — the system design and outcomes.
- Georgia State University institutional research and Strategic Plan reports — graduation-rate and equity data.
- New York Times, "How Colleges Know You're Not Finishing" (2018) — the 800-factor advising model.

> LENS LESSON GSU is the positive counterpart to A-Level (35), Robodebt (36), and educational algorithmic bias (37). The same technical capability — a predictive model — was deployed under a different design constraint and produced an equity outcome rather than an equity harm. The case is the strongest available evidence that the **construct definition** and the **human-loop architecture** determine whether a predictive model produces good or harm, not the model itself.

CASE 113 EDUCATION AT SCALE 1990S – PRESENT

Cognitive Tutor / Carnegie Learning

Randomized controlled trials showed significant learning gains; RAND study found positive effects on Algebra I achievement; adopted across 3,000+ schools

THE FULL CASE, IN FIVE BEATS

Background	Earlier tutoring systems rested on intuition about learning rather than validated theory or controlled trials
The Intervention	Carnegie Learning built Cognitive Tutor from Anderson's ACT-R architecture with Bayesian knowledge tracing
How It Worked	A decomposable skill model, mastery measurement, and an adaptive interface concentrate practice where weakness sits
The Evidence	RAND's multi-site evaluation found significant Algebra I gains; the program scaled to 3,000-plus schools
What Transferred	Pipeline works for tractable, decomposable problems; ill-structured operational domains remain the open frontier

CONNECTIVITY

COURSES LEN 1, LEN 4, LEN 9

KEY REFERENCES

- Anderson, J., Corbett, A., Koedinger, K. & Pelletier, R. (1995), "Cognitive Tutors: Lessons Learned," Journal of the Learning Sciences — the ACT-R basis and design.
- Koedinger, K. & Corbett, A. (2006), Cambridge Handbook of the Learning Sciences — knowledge tracing and adaptive instruction.
- RAND Corporation Algebra I evaluation — the statistically significant achievement effects.

> LENS LESSON Cognitive Tutor is the canonical evidence that the learning- engineering process exists, works, and produces measurable benefits at scale — when the problem is tractable. It is also the canonical case for what tractability looks like: well-defined domain, decomposable skill model, instrumentable interface, rigorous evaluation. The frontier for the discipline is whether the same pipeline can deliver in the operational, ill-structured domains where capability matters most.

CASE 114 HEALTHCARE INDUSTRIAL 1982

Tylenol Recall

Foundational U.S. corporate crisis-management case; produced tamper-evident packaging regulation and modern recall practice

THE FULL CASE, IN FIVE BEATS

Background	The 1943 Credo pre-specified customers ahead of shareholders, ranking the trade-off crisis pressure inverts
The Intervention	J&J recalled 31 million bottles nationwide and engaged openly with regulators despite localized tampering
How It Worked	Pre-committed values moved the hardest decision out of the moment of maximum pressure
The Evidence	Market share recovered within a year; trust repaid the hundred million spent protecting customers
What Transferred	Tamper-evident packaging became standard and pre-committed values emerged as the deeper institutional principle

CONNECTIVITY

COURSES LEN 10, LEN 7, LEN 6

KEY REFERENCES

- Kaplan, T. (2014), The Tylenol Crisis — the recall and corporate response.
- James Burke (J&J CEO), in Lasting Leadership (Wharton) — the Credo quote.
- Greyser, S., Johnson & Johnson: The Tylenol Tragedy (HBS case, 1992) — market recovery and crisis management.

> LENS LESSON Tylenol is the canonical positive case for institutional response to crisis. The capability that was load-bearing was the pre-specified institutional commitment in the Credo. The crisis decision had been made decades earlier; in 1982 it was executed. The case is the strongest evidence in the business-ethics dataset that values must be pre-committed in writing to be operational under stress.

CASE 115 AVIATION 1976 – PRESENT

Aviation Safety Reporting System (ASRS)

NASA-administered confidential reporting system; more than 2M reports received; foundational architecture for evidence-driven aviation safety

THE FULL CASE, IN FIVE BEATS

Background	Valuable safety data lives with operators; punitive systems give them strongest reason to stay silent
The Intervention	FAA and NASA created a confidential channel in 1976 conferring immunity for reported conduct
How It Worked	A neutral third party paired with immunity made non-punitive protection credible enough to trust
The Evidence	Over two million reports surfaced automation surprise, runway incursions, and fatigue before accidents
What Transferred	Patient safety, maritime CHIRP, rail, and nuclear systems emulated the protected-reporting design pattern

CONNECTIVITY

COURSES LEN 4, LEN 8

KEY REFERENCES

- NASA ASRS Program documentation and annual reports — system design, immunity provision, and report volume.
- Reason, J. (1997), Managing the Risks of Organizational Accidents — non-punitive reporting as a model (paraphrased).
- NASA ASRS technical reports (Connell et al.) — patterns first surfaced via ASRS data.

> LENS LESSON ASRS is the canonical positive case for paired-intervention evidence architecture. The technical artifact (the reporting form and the database) is paired with the cultural commitment to non-punitive use. Either alone fails. Together they have produced the most comprehensive operational-safety dataset in any domain.

CASE 116 HEALTHCARE 1984 – PRESENT

Bristol Heart Babies Reform

Foundational UK case in clinical outcomes transparency; produced specialty-level performance reporting in UK cardiac surgery

THE FULL CASE, IN FIVE BEATS

Background	No UK system routinely compared pediatric cardiac outcomes; dangerously poor units looked ordinary from inside
The Intervention	The Kennedy Inquiry recommended routine risk-adjusted specialty-level outcomes reporting starting with cardiac surgery
How It Worked	Risk adjustment made cultural acceptance possible by ensuring hard cases would not penalize surgeons
The Evidence	The UK became among few countries publishing surgeon-level results; clinicians themselves became data users
What Transferred	The registry model extended across NHS specialties and pairs with bedside protocols like Keystone

CONNECTIVITY

COURSES LEN 4, LEN 7, LEN 3

KEY REFERENCES

- Kennedy, I. (2001), Learning from Bristol: The Report of the Public Inquiry into Children's Heart Surgery at the Bristol Royal Infirmary 1984–1995 — the inquiry and recommendations (paraphrased).
- Society for Cardiothoracic Surgery in GB & Ireland, national outcomes reports — the registry and surgeon-level publication.
- Berwick, D. (2013), A Promise to Learn — A Commitment to Act — the broader NHS-safety reform.

> LENS LESSON Bristol is the canonical UK case for outcomes-transparency as a paired intervention. The technical capability — registry design, statistical risk adjustment, publication architecture — was necessary. The cultural capability — surgeons accepting that their results would be visible and comparable — was equally necessary. The pair has produced one of the most mature specialty-outcomes regimes in any country.

CASE 117 AVIATION 1980S – PRESENT

Singapore Airlines Safety Transformation

Sustained safety record over decades despite challenging operating conditions; among the most safety-invested carriers in commercial aviation

THE FULL CASE, IN FIVE BEATS

Background	Thin aviation margins push capability investment downward; each budget cycle invites trimming the safety margin
The Intervention	From the 1980s Singapore Airlines invested in CRM, simulators, young fleet, and reporting culture
How It Worked	Framing safety as a competitive differentiator tied to brand gave the spend a commercial rationale
The Evidence	SQ006's response chose transparency over defensiveness, becoming a reference case in post-accident learning
What Transferred	Two routes to engineered safety emerge; voluntary investment is harder without crisis as justification

CONNECTIVITY

COURSES LEN 8

KEY REFERENCES

- Aviation Safety Council (Taiwan), SQ006 Accident Investigation Final Report (2002) — the accident and the airline's response.
- Singapore Airlines safety reports (multiple years) — training, fleet, and reporting-culture investment.
- IATA Operational Safety Audit (IOSA) documentation — investment ahead of regulatory minimums (paraphrased).

> LENS LESSON Singapore Airlines is the case for sustained capability investment in a competitive industry. The carrier has chosen safety capability investment as a primary differentiator. The result over decades is a safety record that distinguishes it from peers operating under comparable conditions.

CASE 118 HEALTHCARE CLINICAL AI 2018 – 2022

TREWS / Sepsis Watch

Prospective multi-site evidence of reduced mortality, organ failure, and length of stay when clinicians engaged with ML sepsis alerts in deployed care

THE FULL CASE, IN FIVE BEATS

Background	Sepsis is time-critical and heterogeneous; ML can surface the earliest signal from the EHR trace
The Intervention	Prospective multi-site evidence reports lower mortality, organ failure, and LOS when clinicians engage alerts
How It Worked	The deliverable is not the model — it is the alert design, the clinician role, and the evidence loop
The Evidence	Evidence is observational and prospective, not RCT; benefit collapses without clinician engagement
What Transferred	Delegation of detection can be done as a paired intervention with measured outcomes — the engineering counter

CONNECTIVITY

INDUCED	3.1
LENS	D5/PT6
CLO	CLO-3, CLO-5
COURSES	LEN 2, LEN 4, LEN 7

KEY REFERENCES

- Adams et al. (2022), "Prospective, multi-site study of patient outcomes after implementation of the TREWS machine learning-based early warning system for sepsis," Nature Medicine 28(7):1455–1460, doi:10.1038/s41591-022-01894-0.
- Henry et al. (2022), "Factors driving provider adoption of the TREWS machine learning-based early warning system and its effects on sepsis treatment timing," Nature Medicine 28(7):1447–1454, doi:10.1038/s41591-022-01895-z.
- Sendak et al. (2020), "Real-world integration of a sepsis deep learning technology into routine clinical care: implementation study," JMIR Medical Informatics 8(7):e15182 (Sepsis Watch implementation).

> LENS LESSON TREWS is the strongest current evidence that delegating early sepsis detection to a machine-learning system can improve patient outcomes — when the delegation is engineered as a paired intervention. The capability deliverable is the alert design, the clinician role, and the outcome-grounded evidence loop, not the model.

CASE 119 CLINICAL MEDICINE HEALTH EQUITY 2021

Removing the Race Coefficient from eGFR

A clinical estimating equation that for two decades raised estimated kidney function for Black patients was retired through a governance process; the documented effect on disparities remains unknown

THE FULL CASE, IN FIVE BEATS

Background	For two decades, the standard eGFR equation raised estimated kidney function for Black patients via a race coefficient
The Intervention	Documented downstream effects: later nephrology referral, later transplant wait-listing — the coefficient changed who got seen when
How It Worked	NKF-ASN Task Force ran the revision as a governance process: external panel, patient input, 20+ candidates, published report
The Evidence	2021 CKD-EPI race-free equation adopted nationally; the case is the construct-definition act, not yet a closed outcome proof
What Transferred	Hedge preserved: small all-group bias introduced; disparities effect remains unknown; outcome evidence is the continuing work

CONNECTIVITY

INDUCED	8.1
LENS	D3/PT5
CLO	CLO-3, CLO-4
COURSES	LEN 4, LEN 7, LEN 9

KEY REFERENCES

- Delgado et al. (2021), "A Unifying Approach for GFR Estimation: Recommendations of the NKF-ASN Task Force on Reassessing the Inclusion of Race in Diagnosing Kidney Disease," American Journal of Kidney Diseases 79(2):268–288 (published online 2021; in print vol. 79, 2022), doi:10.1053/j.ajkd.2021.08.003. Cited by online-first year, the year of the Task Force recommendation.
- Inker et al. (2021), "New Creatinine- and Cystatin C-Based Equations to Estimate GFR without Race," New England Journal of Medicine 385(19):1737–1749, doi:10.1056/NEJMoa2102953.
- Eneanya, Yang, & Reese (2019), "Reconsidering the Consequences of Using Race to Estimate Kidney Function," JAMA 322(2):113–114 — the equity argument that motivated the revision.

> LENS LESSON eGFR is the canonical recent instance of construct-definition as an equity capability deliverable. A race coefficient was embedded in a continuous estimating equation for two decades, documented downstream effects on referral and transplant access, and was retired through a governance process. The disparities effect of the change remains unknown; the case is the construct-revision act, not the closed outcome.

CASE 120 MEDICAL DEVICES CLINICAL CARE HEALTH EQUITY 1990 – 2020 – 2025

Pulse Oximetry Across Skin Tones

A bedside device validated on a non-representative population systematically under-detected hypoxemia in Black patients for thirty years; the bias persisted because device validation was never demographically stratified

THE FULL CASE, IN FIVE BEATS

Background	Pulse oximetry depends on tissue absorbance, including melanin; clearance documentation reports aggregate accuracy
The Intervention	Jubran & Tobin 1990 documented the bias; the finding did not change validation practice for thirty years
How It Worked	Sjoding et al. 2020 replicated in two large modern cohorts; ~3× higher occult hypoxemia in Black patients
The Evidence	The bias persisted because validation was never demographically stratified — aggregate accuracy concealed a group-specific failure
What Transferred	FDA 2025 draft guidance corrects the validation architecture; the measured disparities-outcome effect is the continuing work

CONNECTIVITY

INDUCED	8.2
LENS	D3/PT5
CLO	CLO-3, CLO-4
COURSES	LEN 4, LEN 7, LEN 9

KEY REFERENCES

- Sjoding, Dickson, Iwashyna, Gay, & Valley (2020), "Racial Bias in Pulse Oximetry Measurement," New England Journal of Medicine 383(25):2477–2478, doi:10.1056/NEJMc2029240.
- Jubran & Tobin (1990), "Reliability of Pulse Oximetry in Titrating Supplemental Oxygen Therapy in Ventilator-Dependent Patients," Chest 97(6):1420–1425 — original finding, published thirty years earlier.
- FDA (2025), "Pulse Oximeters for Medical Purposes — Non-Clinical and Clinical Performance Testing, Labeling, and Premarket Submission Recommendations: Draft Guidance for Industry and Food and Drug Administration Staff," issued January 7 2025, Docket No. FDA-2023-N-4976; Federal Register notice 2024-31540 — the regulatory corrective-action artifact, language may evolve in final.

> LENS LESSON Pulse oximetry is the validation-architecture instance of the race-construct trio. The bias was published in 1990, persisted for thirty years inside aggregate clearance accuracy, was re-documented in 2020, and reached the regulatory architecture in 2025. The capability deliverable is demographic stratification at validation, not after deployment.

CASE 121 NEUROSCIENCE SCIENCE POLICY 2011 – 2015 – PRESENT

Launching the BRAIN Initiative — Governance of a Grand Challenge

A multi-billion-dollar national research endeavor launched via a position-paper-to-policy iteration sequence, with governance contestation on the public record and a 2021 critical retrospective documenting that the unified-understanding framing exceeded what the science delivered

THE FULL CASE, IN FIVE BEATS

Background	Kavli 2011 symposium → 2012 Neuron position paper → 2013 OSTP/Presidential launch → 2015 BRAIN 2025 working-group plan
The Intervention	Position-paper-to-policy iteration is auditable: every step has a published artifact
How It Worked	Governance contestation on the public record: tool-builders vs. users; central planning vs. distributed creativity
The Evidence	2021 retrospective: unified-understanding framing exceeded delivered science; enthusiasm-evidence gap at field scale
What Transferred	Governance model is the deliverable — distributed working-group vs. top-down single-PI explains BRAIN survives, HBP unravels

CONNECTIVITY

INDUCED	5.1
LENS	D4+D1/PT4
CLO	CLO-1, CLO-4
COURSES	LEN 1, LEN 7, LEN 8

KEY REFERENCES

- Alivisatos, Chun, Church, Greenspan, Roukes, & Yuste (2012), “The Brain Activity Map Project and the Challenge of Functional Connectomics,” Neuron 74(6):970–974, doi:10.1016/j.neuron.2012.06.006.
- Jorgenson et al. (2015), “The BRAIN Initiative: developing technology to catalyse neuroscience discovery,” Phil. Trans. R. Soc. B 370(1668):20140164, doi:10.1098/rstb.2014.0164 — the BRAIN 2025 plan.
- Yuste & Bargmann (2017), “Toward a Global BRAIN Initiative,” Cell 168(6):956–959, doi:10.1016/j.cell.2017.02.023.

> LENS LESSON The BRAIN Initiative is one of the few large-program launches in the corpus whose position-paper-to-policy-to-implementation sequence is fully auditable. Governance choices were litigated in the public record, scope drifted from the founding framing, and a critical ten-year retrospective is part of the case materials. Its teaching value is the contested record, not a clean success or scandal.

CASE 122 HIGHER ED (UK) LEARNING ANALYTICS DATA GOVERNANCE 2014 – 2025

Open University 'Ethical Use of Student Data' and OU Analyse

The first institutional 'Ethical Use of Student Data' policy in higher education (2014); an eight-principle consent architecture co-designed with students that unblocked predictive analytics on hundreds of thousands of learners and supported documented intervention work

THE FULL CASE, IN FIVE BEATS

Background	Predictive learning analytics at distance-scale; the governance objection is credible, not abstract
The Intervention	OU authors first higher-education 'Ethical Use of Student Data' policy in 2014 — eight principles, co-designed with students
How It Worked	OU Analyse operates on top of the consent architecture; Analytics4Action pairs predictions with tutor judgment
The Evidence	2019 evaluation (Herodotou et al., BJET): 559 teachers, 14,000+ students; teacher engagement, not raw prediction, is what tracks with success
What Transferred	Governance objection was about trust — dissolvable by design; pair with SyRI where the objection was correct

CONNECTIVITY

INDUCED	5.1
LENS	D3/PT5
CLO	CLO-3, CLO-4
COURSES	LEN 4, LEN 7, LEN 8

KEY REFERENCES

- Slade & Prinsloo (2013), “Learning Analytics: Ethical Issues and Dilemmas,” American Behavioral Scientist 57(10):1510–1529, doi:10.1177/0002764213479366.
- Open University (2014), “Ethical Use of Student Data for Learning Analytics” — first institutional policy of its kind in higher education.
- Rienties, Boroowa, Cross, Kubiak, Mayles, & Murphy (2016), “Analytics4Action Evaluation Framework: A Review of Evidence-Based Learning Analytics Interventions at the Open University UK,” Journal of Interactive Media in Education 2016(1):2, doi:10.5334/jime.394.

> LENS LESSON The Open University case is the cleanest instance in the sweep of a governance objection dissolved by design. The 2014 ethical-use policy was the enabling engineering; OU Analyse and Analytics4Action operated on top of it; the 2019 evaluation showed teacher engagement tracked with intervention success. The open question — serve vs. surveil — remains contested, and the policy is a living artifact, not a solved problem.

CASE 123 NAVAL ENGINEERING DEFENSE SAFETY CERT 1963 – PRESENT

Navy SUBSAFE — Requirements as a Non-Negotiable Sustainment Deliverable

From 1915 to 1963 the US Navy lost 16 submarines to non-combat causes; since 1963 it has lost one — USS Scorpion, the only post-1963 loss that was not SUBSAFE-certified. The Columbia Accident Investigation Board cited SUBSAFE as a model NASA should emulate

THE FULL CASE, IN FIVE BEATS

Background	USS Thresher lost April 1963 with 129 aboard; investigation traces the gap to certification of the watertight-integrity boundary
The Intervention	SUBSAFE created within 54 days; formal requirements issued by December 20 1963
How It Worked	'Objective Quality Evidence' — verifiable fact, not probabilistic assessment — at every certification step; annual training and recurring audits
The Evidence	Non-combat losses: 16 in the 48 years before; one (Scorpion, uncertified) in the 62 years since; Columbia Accident Investigation Board endorsement
What Transferred	Zero-loss record is correlational across many co-varying factors over decades; hedge preserved

CONNECTIVITY

INDUCED	1.4
LENS	D1/PT3
CLO	CLO-1, CLO-3
COURSES	LEN 1, LEN 5, LEN 7

KEY REFERENCES

- Rear Admiral Paul E. Sullivan, statement to House Science Committee (2003), NASA/Columbia archive — primary congressional record on SUBSAFE.
- MIT Press, "SUBSAFE: An Example of a Successful Safety Program" — book chapter (open access).
- NASA SMA (2006), "USS Thresher Lessons Learned" briefing — safety-message archive.

> LENS LESSON SUBSAFE is the archetype of treating capability requirements as a recurring, auditable, non-waiverable deliverable across the entire system lifecycle. The before/after non-combat-loss record is one of the cleanest in safety engineering — and correlational across many co-varying factors over six decades. The hedge is part of the case.

CASE 124 HEALTHCARE WORKFORCE POLICY PATIENT SAFETY 2004 – 2010

California Nurse Staffing Ratios — Legislating a Capability Requirement

California's mandated minimum nurse-to-patient ratios reduced nurse workload by 1–2 patients per nurse and were modeled to imply 10–14% fewer surgical patient deaths in comparison states if matched — observational, cross-sectional, no California baseline

THE FULL CASE, IN FIVE BEATS

Background	California (2004) — first US state to mandate minimum unit-level nurse-to-patient ratios; written into law and enforced
The Intervention	Aiken et al. 2010 surveyed 22,336 nurses across CA / PA / NJ; California nurses cared for 1–2 fewer patients each
How It Worked	Modeled implication: PA and NJ would have 10.6% and 13.9% fewer surgical deaths at California's medical-surgical ratios
The Evidence	Hedge preserved: observational cross-sectional; no California baseline; contested stakeholder debate on causation
What Transferred	Pair with SUBSAFE (Case 123) — converting stated requirement to engineered requirement, with political-process cost

CONNECTIVITY

INDUCED	1.1
LENS	D1/PT3
CLO	CLO-1, CLO-3
COURSES	LEN 5, LEN 7, LEN 8

KEY REFERENCES

- Aiken, Sloane, Cimiotti, Clarke, Flynn, Seago, Spetz, & Smith (2010), "Implications of the California Nurse Staffing Mandate for Other States," Health Services Research 45(4):904–921, doi:10.1111/j.1475-6773.2010.01114.x.
- Aiken, Clarke, Sloane, Sochalski, & Silber (2002), "Hospital nurse staffing and patient mortality, nurse burnout, and job dissatisfaction," JAMA 288(16):1987–1993 — the workload-mortality relationship the 2010 modeling rests on.
- California Department of Health Services (1999–2004), AB 394 regulatory implementation documentation.

> LENS LESSON California's nurse-ratio mandate is the canonical recent instance of converting a stated capability requirement (adequate staffing) into an engineered, enforced one by law. The workload effect is direct; the modeled mortality estimate is observational and cross-sectional with no California baseline. The hedge is the case.

CASE 125 ANESTHESIOLOGY PATIENT SAFETY MEDICAL DEVICES 1986 – PRESENT

Anesthesia Monitoring Standards and the APSF — The Mortality Decline

Mandatory pulse oximetry and capnography monitoring (Harvard 1986; ASA 1986–87) converted silent hypoxemia and esophageal intubation from undetectable failures into monitored, recoverable ones; anesthesia-related mortality fell dramatically over subsequent decades — multifactorial decline

THE FULL CASE, IN FIVE BEATS

Background	Early 1980s anesthesia crisis: silent hypoxemia and esophageal intubation structurally undetectable until catastrophic
The Intervention	1982 ABC special + malpractice-insurance crisis + APSF founding (1985) force institutional change
How It Worked	Harvard standards (Eichhorn JAMA 1986): minimum monitoring with continuous pulse oximetry and capnography; ASA adopts 1986–87
The Evidence	Anesthesia mortality falls dramatically over subsequent decades; malpractice premiums decline alongside
What Transferred	Hedge preserved: decline is multifactorial; device standards still fail in documented edge cases (cf. Case 120 pulse oximetry across skin tones)

CONNECTIVITY

INDUCED	3.1
LENS	D3/PT5
CLO	CLO-3, CLO-5
COURSES	LEN 2, LEN 5, LEN 7

KEY REFERENCES

- Eichhorn, Cooper, Cullen, Maier, Philip, & Seeman (1986), “Standards for Patient Monitoring During Anesthesia at Harvard Medical School,” JAMA 256(8):1017–1020.
- Eichhorn (1989), “Prevention of intraoperative anesthesia accidents and related severe injury through safety monitoring,” Anesthesiology 70(4):572–577.
- Anesthesia Patient Safety Foundation (1985 – present), founding documents and the APSF Newsletter — institutional-history record of the broader change effort.

> LENS LESSON The Harvard / ASA / APSF anesthesia-monitoring intervention is the canonical cue/alert-design success: silent intraoperative hypoxemia and esophageal intubation were converted from undetectable failures into monitored, recoverable ones, and the specialty’s mortality and malpractice record moved over subsequent decades. The decline is multifactorial; the device standards still fail in documented edge cases (Case 120).

CASE 126 RAIL TRANSPORT SAFETY REPORTING SAFETY CULTURE 1996 – PRESENT

CIRAS — Confidential Incident Reporting for UK Rail

Between 2008 and 2012 the UK rail Confidential Incident Reporting and Analysis System received 2,228 reports — 45% led to tangible safety improvements and about 33% contained important safety information (program self-report); directly influenced a confidential reporting system in the US

THE FULL CASE, IN FIVE BEATS

Background	ScotRail pilot 1995–1997 — interview-based confidential reporting; insulated from employer disciplinary process
The Intervention	Ladbroke Grove crash 1999 (31 deaths); CIRAS mandated across UK mainline rail in 2000
How It Worked	Independent unit within RSSB from 2008; the independence is the credible commitment, written into operating structure
The Evidence	Operating program reports 2,228 reports 2008–2012, 45% led to tangible safety improvements (program self-report)
What Transferred	Non-aviation depth for the C4 competency; same structural form as ASRS/CRM (Cases 115, 103)

CONNECTIVITY

INDUCED	4.2
LENS	D4/PT2
CLO	CLO-4
COURSES	LEN 4, LEN 7, LEN 8

KEY REFERENCES

- Davies, Wright, Courtney, & Reid, “Confidential Incident Reporting on the UK Railways: The CIRAS System,” Cognition, Technology & Work, doi:10.1007/PL00011494.
- Rail Safety and Standards Board (RSSB), CIRAS program documentation 2008 – present — operating-program publications.
- University of Strathclyde, CIRAS impact case study — the operating-program-self-report on outcomes between 2008 and 2012.

> LENS LESSON CIRAS is the non-aviation companion to ASRS and CRM in the corpus. The pattern — confidential channel paired with credible commitment (institutional independence from discipline) — works in rail as it does in aviation. The operating outcome statistics are program self-report and deserve their tier acknowledgement; the institutional design is the audited finding.

CASE 127 WORKFORCE DEV STEM TRAINING EQUITY 2017 – 2023 (SIX CYCLES)

CIRCUIT — A Scalable, Equity-Centered Research Workforce Model

An eight-pillar cohort model that in 2022 supported over 100 undergraduate, graduate, and ROTC students from 'trail-blazing' backgrounds (first-generation, low-income, limited prior research access); peer-reviewed program description with longitudinal student outcomes across six cycles

THE FULL CASE, IN FIVE BEATS

Background	Eight-pillar program model published in peer-reviewed ASEE 2023 paper; longitudinal outcomes over six program cycles
The Intervention	2022 program supported >100 undergraduate, graduate, and ROTC students from 'trailblazing' backgrounds
How It Worked	Model is operationalized: holistic recruiting, targeted training, high-resolution assessment, diverse mentorship, partnerships
The Evidence	Honest framing preserved: self-authored multi-cycle program evaluation at a single program; external comparative would strengthen causal claim
What Transferred	Pair with Case 185 — building capability (this case) vs. deploying it against automation failure (proofreading); both carry COI render

CONNECTIVITY

INDUCED	1.2
LENS	D2/PT4
CLO	CLO-2, CLO-3
COURSES	LEN 1, LEN 2, LEN 5

KEY REFERENCES

- Cervantes, Floryanzia, Sharp, Gray-Roncal, & Johnson (2023), “Empowering Trailblazers toward Scalable, Systematized, Research-Based Workforce Development,” ASEE Annual Conference, doi:10.18260/1-2--43271.
- CIRCUIT program documentation (JHU Hub, 2017 – present) — institutional program description.
- National Academies / NSF research on undergraduate research experience effects on STEM persistence (broader literature against which the case sits).

> LENS LESSON CIRCUIT is a peer-reviewed eight-pillar workforce-development program for STEM trainees from trailblazing backgrounds, with longitudinal outcomes over six program cycles. The strongest honest framing is self-authored multi-cycle program evaluation; an external comparative would add the causal half. COI under the title — editor is senior author — is binding.

CASE 128 CLINICAL/TRANSLATIONAL RESEARCH TEAM SCIENCE HEALTHCARE WORKFORCE 2020 – 2022

Team Science Training for Clinical and Translational Scientists

Colorado CTSA team-science training (N=221, pre/post) reported statistically significant improvement ($p<0.05$) across all three TEAMS instrument competencies — Team Planning, Managing a Team, and Interpersonal Relations — with the largest gains in structured/tool-based domains and the smallest in Interpersonal Relations

THE FULL CASE, IN FIVE BEATS

Background	Colorado CTSA structured team-science program; N=221 pre/post evaluation 2020–2022
The Intervention	TEAMS instrument validated; three components — Team Planning, Managing a Team, Interpersonal Relations
How It Worked	Statistically significant improvement ($p<0.05$) across all three; largest gains in structured/tool-based, smallest in interpersonal
The Evidence	Honest limit: self-reported perceived competency, no longitudinal tracking, no comparison to non-trained control
What Transferred	Operationalizes 'collaboration as unit of measurement' — the Domain 3 sub-competency

CONNECTIVITY

INDUCED	4.3
LENS	D4/PT4
CLO	CLO-3, CLO-4
COURSES	LEN 4, LEN 7, LEN 8

KEY REFERENCES

- Colorado CTSA, “Team science training for clinical and translational scientists: An assessment of effectiveness,” Journal of Clinical and Translational Science, PMC12392353.
- Falk-Krzesinski et al. (2010), “Mapping a research agenda for the science of team science,” Research Evaluation — broader team-science literature backdrop.
- National Research Council (2015), Enhancing the Effectiveness of Team Science — the IOM/National Academies team-science synthesis.

> LENS LESSON The Colorado CTSA team-science training program is one of the few cases in the corpus that operationalizes “collaboration as a unit of measurement.” The TEAMS instrument resolves the capability into three components; structured training moved all three with $p<0.05$ but moved interpersonal relations least. The honest limit — self- reported perceived competency, no longitudinal tracking, no control — survives into the case.

CASE 129 HEALTH PROFESSIONS EDUCATION · INTERPROFESSIONAL COLLABORATION · PATIENT SAFETY 2013 – 2015

Interprofessional Education and the Evidence Gap

Decades-long well-funded movement to educate health professionals together for collaborative care; Cochrane 2013 found only 15 studies between 1999 and 2011 met inclusion criteria; IOM 2015 made the gap the central finding — 'paucity of high-quality research' linking IPE to measurable changes in practice and patient outcomes

THE FULL CASE, IN FIVE BEATS

The Shift	IPE — decades-long well-funded movement premised on training health professionals together for collaborative care
What Is Emerging	Reeves et al. Cochrane 2013: only 15 studies from 1999–2011 met inclusion; evidence base thin for linking IPE to practice and patient outcomes
The Capability Question	IOM 2015 makes the gap the central finding: 'paucity of high-quality research'; proposes a conceptual model for doing the measurement
Early Evidence	Canonical enthusiasm-evidence gap case — field instruments enthusiasm faster than outcomes; basis for Domain 3 sub-competency
Open Problems	Pair with Case 128 (team-science training) and 123 (implementation-science training) — collaboration measurement is possible at program scale, absent at field scale

CONNECTIVITY

INDUCED	2.1
LENS	D3/PT5
CLO	CLO-3
COURSES	LEN 4, LEN 7, LEN 8

KEY REFERENCES

- Reeves, Perrier, Goldman, Freeth, & Zwarenstein (2013), "Interprofessional education: effects on professional practice and healthcare outcomes (update)," Cochrane Database of Systematic Reviews, doi:10.1002/14651858.CD002213.pub3.
- Institute of Medicine (2015), Measuring the Impact of Interprofessional Education on Collaborative Practice and Patient Outcomes, National Academies Press, NCBI NBK338352.
- WHO (2010), Framework for Action on Interprofessional Education and Collaborative Practice — the international policy backdrop.

> LENS LESSON Interprofessional Education is the canonical enthusiasm-evidence-gap case in the corpus. A decades-long, well-funded movement; the strongest s [illegible] oduces measurable changes in practice and patient outcomes — is structurally thin. The IOM 2015 conceptual model is the proposed evidence architecture; the field's task is to build to it.

CASE 130 IMPLEMENTATION SCIENCE · TRANSLATIONAL WORKFORCE · EDUCATION AT SCALE 2020S

Implementation-Science Training — Stated Goals Outrunning Operational Practice

Survey of CTSA-funded TL1 training programs (N=50): most name collaboration, team science, and multi/inter/cross-disciplinary training as goals, but far fewer (14–24%) embed competency-based training and assessment, program evaluation, or experiential learning

THE FULL CASE, IN FIVE BEATS

The Shift	Implementation science = moving validated evidence into operational practice; CTSA TL1/T32 is the US training mechanism
What Is Emerging	Survey N=50 CTSA-funded TL1 programs: most name collaboration / team science / multi-disc training as goals
The Capability Question	Far fewer (14–24%) embed competency-based training and assessment, program evaluation, or experiential learning
Early Evidence	Same enthusiasm-ahead-of-evidence pattern as IPE (Case 129) at smaller scale — operational practices lag stated goals
Open Problems	Workforce-training counterpart to Case 87 ('17-year gap'); pair with Cases 128, 129 in the multidisciplinary-translation trio

CONNECTIVITY

INDUCED	6.4
LENS	D4/PT4
CLO	CLO-3, CLO-4
COURSES	LEN 4, LEN 7, LEN 8

KEY REFERENCES

- CTSA T32/TL1 program-goals study (2021), Journal of Clinical and Translational Science, PMC8826009.
- Morris, Wooding, & Grant (2011), "The answer is 17 years, what is the question: understanding time lags in translational research," Journal of the Royal Society of Medicine — the original 17-year-gap source for Case 87.
- Brownson, Colditz, & Proctor (2018), Dissemination and Implementation Research in Health (2nd ed.) — the broader implementation-science synthesis.

> LENS LESSON The CTSA TL1 program survey is the workforce-training instance of the enthusiasm-evidence gap pattern: most programs name competency-based training, assessment, and program evaluation as goals; the operational practices that would deliver on those goals are present in 14–24% of programs. The implementation-science workforce is the recovery mechanism for the 17-year research-to-practice gap; the case names the gap inside the recovery mechanism itself.

CASE 131 DEFENSE HUMAN FACTORS STANDARDS 2020 REVISION (SERIES SINCE 1968)

MIL-STD-1472H — Human Engineering as a Binding Acquisition Standard

DoD Design Criteria Standard MIL-STD-1472H, the 2020 revision of a series dating to 1968, converts human-factors and human-engineering findings into binding design criteria across DoD acquisition — controls, displays, anthropometry, workspace, environment, hazards — invoked by acquisition programs as a mandatory or tailored design specification

THE FULL CASE, IN FIVE BEATS

Background	MIL-STD-1472 series 1968 – present; eight revisions through H (Sept 15, 2020)
The Intervention	Design Criteria Standard: controls, displays, anthropometry, workspace, environment, hazards — binding, not advisory
How It Worked	Converts human-factors findings into engineered requirements an acquisition contract can specify
The Evidence	Structural analog of SUBSAFE (Case 123) at the human-engineering scale — requirements-as-deliverable form
What Transferred	Necessary but not sufficient: standard is the mechanism, program tailoring and verification determine the outcome

CONNECTIVITY

INDUCED	1.1
LENS	D1/PT3
CLO	CLO-1, CLO-4
COURSES	LEN 1, LEN 5, LEN 7

KEY REFERENCES

- Department of Defense (2020), MIL-STD-1472H “Department of Defense Design Criteria Standard: Human Engineering,” 15 September 2020 — replaces MIL-STD-1472G (2012).
- Department of Defense (2012), MIL-STD-1472G — the prior revision; revision history documents the 1968 origin and intermediate letters.
- Chapanis, A. (1965), “Man-Machine Engineering” — foundational text for the discipline the standard codifies.

> LENS LESSON MIL-STD-1472H is the structural archetype of converting a body of human-factors research into binding engineered requirements an acquisition contract can specify. The standard is the mechanism by which usability becomes a contractable, auditable deliverable rather than design advice. It is necessary but not sufficient; tailoring and verification determine whether the contract is honored.

CASE 132 EDUCATION AT SCALE MACHINE LEARNING MEMORY AND SKILL 2016

Duolingo Half-Life Regression — Spaced Repetition at Consumer Scale

Settles & Meeder (ACL 2016) introduced Half-Life Regression (HLR), a trainable spaced-repetition model that learns per-item forgetting rates from large-scale learner data; HLR was deployed in Duolingo's production review-scheduling system and the published evaluation reports improvements over heuristic schedulers (Leitner, Pimsleur) on Duolingo's own predictive metric and on a 14-day return-engagement metric

THE FULL CASE, IN FIVE BEATS

Background	Spaced-repetition theory from Ebbinghaus (1885); operational heuristics from Leitner (1972), Pimsleur (1967), SuperMemo (1985)
The Intervention	Settles & Meeder (ACL 2016) — Half-Life Regression learns per-item half-life from practice data using item features and history
How It Worked	Deployed in Duolingo's production review scheduler; trained on hundreds of millions of trials
The Evidence	Reported improvements on Duolingo's predictive metric and on 14-day learner return rate vs. Leitner / Pimsleur / logistic baselines
What Transferred	Hedges preserved: single-vendor study; 14-day return is engagement proxy not proficiency assessment; cross-domain generalization is structural argument not replication

CONNECTIVITY

INDUCED	2.3
LENS	D2/PT4
CLO	CLO-2, CLO-3
COURSES	LEN 2, LEN 5, LEN 8

KEY REFERENCES

- Settles, B., & Meeder, B. (2016), “A Trainable Spaced Repetition Model for Language Learning,” Proceedings of the 54th Annual Meeting of the Association for Computational Linguistics, pp. 1848–1858, doi:10.18653/v1/P16-1174.
- Ebbinghaus, H. (1885), Über das Gedächtnis — the empirical forgetting-curve foundation HLR formalizes.
- Pimsleur, P. (1967), “A memory schedule,” Modern Language Journal 51(2):73–75 — graduated-interval recall heuristic.

> LENS LESSON Half-Life Regression is one of the few peer-reviewed spaced-repetition deployments at consumer scale with reported behavioral outcomes. The structural argument — learn the half-life rather than fix it by heuristic — is strong; the evidence is single-vendor, the outcome is an engagement proxy not a proficiency assessment, and the cross-domain generalization rests on the structural argument rather than replication.

CASE 133 AVIATION SAFETY ENGINEERING HUMAN FACTORS 1996 – 2002

EGPWS / TAWS — Closing the CFIT Category in Commercial Aviation

Honeywell's Enhanced Ground Proximity Warning System (EGPWS, 1996), mandated by the FAA as Terrain Awareness and Warning System (TAWS) for US-registered turbine aircraft beginning in 2001 and broadly worldwide by 2002, converted controlled flight into terrain (CFIT) — historically one of the largest categories of commercial-aviation fatalities — into a category whose rate in equipped fleets has fallen sharply; CFIT events on properly equipped and operating airliners are now rare

THE FULL CASE, IN FIVE BEATS

Background	CFIT historically among the largest commercial-aviation fatality categories; Erebus 1979 (257), Cali 1995 (159) canonical
The Intervention	GPWS (Bateman, 1970s) reduced CFIT but produced late warnings and was blind to terrain ahead of the aircraft
How It Worked	EGPWS (Honeywell, 1996) added digital terrain database + position; forward-looking alerts hours before terrain contact
The Evidence	FAA TAWS mandate March 2001 (full by 2005); ICAO and most national regulators follow
What Transferred	CFIT in EGPWS-equipped fleets falls sharply; residual events involve disabled equipment, procedure deviation, or terrain outside database

CONNECTIVITY

INDUCED	3.1
LENS	D3/PT5
CLO	CLO-3, CLO-5
COURSES	LEN 3, LEN 5, LEN 7

KEY REFERENCES

- Bateman, C. D. (1999), "The Introduction of Enhanced Ground Proximity Warning Systems (EGPWS) into Civil Aviation Operations Around the World," in Proceedings of the 11th Annual European Aviation Safety Seminar, Flight Safety Foundation, pp. 259–274 — developer history.
- Federal Aviation Administration (2000), 14 CFR §§ 91.223, 121.354, 135.154 — Terrain Awareness and Warning System (TAWS) equipage requirement.
- Flight Safety Foundation (1998 – 2000), CFIT / ALAR Task Force reports — operational and outcome analyses motivating mandate.

> LENS LESSON EGPWS / TAWS is the canonical cue/alert intervention at fleet scale. The forward-looking terrain alert converted a failure mode in which the crew's terrain perception was the limiting variable into a monitored, recoverable mode. CFIT in equipped fleets has become rare; residual events typically involve inhibited equipment or procedure deviation, and the hedge is the case.

CASE 134 AVIATION AUTOMATION HUMAN FACTORS 1981 – 2008 (TCAS II VERSION 7.1)

TCAS — Coordinated Collision Avoidance and the Überlingen Lesson

TCAS II — the Traffic Alert and Collision Avoidance System — provides cockpit traffic display and coordinated Resolution Advisories (RAs) between aircraft on conflicting trajectories; mandated on US air-carrier and on most international turbine aircraft, TCAS converted mid-air collision in commercial aviation from a recurring fatality category to a rare event; the 2002 Überlingen mid-air (71 dead) exposed a specific coordination failure mode and drove the 2008 release of TCAS II Version 7.1 with the 'level off, level off' reversal logic

THE FULL CASE, IN FIVE BEATS

Background	TCAS II mandated on US air-carrier (early 1990s) and on most international turbine aircraft (ICAO); RAs coordinated via Mode S data link
The Intervention	Outcome: mid-air collision in TCAS-equipped fleets becomes rare through the 1990s and 2000s
How It Worked	Überlingen mid-air July 1 2002 (71 dead, including 52 children) — Tu-154 followed ATC, B757 followed TCAS RA; both descended
The Evidence	BFU finding: RA precedence over ATC insufficiently clear; ATC single-controller / degraded-console context a systemic failure
What Transferred	TCAS II Version 7.1 (2008): 'level off, level off' reversal logic; clarified RA precedence over conflicting ATC instructions

CONNECTIVITY

INDUCED	3.1
LENS	D3/PT5
CLO	CLO-3, CLO-5
COURSES	LEN 3, LEN 5, LEN 7

KEY REFERENCES

- RTCA (2008), DO-185B "Minimum Operational Performance Standards for Traffic Alert and Collision Avoidance System II (TCAS II)" — Version 7.1 with reversal logic.
- Bundesstelle für Flugunfalluntersuchung (BFU) (2004), AX001-1-2/02 — Investigation Report on the mid-air collision on 1 July 2002 near Überlingen.
- Eurocontrol (2008 – 2011), ACAS II Bulletin and Programme for the Harmonised Implementation of Satellite Navigation — V7.1 mandate timing in Europe.

> LENS LESSON TCAS is among the strongest aviation automation interventions in the outcome record; the Überlingen coordination failure mode is part of the case rather than smoothed away. The human–automation–ATC triangle has to be designed coherently; V7.1's reversal logic and the clarified precedence rule are the iterative-design response to the documented edge case.

CASE 135 HEALTHCARE PATIENT SAFETY HUMAN FACTORS 2024 – 2025

Sterile-Cockpit Ward Rounds — Adapting an Aviation Principle to Clinical Handoff

A clinical adaptation of the aviation 'sterile cockpit' rule — no non-essential communication or interruptions during high-workload phases — applied to ward rounds and handoffs; the published evaluation reports reductions in interruption frequency and improvements in information-transfer measures during the protected window, on a single-unit single-study evidence base whose generalization the authors are explicit about

THE FULL CASE, IN FIVE BEATS

Background	Aviation sterile cockpit (FAR 121.542, 1981) — no non-essential communication below 10,000 ft; structural template for clinical adaptation
The Intervention	Clinical handoff and ward rounds carry analogous interruption-driven information loss; workflow context as design variable
How It Worked	PMC12515027 intervention: defined protected window with operational scoping (triage responsibility, permitted interruption classes, signaling)
The Evidence	Reported outcomes: reduced interruption frequency and improved information-transfer measures during the protected window
What Transferred	Hedges: single-unit single-study evidence; patient-outcome inference via established handoff-quality link, not direct measurement; sustainability not yet established

CONNECTIVITY

INDUCED	3.2
LENS	D5/PT5
CLO	CLO-3, CLO-4
COURSES	LEN 4, LEN 5, LEN 7

KEY REFERENCES

- Treloar, E., Herath, M., Altree, M., Potter, S., Penhall, M., Walsh, D., Kennedy, L., Bruening, M., Edwards, S., Ey, J. D., Bradshaw, E. L., & Maddern, G. J. (2025), “A Simple Solution for a Complex Problem: The ‘Sterile Cockpit’ to Improve Ward Rounds,” World Journal of Surgery 49(10):2769–2776, doi:10.1002/wjs.70074, PMID:40930848, PMCID:PMC12515027 — the cited adaptation study.
- Federal Aviation Administration, 14 CFR § 121.542 (codified 1981) — origin of the aviation sterile-cockpit rule.
- Starmer, A. J. et al. (2014), “Changes in medical errors after implementation of a handoff program,” New England Journal of Medicine 371(19):1803–1812 — I-PASS handoff intervention; structural cousin in the structured-information half of handoff capability.

> LENS LESSON The sterile-cockpit ward-rounds case is the worked example of cross-domain adaptation discipline at small scale: an aviation safety-culture rule carried into clinical care with its operational scoping translated and its cultural half preserved. The single-unit evidence is direct on interruption frequency and information transfer; the patient-outcome inference rests on the established handoff-quality link, and the multi-site replication is the explicit next deliverable.

CASE 136 AVIATION SAFETY INDUSTRY COORDINATION TRAINING 1992 – 2000S

FSF CFIT and ALAR Task Forces — Industry-Level Institution Building After a Spike

After Controlled Flight Into Terrain emerged as the leading cause of commercial-jet fatalities through the late 1980s, the Flight Safety Foundation convened industry-wide task forces that produced the CFIT Checklist, the ALAR Tool Kit, and the institutional momentum behind Terrain Awareness and Warning System (TAWS) mandates; CFIT and ALAR accident rates fell sharply over the subsequent decade

THE FULL CASE, IN FIVE BEATS

Background	CFIT was the leading cause of commercial-jet fatalities through the late 1980s — serviceable aircraft, controlled flight, terrain encountered too late to recover
The Intervention	FSF CFIT Task Force (1992) produces the CFIT Checklist — cross-operator self-assessment, distributed without restriction
How It Worked	FSF ALAR Task Force (1996); ALAR Tool Kit released 1998 covers stabilized approach, runway excursion, monitored approach, crew procedure
The Evidence	FAA TAWS mandate 2002; ICAO TAWS effective 2007 — regulatory action sits downstream of task-force momentum, not upstream
What Transferred	CFIT and ALAR accident rates fall sharply over the subsequent decade; hedge preserved — decline is multifactorial (TAWS, training, procedure, fleet turnover)

CONNECTIVITY

INDUCED	6.1
LENS	D4/PT4
CLO	CLO-4
COURSES	LEN 4, LEN 7, LEN 8

KEY REFERENCES

- Flight Safety Foundation, “Killers in Aviation: FSF Task Force Presents Facts About Approach-and-landing and Controlled-flight-into-terrain Accidents,” Flight Safety Digest (1998–1999).
- Flight Safety Foundation, ALAR Tool Kit (1998) — distributed multi-element package of briefing notes, training aids, and risk-assessment instruments.
- Khatwa & Helmreich, “Analysis of critical factors during approach and landing in accidents and normal flight,” Flight Safety Digest (1998) — the analytical basis of the ALAR Task Force scope.

> LENS LESSON The FSF CFIT and ALAR task forces are the canonical case of industry-level institution building after a catastrophe-class spike. An independent foundation convened the cross-operator working bodies, released structured diagnostic tools without competitive restriction, and sustained momentum to a regulatory mandate. The accident-rate decline is multifactorial; the institutional form is what the case is teachable on.

CASE 137 MEDICAL EDUCATION PATIENT SAFETY WORKFORCE DISSEMINATION 2009 – 2020S

Barsuk SBML — Simulation-Based Mastery Learning Dissemination from Northwestern to the VA

Barsuk et al. (Northwestern/Feinberg) demonstrated that simulation-based mastery learning for central venous catheter (CVC) insertion reduced catheter-related bloodstream infection rates and procedural complications, with cost-effectiveness shown in a single-hospital evaluation; the program was subsequently disseminated nationally to Department of Veterans Affairs medical centers

THE FULL CASE, IN FIVE BEATS

Background	CVC insertion is high-volume, high-consequence; complication profile (pneumothorax, arterial puncture, CRBSI) partly attributable to procedural skill
The Intervention	Barsuk et al. (Northwestern/Feinberg, 2009 onward) — simulation-based mastery learning: practice to a defined standard, not a clock
How It Worked	Single-center evidence: fewer needle passes, fewer arterial punctures, lower CRBSI rates; cost-saving at the hospital level (Cohen et al. 2010)
The Evidence	Disseminated nationally to VA medical centers — cross-institutional transfer is the C6.4 structural feature
What Transferred	Hedge preserved: multi-site dissemination outcome literature thinner than single-center evidence; pair with cases 121, 122, 123 as cross-domain workforce evidence

CONNECTIVITY

INDUCED 6.4
LENS D2/PT4
CLO CLO-2, CLO-3
COURSES LEN 2, LEN 5, LEN 7

KEY REFERENCES

- Barsuk, Cohen, Feinglass, McGaghie, & Wayne (2009), “Use of simulation-based education to reduce catheter-related bloodstream infections,” Archives of Internal Medicine 169(15):1420–1423, doi:10.1001/archinternmed.2009.215.
- Cohen, Feinglass, Barsuk, Barnard, O’Donnell, McGaghie, & Wayne (2010), “Cost savings from reduced catheter-related bloodstream infection after simulation-based education for residents in a medical intensive care unit,” Critical Care Medicine 38(9):1853–1859.
- Barsuk, McGaghie, Cohen, Balachandran, & Wayne (2009), “Use of simulation-based mastery learning to improve the quality of central venous catheter placement in a medical intensive care unit,” Journal of Hospital Medicine 4(7):397–403.

LENS LESSON Barsuk SBML for CVC insertion is the canonical small-tier intervention case for cross-institutional dissemination of a proven mechanism. The single-center evidence is controlled, comparison and cost-effective; the VA national dissemination is the cross-institutional transfer step. The multi-site outcome evidence is thinner than the single-center evidence — the live frontier of the dissemination case.

CASE 138 AVIATION SAFETY STRUCTURAL ENGINEERING REGULATORY RULEMAKING 1988 – 2010S

FAA Aging-Aircraft Program — Widespread Fatigue Damage and the Part 26 Rule

After Aloha Airlines 243 (1988) exposed widespread fatigue cracking in an aging Boeing 737-200, the FAA's Aging Aircraft Program and the Airworthiness Assurance Working Group produced two decades of structural inspection programs culminating in 14 CFR Part 26 (2010), which requires manufacturers to establish a Limit of Validity for each model and embed widespread fatigue damage protection into the maintenance program

THE FULL CASE, IN FIVE BEATS

Background	Aloha 243 (April 28, 1988): 737-200 at 89,680 cycles loses 18 feet of upper fuselage; WFD identified as the failure mode
The Intervention	FAA stands up Aging Aircraft Program; AAWG operates through 1990s–2000s producing per-model structural inspection programs
How It Worked	14 CFR Part 26 finalized 2007, WFD provisions effective 2010 — Limit of Validity per type; WFD prevention embedded in maintenance program
The Evidence	Two structural elements: LOV as operational service goal protected by maintenance, plus WFD prevention as ongoing program (not one-time inspection)
What Transferred	Closes the induced C7 (system change / aging assumptions) zero-state — sustained two-decade rulemaking pulled an aging fleet back under structured airworthiness governance

CONNECTIVITY

INDUCED 7.3
LENS D1/PT3
CLO CLO-1, CLO-4
COURSES LEN 1, LEN 7, LEN 8

KEY REFERENCES

- NTSB (1989), Aircraft Accident Report AAR-89/03, Aloha Airlines Flight 243, Boeing 737-200, N73711.
- FAA, 14 CFR Part 26, “Continued Airworthiness and Safety Improvements for Transport Category Airplanes,” Final Rule (2007), WFD provisions effective 2010.
- Airworthiness Assurance Working Group reports (1990s–2000s) — per-model structural inspection program record.

LENS LESSON The FAA Aging Aircraft Program and Part 26 are one of the v2 sweep’s clearest closes of the C7 zero-state. Aloha 243 exposed a regime the original certification did not cover; the AAWG operated for two decades; Part 26 codified Limit of Validity and embedded WFD prevention into the maintenance program itself. The sustained two-decade rulemaking is the deliverable, not the rule alone.

CASE 139 AVIATION INFRASTRUCTURE AIR TRAFFIC MANAGEMENT REGULATORY TRANSITION 2003 – 2020

FAA NextGen and the ADS-B Out Transition

The FAA's Next Generation Air Transportation System (NextGen) shifted US air-traffic management from radar-based surveillance to a satellite-based architecture; the ADS-B Out mandate effective January 1, 2020 required equipage across the controlled-airspace fleet, achieving substantial compliance — with documented schedule slippage and benefit-realization gaps preserved as load-bearing hedges

THE FULL CASE, IN FIVE BEATS

Background	Pre-NextGen US air-traffic management rested on radar and voice; incremental modernization inside the radar-paradigm approached its limits by early 2000s
The Intervention	NextGen launched 2003 (Vision 100 Act); ADS-B Out is the load-bearing equipage-mandate piece of the broader program
How It Worked	ADS-B Out final rule published 2010; January 1, 2020 compliance deadline; substantial compliance reported at the deadline
The Evidence	Load-bearing hedge: GAO / DOT IG documented significant schedule slippage and benefit-realization gaps across the broader NextGen program
What Transferred	Closes C7 (aging-infrastructure transition) zero-state alongside Cases 138, 140, 141 — the instance where the hedges are largest

CONNECTIVITY

INDUCED	7.1
LENS	D1/PT4
CLO	CLO-1, CLO-4
COURSES	LEN 1, LEN 7, LEN 8

KEY REFERENCES

- FAA, "Automatic Dependent Surveillance – Broadcast (ADS-B) Out Performance Requirements to Support Air Traffic Control (ATC) Service," Final Rule, 14 CFR Part 91 (2010).
- Vision 100 — Century of Aviation Reauthorization Act, Public Law 108-176 (2003) — NextGen program statutory basis.
- Government Accountability Office, "NextGen Air Transportation System" report series (2010s) — sustained external audit record on schedule slippage and benefit-realization gaps.

> LENS LESSON NextGen / ADS-B is one of the largest planned aging- infrastructure transitions in the recent regulatory record. The equipage mandate executed at the January 2020 deadline. The broader NextGen program has documented significant schedule slippage and benefit-realization gaps in sustai [illegible] a [illegible] wledged hedges in the v2 aging-system quartet.

CASE 140 SOFTWARE SUSTAINMENT CRITICAL INFRASTRUCTURE FEDERAL PROGRAM MANAGEMENT 1996 – 2000

Y2K Remediation — The Aging-System Transition That Worked Because It Was Believed

The US federal government and the broader public and private sectors invested an estimated 100 billion dollars (US) over four years remediating two-digit-year date handling in legacy systems; the January 1, 2000 rollover passed with minimal disruption to critical infrastructure — the success contributed to the durable counterfactual debate about whether the threat justified the spending

THE FULL CASE, IN FIVE BEATS

Background	Y2K problem: decades of two-digit year fields in legacy code, ambiguous at the 1999/2000 rollover, with a hard immovable deadline
The Intervention	Federal-program-management response from 1996 onward: line-item inventory, OMB quarterly reporting, GAO sustained external audit
How It Worked	Parallel private-sector effort across financial, utility, telecom, industrial operators; total US investment estimates around \$100 billion
The Evidence	January 1, 2000 rollover passed with minimal disruption to critical infrastructure; widely characterized as a major program-management success
What Transferred	Counterfactual debate preserved: would the rollover have passed similarly with less spending? — structurally unobservable; the case is teachable on the institutional discipline

CONNECTIVITY

INDUCED	7.1
LENS	D1/PT4
CLO	CLO-1, CLO-4
COURSES	LEN 1, LEN 7, LEN 8

KEY REFERENCES

- Government Accountability Office, Year 2000 Computing Challenge report series (1996–2000), particularly GAO/AIMD-99-225, GAO/T-AIMD-00-30, and GAO/AIMD-00-1 — line-item federal-program-management record.
- Office of Management and Budget, quarterly reports on federal Y2K remediation status (1997–1999) — program-self-report tier.
- President's Council on Year 2000 Conversion, The Journey to Y2K final report (2000) — institutional retrospective of the federal coordination effort.

> LENS LESSON Y2K remediation is the canonical case of a believed-and- treated aging-system transition. Line-item inventory, OMB quarterly reporting, GAO sustained audit, an immovable deadline, and a sustained four-year program-management cadence converted a foreseeable legacy-software failure into a transition the rollover passed quietly. The counterfactual debate the success produced is part of the case.

CASE 141 NUCLEAR ENGINEERING CONTROL-ROOM HUMAN FACTORS RESEARCH AND DEVELOPMENT 2010 – PRESENT

INL / LWRS Control-Room Modernization — Sustainment Research for an Aging Fleet

The US Department of Energy's Light Water Reactor Sustainability (LWRS) program, executed at Idaho National Laboratory in partnership with utilities, has produced a structured research-and-pilot record for modernizing analog control-room instrumentation in the existing US nuclear fleet — pilot-scale evidence covering hybrid digital/analog operator interfaces, human-factors validation, and qualification pathways for digital instrumentation

THE FULL CASE, IN FIVE BEATS

Background	US commercial nuclear fleet dominated by reactors originally licensed 1970s–80s with analog control-room instrumentation
The Intervention	NRC regulatory environment (RG 1.180, BTP 7-19, SRP Ch. 7) makes the qualification path for safety-related digital I&C deliberately stringent
How It Worked	DOE LWRS program executed at INL in partnership with utilities — federally-funded research-and-pilot work across multi-decade horizon
The Evidence	Research line covers hybrid digital/analog operator interfaces, human-factors validation in full-scope simulators, qualification-pathway research
What Transferred	Hedge preserved: LWRS observations are pilot-scale; operational-fleet evidence at scale is forward-looking; closes C7 zero-state with Cases 138, 139, 140

CONNECTIVITY

INDUCED	7.1
LENS	D5/PT4
CLO	CLO-1, CLO-5
COURSES	LEN 1, LEN 7, LEN 8

KEY REFERENCES

- Idaho National Laboratory, Light Water Reactor Sustainability (LWRS) program annual reports (2010 – present) — primary research-and-pilot record.
- Nuclear Regulatory Commission, Regulatory Guide 1.180, “Guidelines for Evaluating Electromagnetic and Radio-Frequency Interference in Safety-Related Instrumentation and Control Systems.”
- Nuclear Regulatory Commission, Branch Technical Position 7-19, “Guidance for Evaluation of Diversity and Defense-in-Depth in Digital Computer-Based Instrumentation and Control Systems.”

> LENS LESSON LWRS is the structured sustainment-research case — federally- funded research at INL in partnership with utilities and the regulator, producing the [illegible] ons across a multi- decade horizon. The observations are pilot-scale; the operational-fleet evidence at scale is forward-looking. The hedge is part of the case.

CASE 142 DEFENSE WORKFORCE L&D INTELLIGENT TUTORING 2009 – 2014

DARPA Digital Tutor — Compressing Years of IT Expertise into 16 Weeks

An IDA independent assessment found that, after 16 weeks of Digital Tutor instruction, US Navy IT graduates with no prior IT experience outscored fleet Information Technology Specialists with an average 7.2 years of experience on a knowledge test, with an effect size of 4.30, and outperformed them on most troubleshooting and design tasks

THE FULL CASE, IN FIVE BEATS

Background	DARPA Digital Tutor — intelligent tutoring system modelled on expert one-on-one human tutoring; 16-week pipeline for US Navy IT rating
The Intervention	IDA independent evaluation (Fletcher & Morrison, IDA D-4686): Digital Tutor graduates vs. fleet ITs with 7.2 years' average experience
How It Worked	Knowledge test effect size 4.30 in favor of Digital Tutor; Digital Tutor cohort outperforms fleet on most troubleshooting/design tasks (Security the exception)
The Evidence	Report concludes the effort 'appears to have achieved its goals'
What Transferred	Hedges preserved: knowledge accounts for ~40% of practical-exercise variance, 'an enabler of performance rather than a direct measure'; architecture detail too scant to reproduce

CONNECTIVITY

INDUCED	1.2
LENS	D2/PT4
CLO	CLO-2, CLO-3
COURSES	LEN 1, LEN 2, LEN 5

KEY REFERENCES

- Fletcher, J. D., & Morrison, J. E. (2014). DARPA Digital Tutor: Assessment Data. IDA Document D-4686. https://apps.dtic.mil/sti/tr/pdf/AD1002362.pdf — independent evaluation that the case rests on.
- Defense Advanced Research Projects Agency, Digital Tutor program documentation — program description and design rationale.
- Fletcher, J. D. (2009). From behaviorism to constructivism: a philosophical journey from drill and practice to situated learning. — methodological grounding for the Digital Tutor’s tutorial discipline.

> LENS LESSON DARPA’s Digital Tutor is the cleanest available evidence that the capability envelope of a training pipeline can be re-specified — from years of seat time to 16 weeks of tutorial-discipline instruction — against an operational comparison the program is built to compete with. The hedges (knowledge as enabler, architecture detail scant) are part of what makes the result interpretable.

CASE 143 CLINICAL CARE PATIENT SAFETY HEALTH IT 2010

Bar-Code Medication Administration — Cue at the Point of Care

A before-and-after study at an academic medical center associated bar-code electronic medication administration (bar-code eMAR) with a 41.4% reduction in nontiming administration errors and a 50.8% reduction in potential adverse drug events; transcription errors on order documentation fell 80.3%; a later single-site rollout reported a 55.4% reduction in actual patient-harm events

THE FULL CASE, IN FIVE BEATS

Background	Wrong-drug / wrong-patient administration: the cue the bedside nurse needs to catch the mismatch is structurally absent in the unaided workflow
The Intervention	Bar-code eMAR supplies the cue in hardware: medication scan + wristband scan gated by the electronic record at the moment of administration
How It Worked	Poon et al. NEJM 2010 — 41.4% reduction in nontiming administration errors, 50.8% in potential adverse drug events, 80.3% in transcription errors
The Evidence	Later single-site rollout (PMC6257885) — 55.4% reduction in actual patient-harm events
What Transferred	Hedge preserved: before-and-after / observational design, not a randomized trial; significant workflow changes were required

CONNECTIVITY

INDUCED	3.1
LENS	D3/PT5
CLO	CLO-3, CLO-5
COURSES	LEN 2, LEN 5, LEN 7

KEY REFERENCES

- Poon, E. G., Keohane, C. A., Yoon, C. S., Ditmore, M., Bane, A., Levtzion-Korach, O., Moniz, T., Rothschild, J. M., Kachalia, A. B., Hayes, J., Churchill, W. W., Lipsitz, S., Whittemore, A. D., Bates, D. W., & Gandhi, T. K. (2010). Effect of bar-code technology on the safety of medication administration. New England Journal of Medicine, 362(18):1698–1707. doi:10.1056/NEJMsa0907115 — the case's primary evaluation.
- Bonkowski, J., Carnes, C., Melucci, J., Mirtallo, J., Prier, B., Reichert, E., Moffatt-Bruce, S., & Weber, R. J. (2013). Effect of barcode-assisted medication administration on emergency department medication errors. Academic Emergency Medicine, 20(8):801–806 — adjacent transfer evidence.
- Thompson, K. M., Swanson, K. M., Cox, D. L., Kirchner, R. B., Russell, J. J., Wermers, R. A., Storlie, C. B., Johnson, M. G., & Naessens, J. M. (2018), "Implementation of Bar-Code Medication Administration to Reduce Patient Harm," Mayo Clinic Proceedings: Innovations, Quality & Outcomes 2(4):342–351, doi:10.1016/j.mayocpiqo.2018.09.001, PMID:30560236, PMCID:PMC6257885 — later single-site rollout reporting a 55.4% reduction in actual patient-harm events

LENS LESSON Bar-code medication administration is the canonical small-tier cue/alert intervention at the point of care. The cue lands at the bedside befor[illegible]tions (41.4%, 50.8%, 80.3%, and a 55.4% transfer figure) rest on a before-and-after / observational design that the case preserves verbatim.

CASE 144 CLINICAL CARE PATIENT SAFETY TEAM COMMUNICATION 2014

I-PASS Handoff Bundle — Structuring the Human-to-Human Transfer

Across nine pediatric residency programs, implementation of the I-PASS handoff bundle (mnemonic + training + faculty development + sustainability campaign) was associated with a 23% relative reduction in preventable adverse events and a drop in medical errors, without negatively affecting resident workflow — the study design 'precludes definitively establishing a causal link'

THE FULL CASE, IN FIVE BEATS

Background	Shift-change handoff: structural risk point; loss of safety-critical information is documented failure mode
The Intervention	I-PASS bundle — mnemonic (Illness severity, Patient summary, Action list, Situation awareness, Synthesis by receiver) + trainee education + faculty development + sustainability campaign
How It Worked	Starmer et al. NEJM 2014: 23% relative reduction in preventable adverse events across nine pediatric residency programs; drop in medical errors; no negative effect on resident workflow
The Evidence	Hedge preserved verbatim: 'our study design precludes definitively establishing a causal link'
What Transferred	Published correspondence cautions against implementing the mnemonic alone without the full bundle

CONNECTIVITY

INDUCED	3.3
LENS	D2/PT5
CLO	CLO-2, CLO-4
COURSES	LEN 2, LEN 4, LEN 7

KEY REFERENCES

- Starmer, A. J., Spector, N. D., Srivastava, R., West, D. C., Rosenbluth, G., Allen, A. D., Noble, E. L., Tse, L. L., Dalal, A. K., Keohane, C. A., Lipsitz, S. R., Rothschild, J. M., Wien, M. F., Yoon, C. S., Zigmont, K. R., Wilson, K. M., O'Toole, J. K., Solan, L. G., Aylor, M., Bismilla, Z., Coffey, M., Mahant, S., Blankenburg, R. L., Destino, L. A., Everhart, J. L., Patel, S. J., Bale, J. F., Spackman, J. B., Stevenson, A. T., Calaman, S., Cole, F. S., Balmer, D. F., Hepps, J. H., Lopreiato, J. O., Yu, C. E., Sectish, T. C., & Landrigan, C. P. (2014). Changes in medical errors after implementation of a handoff program. New England Journal of Medicine, 371(19):1803–1812. doi:10.1056/NEJMsa1405556 — the case's primary evaluation.
- Starmer, A. J., et al. (2014). Implementation correspondence and follow-up. New England Journal of Medicine — the published correspondence cautioning against implementing the mnemonic alone.
- Sectish, T. C., et al. (2010). Establishing a multisite education and research project: The I-PASS Study Group. Academic Medicine — the I-PASS Study Group methodology and design rationale.

LENS LESSON I-PASS is the structured-handoff intervention the patient-safety literature anchors on — a 23% relative reduction in preventable adverse events across nine residency programs, with no negative effect on resident workflow. The hedge that survives verbatim is the authors' own: the study design "precludes definitively establishing a causal link," and the bundle, not the mnemonic alone, is what the evidence is about.

CASE 145 SURGERY CLINICAL OUTCOMES ASSESSMENT 2013

Surgical Skill Video Peer-Rating Predicts Complications

Twenty bariatric surgeons each submitted a representative gastric-bypass video, rated blind by at least 10 peers; skill ratings ranged 2.6–4.8; the bottom skill quartile had a higher complication rate (14.5%) than the top quartile across a registry of 10,343 patients, and greater skill was associated with fewer reoperations, readmissions, and emergency department visits

THE FULL CASE, IN FIVE BEATS

Background	Twenty bariatric surgeons each submit a representative laparoscopic gastric bypass video; rated blind by ≥10 peers on a structured scale
The Intervention	Skill ratings range 2.6–4.8 with inter-rater reliability adequate for rank-ordering
How It Worked	Linked to Michigan registry of 10,343 patients: bottom skill quartile complication rate 14.5%; greater skill → fewer reoperations, readmissions, ED visits
The Evidence	Authors call findings preliminary; skill-versus-volume confound named explicitly — low-skill surgeons did fewer cases and operated more slowly
What Transferred	Multi-anchor: 2.1 primary, 1.1 and 6.2 alternates; editor may move

CONNECTIVITY

INDUCED	2.1
LENS	D3/PT5
CLO	CLO-3, CLO-5
COURSES	LEN 2, LEN 5, LEN 7

KEY REFERENCES

- Birkmeyer, J. D., Finks, J. F., O'Reilly, A., Oerline, M., Carlin, A. M., Nunn, A. R., Dimick, J., Banerjee, M., & Birkmeyer, N. J. O., for the Michigan Bariatric Surgery Collaborative. (2013). Surgical skill and complication rates after bariatric surgery. New England Journal of Medicine, 369(15):1434–1442. doi:10.1056/NEJMsa1300625 — the case's primary evaluation.
- Birkmeyer, J. D., Stukel, T. A., Siewers, A. E., Goodney, P. P., Wennberg, D. E., & Lucas, F. L. (2003). Surgeon volume and operative mortality in the United States. New England Journal of Medicine, 349(22):2117–2127 — the volume-outcome literature the confound rests against.
- Vassiliou, M. C., et al. (2005). The MISTELS program — global objective assessment of laparoscopic skills. Surgical Endoscopy — the surgical-skill assessment literature the rating scale derives from.

LENS LESSON The Birkmeyer skill–outcomes study is the first large- registry evidence that operative skill, measured directly from operative video by blind peer … preliminary; the skill-versus-volume confound is explicit; the multi-anchor (2.1 primary, 1.1 and 6.2 alternates) is the editor's call.

CASE 146 NURSING EDUCATION CLINICAL SIMULATION REGULATION 2014

NCSBN National Simulation Study — Licensing the 50% Substitution Rule

A longitudinal RCT randomized students across multiple US nursing programs to control, 25%, or 50% simulation substitution for traditional clinical hours; 660+ took the NCLEX with no statistically significant differences in clinical competency, nursing knowledge, or NCLEX pass rates — the number of nursing regulatory boards permitting up to 50% simulation substitution increased more than 20-fold from 2014 to 2022

THE FULL CASE, IN FIVE BEATS

Background	NCSBN longitudinal RCT — students at multiple US nursing programs randomized to control, 25%, or 50% simulation substitution for traditional clinical hours
The Intervention	Hayden et al. 2014 (_J Nursing Regulation_): no statistically significant differences in clinical competency, nursing knowledge, or NCLEX pass rates across groups; 660+ took the NCLEX
How It Worked	Result holds only 'under conditions comparable to those described in the study' (high-quality simulation, trained faculty, structured debriefing)
The Evidence	Institutional transfer: number of nursing regulatory boards permitting up to 50% substitution increased 20-fold from 2014 to 2022 — single evidence base propagating across the regulatory field
What Transferred	Pair with Case 128 (Colorado CTSA team-science) at the cross-domain workforce-intervention layer

CONNECTIVITY

INDUCED	6.1
LENS	D2/PT4
CLO	CLO-2, CLO-3
COURSES	LEN 2, LEN 5, LEN 7

KEY REFERENCES

- Hayden, J. K., Smiley, R. A., Alexander, M., Kardong-Edgren, S., & Jeffries, P. R. (2014). The NCSBN National Simulation Study: A longitudinal, randomized, controlled study replacing clinical hours with simulation in prelicensure nursing education. Journal of Nursing Regulation, 5(2 Suppl):S1–S64. https://www.journalofnursingregulation.com/article/s2155-8256(15)30062-4/fulltext — the case's primary study.
- Smiley, R. A. (2023). Survey of simulation use in prelicensure nursing programs and the regulatory landscape 2014–2022. Journal of Nursing Regulation. doi:10.1016/S2155-8256(23)00086-8 — the 2023 follow-up documenting the regulatory propagation.
- Jeffries, P. R. (2012). Simulation in Nursing Education: From Conceptualization to Evaluation (2nd ed.). National League for Nursing — the simulation-quality framework the NCSBN study's quality conditions rest on.

LENS LESSON The NCSBN National Simulation Study is the unusual case of a single, well-designed, regulator-commissioned study propagating a substantial workforce-capability change across an entire decentralized regulatory field. The null result on NCLEX is conditional on the quality conditions the study names — high-quality simulation, trained faculty, structured debriefing — and the qualifying language has to travel with the result.

CASE 147 MEDICAL EDUCATION RETENTION WORKFORCE L&D 2007 – 2009

Spaced Education RCTs in Medical Training

A multi-institution RCT of urology residents across the US and Canada randomized participants to bolus versus spaced-pattern email delivery of validated study questions; spaced education improved acquisition and retention of medical knowledge, and a follow-up showed the learning benefit persisting for two years

THE FULL CASE, IN FIVE BEATS

Background	Spacing — one of the most robust findings in basic learning-science research; transfer to workplace-L&D in medical trainees is the empirical question
The Intervention	Kerfoot et al. _J Urology_ 2007 — multi-institution RCT, urology residents across US/Canada; validated study questions delivered by email in bolus vs. spaced pattern
How It Worked	Spaced-education condition improved acquisition and retention of medical knowledge against bolus comparison
The Evidence	Kerfoot 2009 follow-up (_J Urology_ 181:2671) — learning benefit persisted for 2 years
What Transferred	Scope discipline: speaks to email-delivered validated-question delivery pattern, not spacing in general; strongest randomized strength in the small-tier batch

CONNECTIVITY

INDUCED	2.3
LENS	D2/PT4
CLO	CLO-2, CLO-3
COURSES	LEN 2, LEN 3, LEN 5

KEY REFERENCES

- Kerfoot, B. P., DeWolf, W. C., Masser, B. A., Church, P. A., & Federman, D. D. (2007). Spaced education improves the retention of clinical knowledge by medical students: A randomised controlled trial. Journal of Urology, 177(4):1481–1487. doi:10.1016/j.juro.2006.11.074 — the case's primary RCT.
- Kerfoot, B. P. (2009). Learning benefits of on-line spaced education persist for 2 years. Journal of Urology, 181(6):2671 — the two-year persistence follow-up.
- Cepeda, N. J., Pashler, H., Vul, E., Wixted, J. T., & Rohrer, D. (2006). Distributed practice in verbal recall tasks: A review and quantitative synthesis. Psychological Bulletin, 132(3):354–380 — the basic-literature spacing-effect review the workplace-L&D RCT sits against.

> LENS LESSON Kerfoot et al. is the methodologically clean small-tier closed-loop spaced-education RCT in workplace medical training, with two-year persistence in the follow-up. It is the strongest randomized strength in the small-tier batch. The scope discipline is what makes it interpretable: the result speaks t[illegible]ral.

CASE 148 SURGERY SKILL ASSESSMENT HUMAN-MOTION ANALYSIS 2009 – 2016

Language of Surgery / JIGSAWS — Decomposing Skill into Measurable Units

JHU's Language of Surgery project treated surgical motion as language — decomposing tasks into gesture and sub-gesture motion primitives — and released JIGSAWS, a public da Vinci kinematic/video/gesture/skill-rating dataset that became a standard benchmark; experts used fewer gestures (26.29 vs 31.30) and fewer gesture errors than novices for a knot-tying task

THE FULL CASE, IN FIVE BEATS

Background	Language of Surgery (JHU, Hager et al.) treats surgical motion as language; decomposes task into surgemes and dexemes
The Intervention	JIGSAWS released as a public da Vinci dataset with synchronized kinematic, video, gesture, and skill-rating tracks
How It Worked	Vedula et al. 2016: experts use fewer gestures (26.29 vs 31.30) and fewer gesture errors than novices on knot-tying
The Evidence	Open question preserved: whether automated motion-level feedback accelerates skill acquisition or improves patient outcomes
What Transferred	Pair with Case 145 (Birkmeyer) — Birkmeyer shows skill matters; this case shows skill is decomposable and machine-measurable

CONNECTIVITY

INDUCED	2.1
LENS	D3/PT5
CLO	CLO-2, CLO-3
COURSES	LEN 2, LEN 5, LEN 7

KEY REFERENCES

- Vedula, Ishii, & Hager (2016), "Analysis of the Structure of Surgical Activity for a Suturing and Knot-Tying Task," PLOS One 11(3):e0149174, doi:10.1371/journal.pone.0149174.
- Gao, Vedula, Reiley, Ahmidi, Varadarajan, Lin, Tao, Zappella, Bejar, Yuh, Chen, Vidal, Khudanpur, & Hager (2014), "JHU-ISI Gesture and Skill Assessment Working Set (JIGSAWS): A Surgical Activity Dataset for Human Motion Modeling," MICCAI Workshop — JIGSAWS dataset release paper.
- Reiley, Lin, Yuh, & Hager (2011), "Review of methods for objective surgical skill evaluation," Surgical Endoscopy 25(2):356–366 — situates the decomposition within the broader skill-assessment literature.

> LENS LESSON Language of Surgery / JIGSAWS is the corpus's worked example of decomposing a tacit capability — surgical skill — into machine-measurable units, and releasing the measurement infrastructure openly so the field can build on it. The decomposition is established; the downstream question — whether motion-level feedback accelerates skill or improves patient outcomes — is open. The case enables the question rather than answering it.

CASE 149 HIGHER EDUCATION (LATIN AMERICA) LEARNING ANALYTICS CROSS-NATIONAL GOVERNANCE 2017 – 2020

LALA — Building Learning-Analytics Governance Capacity Across Latin America

An EU-funded multi-country project (Chile, Ecuador, Mexico) that explicitly rejected lifting Global-North learning-analytics tools wholesale; structured interviews with administrators and focus groups with students and teachers produced the LALA CANVAS participatory adoption framework, with stakeholders demanding ethical responsibility as a precondition for data-driven feedback

THE FULL CASE, IN FIVE BEATS

Background	LALA (EU Erasmus+, 2017–2020): explicit refusal to lift Global-North LA tools wholesale into Latin American institutions
The Intervention	Structured interviews with administrators and focus groups with students and teachers across Chile, Ecuador, Mexico
How It Worked	Hilliger et al. 2020 (Internet and Higher Education): stakeholders demand ethical responsibility as precondition, not afterthought
The Evidence	Deliverable is the LALA CANVAS — participatory adoption framework converting governance from a deployment gate to a structured set of decisions
What Transferred	Honest limit: adoption-readiness / capacity-building evidence, not yet long-run outcome evidence that deployed systems improved retention

CONNECTIVITY

INDUCED	5.1
LENS	D4/PT4
CLO	CLO-4, CLO-5
COURSES	LEN 4, LEN 7, LEN 8

KEY REFERENCES

- Hilliger, Ortiz-Rojas, Pesántez-Cabrera, Scheihing, Tsai, Muñoz-Merino, Broos, Whitelock-Wainwright, & Pérez-Sanagustín (2020), "Identifying needs for learning analytics adoption in Latin American universities: A mixed-methods approach," The Internet and Higher Education 45:100726, doi:10.1016/j.iheduc.2020.100726.
- LALA project — EU Erasmus+ grant 586120-EPP-1-2017-1-ES (2017–2020) — program documentation and the LALA CANVAS artifact.
- Open University Ethical Use of Student Data policy (2014) and OU Analyse — UK companion governance-by-design case (Case 122).

> LENS LESSON LALA converted learning-analytics adoption from a Global-North template-lift into a participatory process that built local legitimacy across three L[illegible]ble. The honest limit — adoption-readiness evidence, not yet long-run outcome evidence — is what the case carries, and the outcome study is the next one.

CASE 150 EDUCATION (NORWAY) NATIONAL POLICY LEARNING ANALYTICS 2022 – 2023

Norway's National Expert Commission on Learning Analytics

Rather than let learning analytics diffuse unregulated or block it, Norway's Ministry of Education convened a national Expert Commission to investigate the pedagogical, ethical, legal, and privacy issues and establish a regulatory foundation before sector-wide deployment; interim report June 2022, final report 2023 (NOU), with central dilemmas explicitly framed

THE FULL CASE, IN FIVE BEATS

Background	Norway's Ministry of Education convenes national Expert Commission on Learning Analytics in 2022 — sector-scale governance-first response
The Intervention	Mandate covers pedagogical, ethical, legal, and privacy dimensions across the whole education sector
How It Worked	Interim report June 2022, final report 2023 (NOU) names central dilemmas: predictive support vs gatekeeping, transparency vs model complexity, benchmarking vs data protection
The Evidence	Honest limit: process-level evidence (artifact exists, dilemmas named); downstream sector outcomes not yet documented — governance-process success, not yet deployment-outcome success
What Transferred	Pair with OU (Case 122, institutional) and SyRI (Case 64, judicial); national-scale governance-architecture mode in the non-US LA triple

CONNECTIVITY

INDUCED	5.4
LENS	D4/PT4
CLO	CLO-4, CLO-5
COURSES	LEN 4, LEN 7, LEN 8

KEY REFERENCES

- Norwegian Expert Commission on Learning Analytics, interim report to the Minister of Education and Research (June 2022).
- Norwegian Expert Commission on Learning Analytics, final NOU report (2023), Norges offentlige utredninger.
- Misiejuk & Wasson (2023), "Learning analytics in Norway: A national perspective," Journal of Learning Analytics — secondary academic synthesis of the commission and its dilemmas.

> LENS LESSON Norway's national Expert Commission is the governance-architecture-at-national-scale instance: a country constructing the regulatory and pedagogical foundation before sector-scale learning-analytics deployment, rather than after diffusion or via judicial halt. The artifact exists and the dilemmas are named; downstream sector outcomes are not yet documented. Process-level success; deployment-outcome evidence is the next decade's work.

CASE 151 HIGHER EDUCATION (BRAZIL) LEARNING ANALYTICS INSTITUTIONAL GOVERNANCE 2024

MMALA — A Maturity Model for Responsible Learning-Analytics Adoption (Brazil)

MMALA is a maturity model spanning infrastructure, human resources, ethics, and pedagogy; expert evaluation rated it comprehensive and suitable; three-institution validation exercise in Brazilian universities found it could outline essential practices and support self-assessment for scaling — instrument for responsible adoption, downstream learning outcome open

THE FULL CASE, IN FIVE BEATS

Background	MMALA (Freitas et al. 2024, JLA): maturity model for adopting LA across infrastructure, human resources, ethics, pedagogy
The Intervention	Each dimension resolved into maturity levels — structured self-assessment, not a single overall readiness score
How It Worked	Validation: expert evaluation (comprehensive, suitable) + three-institution exercise at Brazilian universities (usable, actionable)
The Evidence	Honest limit: expert opinion + three-institution validation; not yet longitudinal outcomes of institutions that used MMALA to adopt LA
What Transferred	Institutional-instrument layer of the non-US LA governance set — pair with OU, LALA, Norway, African data privacy

CONNECTIVITY

INDUCED	5.4
LENS	D3/PT4
CLO	CLO-3, CLO-4
COURSES	LEN 4, LEN 7, LEN 8

KEY REFERENCES

- Freitas, Mello, Gasevic, Costa, & Andrade (2024), “MMALA: Developing and Evaluating a Maturity Model for Adopting Learning Analytics,” Journal of Learning Analytics 11(1):67–86.
- Open University Ethical Use of Student Data policy (2014) — institutional-policy companion (Case 122).
- Hilliger et al. (2020), Internet and Higher Education — LALA participatory adoption companion (Case 149).

> LENS LESSON MMALA is the institutional-instrument instance of governance for learning-analytics adoption: a structured maturity model across infrastructure, human resources, ethics, and pedagogy. The validation evidence — expert evaluation and three-institution pilot — is what the case claims; the downstream effect on student learning is the next study, and the case is honest about that gap.

CASE 152 AUTONOMOUS VEHICLES TRANSPORTATION SAFETY PUBLIC-SECTOR GOVERNANCE 2023 / 2025

Waymo's Safety Case Framework — Governance Objection Dissolved by Designed Artifact

After a California court let Waymo withhold trade-secret-laden safety data from a DMV public-disclosure request, the company answered the governance objection with a published, structured safety case framework — and in November 2025 commissioned the first independent third-party audits of both the safety case and the remote-assistance program

THE FULL CASE, IN FIVE BEATS

Background	2022 court let Waymo withhold trade-secret safety data from a public DMV disclosure — public-trust gap with no disclosure-side answer
The Intervention	2023 response: structured safety case framework — claims/sub-claims/evidence types public; trade-secret items remain proprietary
How It Worked	November 2025 independent third-party audits of safety case and remote-assistance program — the audits disclosed, not the underlying data
The Evidence	Practice-synthesis tier: Waymo whitepaper + Montreal AI Ethics Institute + 2025 audit summaries; future validation ongoing
What Transferred	Pairs with Case 70 (Cruise foil) and Case 153 (CPUC permit framework); exercises NEW CLO Delegation with revocation

CONNECTIVITY

INDUCED	5.1
LENS	D4/PT6
CLO	CLO-4, CLO-5, NEW CLO Delegation-with-revocation
COURSES	LEN 4, LEN 8, LEN 9

KEY REFERENCES

- Waymo (2023), “A Blueprint for AV Safety: Waymo’s Toolkit For Building a Credible Safety Case,” whitepaper.
- Waymo (November 2025), “Independent Audits of Waymo’s Safety Case and Remote Assistance Programs,” summary release.
- Montreal AI Ethics Institute (2023), summary and analysis of the Waymo safety case framework.

> LENS LESSON Waymo’s safety case framework is the AV-domain instance of the OU-Analyse / inBloom move: a governance objection dissolved by a designed artifact, not by disclosure of the contested data. Evidence-tier flag is practice-synthesis; the artifact pattern is teachable and the third-party audit posture pushes some elements toward investigation-grade, but the synthesis as a whole is practitioner-authored and future validation is ongoing.

CASE 153 AUTONOMOUS VEHICLES PUBLIC-UTILITY GOVERNANCE ACCESSIBILITY 2020 – 2024

CPUC AV Passenger-Service Permits — Conditions as a Designed Objection-Dissolver

California's Public Utilities Commission built an AV passenger-service permit framework whose conditions — time-of-day limits, weather restrictions, fleet caps, geographic carve-outs, mandatory Passenger Safety Plan for riders with disabilities — are explicitly designed to operationalize the governance objections that would otherwise block deployment outright

THE FULL CASE, IN FIVE BEATS

Background	CPUC and California DMV regulate AV passenger service in a split-jurisdiction regime; binary deploy/don't-deploy risked either blocking deployment or losing governance handle
The Intervention	Permit conditions: time-of-day limits, weather restrictions, fleet caps, geographic carve-outs, mandatory Passenger Safety Plan for riders with disabilities
How It Worked	August 2024 update: DMV authority strengthened to impose targeted operational restrictions for safety during deployment lifecycle
The Evidence	Structural complement to Case 152 (Waymo deployer-side safety case) and inverse-outcome companion of Case 70 (Cruise revocation)
What Transferred	Practice-synthesis tier: program guidance and permit decisions documented; no peer-reviewed evaluation of equity-of-service outcomes yet — future validation ongoing

CONNECTIVITY

INDUCED	5.1
LENS	D4/PT6
CLO	CLO-4, CLO-5
COURSES	LEN 4, LEN 8, LEN 9

KEY REFERENCES

- California Public Utilities Commission, “Autonomous Vehicle Passenger Service Programs,” CPUC program page and August 2024 application guidance.
- CPUC permit decisions for Cruise and Waymo, 2020–2024.
- California Department of Motor Vehicles, AV regulatory program — strengthened safety-restriction authority, 2024.

> LENS LESSON The CPUC permit framework is the regulator-side counterpart to the Waymo safety case (Case 152): conditions operationalize objections rather than blocking deployment. Evidence-tier flag is practice-synthesis; the regulatory architecture is documented in CPUC decisions, but no peer-reviewed evaluation of the equity-of-service goals yet exists, and future validation is ongoing.

CASE 154 CORPORATE L&D TRAINING EVALUATION WORKFORCE DEV 2005 – PRESENT

Brinkerhoff Success Case Method — Tails as the Evaluation Instrument

When ROI-style evaluation of corporate training is intractable, Brinkerhoff's Success Case Method samples the tails of the outcome distribution — the highest- and lowest-impact participants — and reconstructs the system conditions that made the program work for some and fail for others; deployed at Cargill, Ford, Merck, World Bank, ICRC

THE FULL CASE, IN FIVE BEATS

Background	Corporate L&D evaluation problem: Levels 3 and 4 require data the training org cannot access; average effects flatter most programs (Case 99)
The Intervention	SCM: sample the highest- and lowest-impact participants; study in detail; reconstruct the system conditions around each
How It Worked	Tails carry decision-grade information — success cases prove the program *can* work; failure cases name what the surrounding system has to provide for transfer
The Evidence	Deployed at Cargill, Ford, Merck, World Bank, ICRC; method peer-reviewed; per-firm impact data live in practitioner channels
What Transferred	Operational complement to Blume's environment-as-decisive-variable finding (Case 100); exercises NEW CLO Judgment under inadequate evidence

CONNECTIVITY

INDUCED	2.1
LENS	D3/PT5
CLO	CLO-3, CLO-2
COURSES	LEN 4, LEN 7, LEN 8

KEY REFERENCES

- Brinkerhoff, R. O. (2005), “The Success Case Method: A Strategic Evaluation Approach to Increasing the Value and Effect of Training,” Advances in Developing Human Resources 7(1):86–101, doi:10.1177/1523422304272172.
- Brinkerhoff Evaluation Institute deployment list — Cargill, Ford, Merck, World Bank, International Committee of the Red Cross — practitioner channel.
- Kirkpatrick & Kirkpatrick (2006), Evaluating Training Programs — the chain-of-evidence framework SCM operationalizes (paired Case 99).

> LENS LESSON SCM is the practitioner instrument that operationalizes Blume’s environment-as-decisive finding (Case 100) by sampling the tails of the outcome distribution. The method is peer-reviewed; the per-firm impact data at Cargill, Ford, Merck, World Bank, ICRC live in practitioner channels. Evidence-tier flag is practice-synthesis; future validation will continue as more firms publish.

CASE 155 CORPORATE L&D PERFORMANCE CONSULTING LEARNING TRANSFER 2001 – PRESENT

High-Impact Learning System — Engineering the Environment, Not Just the Event

Brinkerhoff & Apking's HILS reframes corporate L&D as a system spanning pre-training (line-manager alignment, work-context preparation), the event itself, and post-training (supervisor support, on-the-job practice) — translating Blume's meta-analytic finding that the work environment dominates transfer into a deployable program model

THE FULL CASE, IN FIVE BEATS

Background	Blume meta-analysis (Case 100) names work environment as decisive transfer variable; HILS is the deployed-program answer
The Intervention	Three phases: pre-training (line-manager alignment, work-context prep); the event; post-training (supervisor support, peer support, practice opportunity)
How It Worked	Corporate deployments report transfer rising from 10–20% baseline to substantially higher figures; per-firm numbers self-reported in practitioner channels
The Evidence	Deployed-program counterpart to Case 154 (SCM as evaluation); together they structure the Level-2/Level-3 seam crossing (Case 99)
What Transferred	Practice-synthesis tier: model documented in Brinkerhoff & Apking, Watershed, L-TEN; effect sizes self-reported; future validation ongoing

CONNECTIVITY

INDUCED	2.4
LENS	D2/PT4
CLO	CLO-2, NEW CLO Judgment-under-inadequate-evidence
COURSES	LEN 2, LEN 4, LEN 7

KEY REFERENCES

- Brinkerhoff, R. O., & Apking, A. M. (2001), High Impact Learning: Strategies for Leveraging Performance and Business Results from Training Investments, Basic Books.
- Watershed LRS practitioner summaries of HILS deployment patterns.
- L-TEN (Life Sciences Trainers and Educators Network) practitioner summaries of HILS in life-sciences L&D.

> LENS LESSON HILS is the deployed-program operationalization of Blume's environment-as-decisive finding (Case 100) and the design-side counterpart of SCM (Case 154). Evidence is practice-synthesis: the model is durable in practitioner literature, per-firm effect sizes are self-reported, future validation continues. The CLO **Judgment under inadequate evidence** is the capability the case asks for.

CASE 156 WORKFORCE DEV NATIONAL L&D POLICY ASIA-PACIFIC 2015 – PRESENT

Singapore SkillsFuture — National Workforce Capability at Scale

Singapore's SkillsFuture pairs individual training credits with employer subsidies, a cross-sector skills framework, and a two-wave outcome survey (TRAQOM, at end-of-course and at six months) — a 2018 MTI study found a 5.8% real wage premium for WSQ-trained workers, with 87% of Work-Study Programme graduates employed full-time within six months

THE FULL CASE, IN FIVE BEATS

Background	SkillsFuture launched 2015: individual training credits + employer subsidies + cross-sector skills framework + Work-Study Programme
The Intervention	TRAQOM: two-wave outcome survey (end-of-course + six months) paired with labor-market data; ambitious national L&D instrument
How It Worked	2024 Year-in-Review: 98% perform-better self-report; 93% pivotal role; 87% WSP graduates employed FT within 6 months; 2018 MTI 5.8% wage premium for WSQ-trained
The Evidence	Honest reading: self-report dominates; no rigorous quasi-experimental causal evaluation; future validation ongoing
What Transferred	Practice-synthesis tier; cross-listed Gap 2 (workforce L&D) + Gap 5 (non-US/UK/EU); pairs with Cases 155 (HILS) and 163 (PEPFAR)

CONNECTIVITY

INDUCED	2.4
LENS	D4/PT4
CLO	CLO-2, CLO-4, CLO Collaboration-measurement
COURSES	LEN 2, LEN 4, LEN 8

KEY REFERENCES

- SkillsFuture Singapore (SSG), Year-in-Review 2024 — program metrics and outcome reporting.
- Ministry of Education (MOE), Singapore, parliamentary replies on TRAQOM, 2020.
- International Labour Organization (ILO), "Investigating an Upskilling Programme in Singapore" — international comparative analysis.

> LENS LESSON SkillsFuture is the non-US national-scale L&D case the corpus needs. The TRAQOM instrument is among the most ambitious national L&D measurement architectures deployed; the headline outcomes are self-report dominant; no rigorous quasi-experimental external evaluation yet exists. Evidence-tier flag is practice-synthesis; future validation is ongoing.

CASE 157 GLOBAL HEALTH HIV CARE TRAINING-MODALITY COMPARISON SUB-SAHARAN AFRICA 2023

PEPFAR HIV Training Across 16 Sub-Saharan African Countries — Modality Comparison Under Disruption

Across 16 PEPFAR-supported Sub-Saharan African countries, pre/post knowledge and self-confidence assessments showed mean increases across all training modalities — in-person (pre-COVID), virtual, and blended — when pandemic disruption forced a delivery-mode switch; the L1–L2 limitation is explicit and the L3/L4 question remains open

THE FULL CASE, IN FIVE BEATS

Background	PEPFAR workforce-capability program; COVID forced modality switch across 16 SSA countries with no prior evidence base on modality equivalence
The Intervention	2023 study: pre/post assessments across knowledge and self-confidence domains showed mean increases regardless of modality (in-person, virtual, blended)
How It Worked	Kirkpatrick L1–L2 limitation explicit: outcomes are knowledge and self-rated confidence, not L3 behavior change or L4 patient outcomes (Case 99)
The Evidence	Rare real-world cross-country modality comparison at meaningful scale; multi-country scope limits single-country confounding
What Transferred	Preprint-tier flag: medRxiv preprint and PMC published; editor citation choice carried explicitly; future validation on L3/L4 ongoing

CONNECTIVITY

INDUCED	2.3
LENS	D2/PT4
CLO	CLO-2, CLO-3, CLO-4
COURSES	LEN 2, LEN 4, LEN 7

KEY REFERENCES

- “Comparing in-person, blended and virtual training interventions; a real-world evaluation of HIV capacity building programs in 16 countries in sub-Saharan Africa,” medRxiv 2023.02.08.23285641 (preprint) → PMC10365303 (published).
- PEPFAR program documentation — workforce-capability training as a load-bearing deliverable across Sub-Saharan African deployment countries.
- Kirkpatrick & Kirkpatrick (2006), Evaluating Training Programs — the chain-of-evidence framework the L1–L2 limitation references (paired Case 99).

> LENS LESSON PEPFAR’s 16-country modality comparison is the L&D evaluation pattern in global health: mean L1–L2 gains across all modalities, with the Kirkpatrick limitation (Case 99) explicit. Evidence-tier flag is preprint-tier — both the medRxiv preprint and the PMC published version are citable — and the L3 / L4 questions remain open. Future validation is ongoing.

CASE 158 HEALTHCARE EHR USABILITY CLINICAL DECISION SUPPORT 2020

Annual-Screening UI Redesign + CDS at University of Missouri Health Care

A multidisciplinary EHR redesign of ambulatory annual-screening prompts (advance directives, depression, falls risk, alcohol/drug misuse), paired with embedded CDS, reported improvements in task time, error rates, System Usability Scale scores, and the downstream screening-rate outcomes the project was scoped to move

THE FULL CASE, IN FIVE BEATS

Background	Ambulatory annual-screening rates for guideline-recommended care; the cue-action gap is the C3 failure mode
The Intervention	Multidisciplinary EHR redesign of screening prompts + embedded CDS at the point of decision
How It Worked	Reported gains: task time, error rate, SUS score, and the downstream screening-rate metric
The Evidence	Evidence tier: HIMSS case-study format; single-institution QI; magnitudes await independent replication
What Transferred	The missing small-tier C3 positive example to set against Therac-25, CPOE, Helios at the big tier

CONNECTIVITY

INDUCED	3.1
LENS	D3/PT5
CLO	CLO-3, CLO-1
COURSES	LEN 3, LEN 4, LEN 8

KEY REFERENCES

- HIMSS (Greater Kansas City chapter), “Usability Redesign Improves Annual Screening Rates in an Ambulatory Setting,” case study, University of Missouri Health Care.
- Co et al. (2019), “Medication safety alert fatigue may be reduced via interaction design and clinical role tailoring: a systematic review,” JAMIA 26(10):1141, doi:10.1093/jamia/ocz095 — adjacent systematic-review evidence on interaction-design redesign.
- Office of the National Coordinator for Health IT, SAFER Guides on CDS design — practitioner-tier guidance the redesign instantiates.

> LENS LESSON The UMHC redesign is the small-tier C3 positive example the corpus needed: cue-and-alert design as a capability deliverable, with both usability and downstream screening-rate gains in the same project. The evidence is practice-synthesis tier (HIMSS case-study format), not a top-tier peer-reviewed article; magnitudes await independent replication and durability tracking. Future validation ongoing.

CASE 159 HEALTHCARE CLINICAL DECISION SUPPORT EHR DESIGN NURSING WORKFLOW 2019 – 2024

Alert-Fatigue Redesign — Cutting EHR Alerts Without Cutting the Safety Signal

Structured EHR alert redesign — fewer alerts, severity reclassification, interruptive-to-passive conversion, role-tailoring — reduced alert burden in published systematic-review and quality-improvement evidence; the 2024 case studies report alert-rate reduction with the underlying safety signal preserved

THE FULL CASE, IN FIVE BEATS

Background	Alert fatigue as the C3.2 failure mode at the EHR — high-firing low-action alerts train clinicians to dismiss
The Intervention	2019 JAMIA systematic review (Co et al.) — interaction design + role tailoring reduce alert burden across optimization studies
How It Worked	2024 QI redesign of four high-firing nursing alerts: quantitative firing analysis + empathy mapping + iterative user feedback
The Evidence	2024 interruptive-to-passive conversion of COVID-precautions alert with dual outcomes: burden + precautions-order timing
What Transferred	Evidence tier: systematic review peer-reviewed, per-site QI publications practice-tier; durability across upgrades open

CONNECTIVITY

INDUCED	3.1
LENS	D3/PT5
CLO	CLO-3, CLO-5
COURSES	LEN 3, LEN 4, LEN 8

KEY REFERENCES

- Co et al. (2019), “Medication safety alert fatigue may be reduced via interaction design and clinical role tailoring: a systematic review,” JAMIA 26(10):1141, doi:10.1093/jamia/ocz095.
- Russ et al. and colleagues (2024), “Navigating Alert Fatigue: A Case Study in Electronic Health Record Alert Design Optimization,” PubMed 39049299 — nursing-workflow QI redesign of four high-firing alerts.
- Authors (2024), “Addressing Alert Fatigue by Replacing a Burdensome Interruptive Alert with Passive Clinical Decision Support,” Applied Clinical Informatics — interruptive-to-passive conversion with dual outcomes.

> LENS LESSON The 2019 JAMIA review plus the 2024 QI projects are the small-tier C3.2 intervention companion the corpus needed — the failure thread (Uber ATG, Robodebt, UK Post Office, Tesla) is redressable by deliberate alert redesign. The systematic review is peer-reviewed; the per-site QI publications are practice-tier; magnitudes and durability open. Future validation ongoing.

CASE 160 HOSPITAL PHARMACY HEALTHCARE WORKFORCE ROLE REDESIGN 2016

Australian Hospital-Pharmacy Technician Role Redesign

Expanded pharmacy-technician scope (final accuracy checking, drugs-of-addiction management, clinical support) reportedly cut average prescription turnaround from 18.5 to 12.3 minutes, increased throughput from 220 to 295 prescriptions per shift, and decreased dispensing errors from 2.1% to 1.2% — throughput and the safety metric moving in the same direction

THE FULL CASE, IN FIVE BEATS

Background	Pharmacist capacity absorbed by dispensing-accuracy checking — clinical-support work crowded out
The Intervention	Project expands pharmacy-technician scope: final accuracy checking, drugs-of-addiction management, clinical support
How It Worked	Reported outcomes: turnaround 18.5→12.3 min; throughput 220→295 per shift; errors 2.1%→1.2%
The Evidence	2021 JPPR cross-sectional survey extends evidence into workforce-acceptance attitudes
What Transferred	Evidence tier: program-report magnitudes; no independent multi-site audit; future validation ongoing

CONNECTIVITY

INDUCED	3.2
LENS	D4/PT3
CLO	CLO-1, CLO-4
COURSES	LEN 4, LEN 5, LEN 8

KEY REFERENCES

- SHPA / Australian hospital-pharmacy network (2016), “Pharmacy Technician and Assistant Role Redesign within Australian Hospitals Project,” outcomes report.
- Anderson et al. (2021), “Perceptions of hospital pharmacists and pharmacy technicians towards expanding roles for hospital pharmacy technicians: a cross-sectional survey,” Journal of Pharmacy Practice and Research, doi:10.1002/jppr.1697.
- Boughen, Sutton, Fenn, & Wright (2017), “Defining the Role of the Pharmacy Technician and Identifying Their Future Role in Medicines Optimisation,” Pharmacy 5(3):40 — UK companion analysis.

> LENS LESSON The Australian pharmacy-technician redesign is a small-tier C3 role-redesign intervention with both throughput and safety moving in the same direction (turnaround, throughput, error rate). Evidence base is program-report plus a peer-reviewed attitudes survey; the operational magnitudes rest on the project’s internal measurement and have not been independently audited at multi-site scale. Future validation ongoing.

CASE 161 AIR TRAFFIC MANAGEMENT LEGACY MODERNIZATION SOFTWARE TRANSFORMATION 2005

Eurocat ATM Pilot Modernization — Small-Tier Verification as the Gateway to Big-Tier Change

A 17,000-line pilot modernization of the Eurocat air-traffic-management system was scoped explicitly to generate safety evidence that an automated transformation was non-distortive of original functionality; that pilot-tier evidence was used to convince customers to accept a system-wide architecture-driven modernization with 100% automated code transformation

THE FULL CASE, IN FIVE BEATS

Background	Safety-critical ATM legacy cannot tolerate big-bang rewrite — operational discontinuity is the safety hazard
The Intervention	Pilot scoped narrowly (17,000 lines) so the verification can be exhaustive against the actual legacy
How It Worked	Pilot succeeds in producing safety-equivalence evidence; customer objection to system-wide change dissolves
The Evidence	Causal claim is dissolution of the objection by the evidence artifact, not yet long-run operational success
What Transferred	Evidence tier: vendor-authored Elsevier chapter; no independent academic evaluation of the safety-equivalence claim

CONNECTIVITY

INDUCED	7.1
LENS	D1/PT1
CLO	CLO-1, CLO-2
COURSES	LEN 1, LEN 2, LEN 6

KEY REFERENCES

- Reus, Geers, & van Deursen (2010), “Modernization of the Eurocat Air Traffic Management System (EATMS),” in Information Systems Transformation: Architecture-Driven Modernization Case Studies (Elsevier / Morgan Kaufmann), Chapter 5.
- Ulrich & Newcomb (eds., 2010), Information Systems Transformation — the host volume on architecture-driven modernization patterns.
- Brodie & Stonebraker (1995), Migrating Legacy Systems — the framing reference on small-step legacy modernization.

> LENS LESSON Eurocat is the small-tier C7 transition-verification intervention the corpus needed: a narrowly scoped pilot designed to produce the safety-equivalence evidence that dissolves customer objection to a system-wide modernization. Evidence is vendor-authored practitioner literature; the long-run operational record of the modernized system is the open question. Future validation ongoing.

CASE 162 NUCLEAR POWER CONTROL-ROOM MODERNIZATION HUMAN FACTORS 2014

INL Turbine-Control Upgrade — Low-Burden Cutover as a Human-Factors Finding

An INL-affiliated study reported that operators were able to use the new digital turbine-control system without extensive additional training or rewriting of operating procedures — i.e., the human-factors verification-and-validation evidence supported a low-burden cutover from the legacy analog control

THE FULL CASE, IN FIVE BEATS

Background	Nuclear control-room modernization — safety case must survive the analog-to-digital transition
The Intervention	INL LWRS program produces per-subsystem human-factors V&V studies as small-tier deliverables
How It Worked	Turbine-control upgrade study reports low-burden cutover — no extensive retraining or procedure rewrite
The Evidence	Small-tier per-subsystem evidence is the building block of the program-level fleet-modernization claim
What Transferred	Evidence tier: INL technical reporting + OSTI conference papers; generalization to other subsystems open

CONNECTIVITY

INDUCED	7.1
LENS	D1/PT1
CLO	CLO-1, CLO-3
COURSES	LEN 1, LEN 3, LEN 6

KEY REFERENCES

- INL / LWRS program (2014), “Human Factors Design, Verification, and Validation for Two Types of Control Room Upgrades at a Nuclear Power Plant,” technical report and conference paper (ResearchGate publication 271728006).
- Idaho National Laboratory, Light Water Reactor Sustainability Program reports on control-room modernization — series available via OSTI.
- Nuclear Regulatory Commission (NUREG-0711), “Human Factors Engineering Program Review Model” — the regulatory framework the V&V deliverables are produced against.

> LENS LESSON The INL turbine-control finding is a small-tier C7 verification-as-deliverable case inside the LWRS nuclear-modernization program: the human-factors V&V evidence supports a low-burden cutover. Evidence is INL technical reporting and OSTI conference papers, not independent academic evaluation; the generalization beyond the studied subsystem is the open question. Future validation ongoing.

CASE 163 E-GOVERNMENT DISTRIBUTED DATA EXCHANGE NATIONAL IT INFRASTRUCTURE 2001 – PRESENT

Estonia X-Road — Continuous Migration as a Governance Pattern (and the No-Legacy Paradox)

By December 2024 Estonia reported effectively 100% digitalization of government services across the X-Road data-exchange layer, with sub-five-minute tax filing for >95% of the population; the load-bearing self-critique is that the country has now created its own legacy system — the very thing the program set out to avoid

THE FULL CASE, IN FIVE BEATS

Background	X-Road launched 2001 as distributed data-exchange answer to fragmented government databases
The Intervention	By Dec 2024: effectively 100% digitalization; sub-five-minute tax filing for >95% of population
How It Worked	Load-bearing self-critique — the no-legacy paradox — surfaces in the peer-reviewed analysis
The Evidence	Success-as-aging is the failure mode; generational replacement of the platform itself is the new modernization problem
What Transferred	Evidence tier: peer-reviewed analysis + program-report + practitioner reflection; future replacement is open

CONNECTIVITY

INDUCED	7.1
LENS	D1/PT1
CLO	CLO-1, CLO-4
COURSES	LEN 1, LEN 5, LEN 6

KEY REFERENCES

- Kalvet, Tiits, & Hinsberg (2024), “Estonia’s Digital Transformation: Mission Mystique and the Hiding Hand,” in Mission Mystique (Oxford University Press, 2024) — peer-reviewed analytical chapter.
- Kotka, Castro, & Kasakov (2021), “A Historical Analysis on Interoperability in Estonian Data Exchange Architecture,” ICEGOV 2021 proceedings, doi:10.1145/3494193.3494209.
- X-Road Global / Nordic Institute for Interoperability Solutions (NIIS) — program documentation and deployment-partner case studies.

> LENS LESSON Estonia X-Road is the C7 case the corpus needed for the success-as-aging failure mode: a 100%-digitalization program whose own success has now created the legacy system its founders set out to avoid. Evidence is mixed — peer-reviewed analytical chapters plus program-report and practitioner reflection; the generational-replacement trajectory is the open empirical question. Future validation ongoing. Non-US/UK case, Gap-5 cross-listed.

CASE 164 DIGITAL IDENTITY WELFARE BIOMETRIC AUTHENTICATION INDIA 2018 – 2025

Aadhaar Exclusion Litigation — Judicial Correction of Biometric Welfare Delegation in India

The Indian Supreme Court's Pragya Prasun ruling (April 2025) declared a fundamental right to inclusive and meaningful digital access and held that exclusion from welfare based on biometric-authentication failure — through no fault of the individual — violates constitutional dignity; the system can continue but alternatives to biometric authentication must be provided

THE FULL CASE, IN FIVE BEATS

Background	Aadhaar at ~1 billion enrolled — biometric authentication delegated for welfare access at point of service
The Intervention	Operational exclusion as load-bearing failure mode — worn fingerprints, missing iris, connectivity loss at the operator interface
How It Worked	Puttaswamy II (2018) — Chandrachud dissent names the structural risk; majority does not directly address it
The Evidence	Pragya Prasun (April 2025) — fundamental right to inclusive digital access; alternatives to biometric auth must be provided
What Transferred	Evidence tier: court judgments investigation-grade; lived-exclusion accounts journalism + advocacy + peer-reviewed comparative analysis

CONNECTIVITY

INDUCED	5.2
LENS	D4/PT5
CLO	CLO-1, CLO-4
COURSES	LEN 5, LEN 8, LEN 9

KEY REFERENCES

- Supreme Court of India (2018), Justice K. S. Puttaswamy (Retd.) v. Union of India (Aadhaar judgment); Justice Chandrachud’s dissent on structural exclusion risk.
- Supreme Court of India (April 2025), Pragya Prasun & Ors. v. Union of India — fundamental right to inclusive digital access; alternatives-must-be-provided remedy.
- IAPP and Access Now analyses (2024–2025), reporting and commentary on the Pragya Prasun ruling and the structural Aadhaar exclusion pattern.

> LENS LESSON Aadhaar exclusion is the non-US automated-welfare-delegation case the corpus needed: an operational exclusion failure mode judicially corrected on dignity grounds by the Pragya Prasun ruling, with the system continuing under an alternatives-must-be-provided remedy. Evidence tier is split — court judgments investigation-grade, lived-exclusion sourcing journalism plus advocacy plus peer-reviewed comparative analysis. Future validation ongoing on implementation of the remedy.

CASE 165 GLOBAL HEALTH MHEALTH MATERNAL AND NEWBORN CARE 2013 – 2018

Rwanda mHealth Maternal Care — Community Health Workers as the Capability Interface

A Rwanda mHealth monitoring system equipped community health workers with mobile decision support for maternal and newborn care; the published evaluation reported increased uptake of maternal and newborn health services tied to the intervention, with the technology extending the CHW's diagnostic-and-referral role rather than replacing it

THE FULL CASE, IN FIVE BEATS

Background	Rwanda Ministry of Health mHealth program (2013–2018) puts mobile decision support in CHW hands for maternal/newborn care
The Intervention	Tool scoped to extend the CHW's diagnostic-and-referral role, not relocate judgment to the center
How It Worked	Musabyimana et al. (2019) report increased uptake of maternal and newborn health services tied to intervention
The Evidence	Preprint-tier evidence flag: one peer-reviewed evaluation; durability and generalization remain open
What Transferred	Pairs with PEPFAR (Sub-Saharan training-modality comparison) as the African workforce-capability evidence base

CONNECTIVITY

INDUCED	6.4
LENS	D4/PT4
CLO	CLO-2, CLO-4
COURSES	LEN 4, LEN 7, LEN 8

KEY REFERENCES

- Musabyimana, A., Lundeen, T., Sayinzoga, F., et al. (2019), “Effect of a community health worker mHealth monitoring system on uptake of maternal and newborn health services in Rwanda,” Global Health Research and Policy, PMC6429813.
- Rwanda Ministry of Health, community health program documentation and CHW scope-of-practice guidance, 2013–2018.
- MIT News (2022), reporting on subsequent AI-augmented maternal-care work in Rwanda — journalism-tier companion to the peer-reviewed evaluation.

> LENS LESSON Rwanda mHealth is a small-tier capability-extension case at the frontline: technology designed around an existing CHW role, with one peer-reviewed evaluation reporting increased uptake of maternal and newborn services. The preprint-tier flag is load-bearing — one study does not close the durability or generalization question, and the broader impact claims rest partly on practitioner reporting. Future validation ongoing.

CASE 166 MEDICAL-DEVICE REGULATION AI/SAMD JAPAN 2014 – PRESENT

Japan PMDA — Pre-Approved Change Management as Regulatory Architecture for AI/SaMD

Japan's 2019 PMD Act amendment introduced the Post-Approval Change Management Protocol (PACMP): a pre-agreed modification scope at initial approval, streamlined review for in-scope changes. The 2025 scoping review of PMDA-approved AI radiology software documents transparency variability — hedge preserved

THE FULL CASE, IN FIVE BEATS

Background	2014 PMD Act lays the groundwork; 2019 amendment introduces conditional early approval and PACMP
The Intervention	PACMP: manufacturer submits pre-agreed modification scope at initial approval; in-scope changes get streamlined review
How It Worked	DASH for SaMD initiative supports faster reviews and earlier application — pace-matched architecture for iteration
The Evidence	2025 scoping review documents transparency variability across PMDA-approved AI radiology software — hedge preserved
What Transferred	Pairs with Epic-Sepsis governance gap and FDA's evolving SaMD policy as the non-US regulator-designed change-control case

CONNECTIVITY

INDUCED	5.4
LENS	D4/PT6
CLO	CLO-4, CLO-5
COURSES	LEN 5, LEN 7, LEN 8

KEY REFERENCES

- Aoki, T., et al. (2025), “Scoping Review of Regulatory Transparency in AI-based Radiology Software: Analysis of PMDA-approved SaMD Products,” medRxiv 2025.10.02.25336333.
- Aoki, T., et al. (2021 → published), “Regulatory-approved Deep Learning/Machine Learning-Based Medical Devices in Japan as of 2020: A Systematic Review,” PMC9931274.
- Pharmaceuticals and Medical Devices Agency of Japan, PMD Act amendment (2019) and DASH for SaMD program documentation.

> LENS LESSON PMDA’s PACMP and DASH for SaMD are the non-US regulatory architecture for AI/SaMD change control: the modification scope is pre-agreed at initial approval, in-scope changes get streamlined review, and the delegation-with-revocation structure is explicit. The 2025 scoping review documents transparency variability across approvals — load-bearing hedge preserved. Preprint-tier flag for the recent systematic analyses; future validation ongoing.

CASE 167

DATA GOVERNANCE · INDIGENOUS DATA SOVEREIGNTY · AUSTRALIA / NZ / US

2019 – 2020 (PRINCIPLES); ONGOING

CARE Principles — Indigenous Data Governance as a Replaced Regime

The CARE Principles (Collective Benefit, Authority to Control, Responsibility, Ethics) were developed by Indigenous Data Sovereignty networks in Aotearoa NZ, Australia, and the US to shift the framing from consultation under existing data regimes to Indigenous Peoples as the data owners and beneficiaries; the Lowitja Journal 2025 documents one of the first detailed implementation case studies

THE FULL CASE, IN FIVE BEATS

Background	CARE Principles (Carroll et al. 2020): Collective Benefit, Authority to Control, Responsibility, Ethics
The Intervention	Developed by Indigenous Data Sovereignty networks in Aotearoa NZ, Australia, US — explicit non-US-centered leadership
How It Worked	Governance-replacement structure, not governance-supplementation — existing privacy regimes named as inadequate
The Evidence	Lowitja Journal 2025 documents Ngangk Yira Institute implementation — first detailed case study at institutional scale
What Transferred	Supports CLO *Fairness beyond omission* — fairness as positive sovereignty, not subtraction of biased features

CONNECTIVITY

INDUCED	5.1
LENS	D4/PT4
CLO	CLO-4
COURSES	LEN 4, LEN 5, LEN 8

KEY REFERENCES

- Carroll, S. R., Garba, I., Figueroa-Rodríguez, O. L., et al. (2020), “The CARE Principles for Indigenous Data Governance,” Data Science Journal 19(1):43, doi:10.5334/dsj-2020-043.
- Ngangk Yira Institute for Change (2025), “Recognising Indigenous data sovereignty and implementing Indigenous data governance at the Ngangk Yira Institute for Change,” The Lowitja Journal, doi:10.1016/j.lowitj.2025.100030.
- Global Indigenous Data Alliance (GIDA), CARE Principles documentation and implementation guidance.

> LENS LESSON CARE is the governance-replacement case for Indigenous data: collective benefit, authority, responsibility, ethics — published peer-reviewed in 2020 with an emerging implementation literature anchored by the Lowitja Journal 2025 Ngangk Yira paper. Practice synthesis tier — the principles are s g.

CASE 168

SOFTWARE ENGINEERING EDUCATION · WORK-BASED LEARNING · REFLECTIVE PRACTICE

2025 (PREPRINT)

Reflective Practice on a Work-Based Software Engineering Program — Longitudinal Capability Development

A 2025 longitudinal preprint study of reflective-practice development across a multi-year work-based software-engineering program — one of the few published instruments aimed at measuring the development of reflective capability rather than only its presence; preprint-tier flag is load-bearing

THE FULL CASE, IN FIVE BEATS

The Shift	2025 arXiv preprint: longitudinal study of reflective-practice development on multi-year work-based SE program
What Is Emerging	Instrument designed to measure development of reflective capability over time, not only presence at a snapshot
The Capability Question	Construct boundary: intra-learner depth-change measurement vs. cross-learner presence measurement
Early Evidence	Preprint-tier flag load-bearing — not yet peer-reviewed at time of drafting; structural signal extracted, not specific magnitudes
Open Problems	Prior art for the editor-commissioned first-person Practice Flywheel accounts: shows the evaluation pathway, not only the genre

CONNECTIVITY

INDUCED	2.3
LENS	D2/PT4
CLO	CLO-2, CLO-3
COURSES	LEN 2, LEN 7, LEN 8

KEY REFERENCES

- “The Development of Reflective Practice on a Work-Based Software Engineering Program: A Longitudinal Study,” arXiv:2504.20956 (2025) — preprint.
- D. Schön, The Reflective Practitioner (1983) — the foundational account of reflection-in-action the genre rests on.
- Boud, Keogh & Walker (eds.), Reflection: Turning Experience into Learning (1985) — reflection as a learning process, and the measurement problem it raises.

> LENS LESSON The 2025 preprint is one of the few published instruments aimed at measuring the development of reflective capability across a multi-year work-based program, not only its presence at a snapshot. Preprint-tier flag load-bearing — not yet peer-reviewed at time of drafting; the case is included on the structural contribution (construct distinction, instrument-design move) rather than specific magnitudes. Future validation ongoing.

CASE 169 K-12 MATHEMATICS INTELLIGENT TUTORING EDUCATION POLICY 2007 – 2014

Cognitive Tutor Algebra I at Scale — Year-One Null, Year-Two Positive

Cluster-randomized 147 high schools across seven states; year-one posttest scores showed no significant difference between Cognitive Tutor and control conditions; year-two posttest scores showed CTAI schools significantly outperforming controls; a one-year evaluation would have published the wrong answer

THE FULL CASE, IN FIVE BEATS

Background 147 high schools, seven states, cluster-randomized to CTAI or current curriculum for two years
The Intervention Year-one posttest: no significant difference; year-two posttest: CTAI significantly outperforms control
How It Worked A one-year evaluation would have published a null on the same intervention; both findings in the same trial
The Evidence Timeline of evaluation is itself a falsifiable design choice; year-two horizon required for deployment-substrate stabilization
What Transferred Deeper-evidence-of v1 Case 113; pair with Case 170 (ASSISTments) and Case 60 (Epic Sepsis horizon discipline)

CONNECTIVITY

INDUCED 2.3
LENS D3/PT4
CLO CLO-2, CLO-3
COURSES LEN 2, LEN 3, LEN 7

KEY REFERENCES

- Pane, J. F., Griffin, B. A., McCaffrey, D. F., & Karam, R. (2014), “Effectiveness of Cognitive Tutor Algebra I at Scale,” Educational Evaluation and Policy Analysis 36(2):127–144, doi:10.3102/0162373713507480.
- RAND Working Paper WR-1050 — addendum to the Pane et al. evaluation.
- Koedinger, K. R., Anderson, J. R., Hadley, W. H., & Mark, M. A. (1997), “Intelligent tutoring goes to school in the big city,” IJAIED — the v1 Case 113 system description Cognitive Tutor builds from.

> LENS LESSON Pane et al. is the at-scale evaluation case for Cognitive Tutor and the worked example of evaluation horizon as a falsifiable design choice. Year one: null. Year two: significantly positive. The case is the deeper-evidence-of update on v1 Case 113 and the curriculum’s primary anchor for the CLO on judgment under inadequate evidence where the inadequacy is the horizon, not the sample size.

CASE 170 K-12 MATHEMATICS HOMEWORK SUPPORT FORMATIVE ASSESSMENT 2014 – PRESENT

ASSISTments — National Replication and Long-Term Follow-Through

Cluster RCT across 46 schools and 3,035 students: 7th-graders assigned to ASSISTments outperformed controls on end-of-year math; largest gains for lower-performing students; minority students benefited more from the intervention; effect persisted into 8th-grade outcomes in 2020 follow-up

THE FULL CASE, IN FIVE BEATS

Background Roschelle et al. 2016 cluster RCT: 46 schools, 3,035 students; ASSISTments-assigned 7th-graders outperformed controls
The Intervention Heterogeneity: largest gains for lower-performing students; minority students benefited more — pre-specified equity-relevant outcome
How It Worked Murphy et al. 2020: 7th-grade effect persisted into 8th-grade outcomes (longitudinal follow-through)
The Evidence Arnold Ventures extension: lower-cost virtual-training adaptation in rural areas, longitudinal through end of 8th grade
What Transferred Pair with Case 169 (Cognitive Tutor horizon), Case 147 (spaced ed RCTs), Case 71 (Engler — equity-relevant inversion of gatekeeping)

CONNECTIVITY

INDUCED 2.3
LENS D2/PT4
CLO CLO-2, CLO-3
COURSES LEN 2, LEN 3, LEN 7

KEY REFERENCES

- Roschelle, J., Feng, M., Murphy, R. F., & Mason, C. A. (2016), “Online Mathematics Homework Increases Student Achievement,” AERA Open 2(4):1–12, doi:10.1177/2332858416673968.
- Murphy, R. et al. (2020), follow-up evaluation extending the 7th-grade effect into 8th-grade outcomes.
- Heffernan, N. T., & Heffernan, C. L. (2014), “The ASSISTments ecosystem,” International Journal of AI in Education 24:470–497 — platform design and research-loop description.

> LENS LESSON ASSISTments is the case in the corpus with the cleanest closed-loop evidence architecture for EdTech: cluster RCT, longitudinal follow-through into the next grade, adaptation arm under different deployment conditions, and a pre-specified equity-relevant heterogeneity finding. The case grounds the closed-loop evaluation anchor in EdTech the same way Case 128 grounds it in team-science training.

CASE 171 HIGHER EDUCATION LEARNING ANALYTICS DISTANCE LEARNING 2019 – 2023

OU Analyse — Predictive Learning Analytics and Teacher Use at Scale

Across 9 courses and 559 teachers (189 with OU Analyse access), teachers' engagement with predictive learning analytics was associated with measurable improvements in student performance for >14,000 students; three-year post-implementation follow-up extends the evidence into sustained adoption and perceptions

THE FULL CASE, IN FIVE BEATS

Background	OU Analyse: predictive-learning-analytics dashboard deployed across the Open University UK's distance-learning operation
The Intervention	Herodotou et al. 2019 BJET: 9 courses, 559 teachers (189 with OUA access), >14,000 students; teacher engagement → measurable improvement
How It Worked	Herodotou et al. 2023 LAK: three-year-post-implementation follow-up — stabilization, sustained adoption stratification, perception evolution
The Evidence	Distinct from Case 122 (OU consent governance); this case is post-deployment teacher-use at multi-cohort scale
What Transferred	Hedges binding: causal attribution bounded (propensity matching, not RCT randomization); multi-institution transfer evidence pending

CONNECTIVITY

INDUCED	5.2
LENS	D3/PT5
CLO	CLO-3, CLO-5
COURSES	LEN 2, LEN 3, LEN 7

KEY REFERENCES

- Herodotou, C., Hlosta, M., Boroowa, A., Rienties, B., Zdrahal, Z., & Mangafa, C. (2019), “Empowering online teachers through predictive learning analytics,” British Journal of Educational Technology 50(6):3064–3079, doi:10.1111/bjet.12853.
- Herodotou, C. et al. (2023), “Predictive Learning Analytics and University Teachers: Usage and perceptions three years post implementation,” LAK ‘23, doi:10.1145/3576050.3576061.
- Herodotou et al. (2019), “A large-scale implementation of predictive learning analytics in higher education: the teachers’ role and perspective,” Educational Technology Research and Development, ERIC EJ1227972 — complementary teacher-perspective paper.

LENS LESSON OU Analyse is the rare successful learning-engineering intervention with both deployment scale and longitudinal teacher-use evidence at journal tier. The 2019 BJET paper establishes the effect on student performance at the institutional deployment; the 2023 LAK follow-up extends the picture into three-year sustained adoption. The delegation-with-revocation structure is operative and teachable. Hedges binding on causal attribution and on multi-institution transfer

CASE 172 LEARNING ANALYTICS ONLINE LEARNING MOBILE/DISTANCE LEARNING 2016 – 2025

The Doer Effect at Scale — Replication, AI-Generated Questions, Non-WEIRD Extension

Van Campenhout et al. (LAK 2023) replicated the doer-effect causal claim across seven courses with 15.2 million data points; L@S 2025 replication held with AI-generated practice questions; LAK 2025 non-WEIRD radio/phone extension found weaker effect for learners with higher prior educational attainment — the load-bearing heterogeneity finding

THE FULL CASE, IN FIVE BEATS

Background	Original doer-effect claim (Koedinger et al. LAK 2016): doing improves learning more than reading; appears causal
The Intervention	Van Campenhout et al. LAK 2023: seven-course replication with 15.2M data points — effect holds in direction and magnitude
How It Worked	L@S 2025 replication: AI-generated practice questions — effect still holds; meaningful given LLM-generated content rising
The Evidence	Butler et al. LAK 2025 non-WEIRD: radio-lecture + mobile-phone practice — effect weaker for higher-prior-attainment learners (heterogeneity is the result)
What Transferred	Closed loop via replication, not single trial; pair with Case 132 (Duolingo half-life) and Case 147 (spaced ed RCTs)

CONNECTIVITY

INDUCED	2.3
LENS	D2/PT4
CLO	CLO-2, CLO-3
COURSES	LEN 2, LEN 3, LEN 7

KEY REFERENCES

- Van Campenhout, R., Jerome, B., Dittel, J. S., & Johnson, B. G. (2023), “The Doer Effect at Scale: Investigating Correlation and Causation Across Seven Courses,” LAK23, doi:10.1145/3576050.3576103.
- Van Campenhout et al. (2025), “Scaling the Doer Effect: A Replication Analysis Using AI-Generated Questions,” L@S ‘25, doi:10.1145/3698205.3729545.
- Butler, D. et al. (2025), “Does the Doer Effect Generalize To Non-WEIRD Populations? Toward Analytics in Radio and Phone-Based Learning,” LAK ‘25, doi:10.1145/3706468.3706505 (also arXiv 2412.20923).

LENS LESSON The doer-effect replication arc is the closed-loop-via- replication case in the corpus. Original claim, seven-course large-N replication, AI-generated-content replication, non- WEIRD-modality extension — four converging pieces of evidence with the prior-attainment heterogeneity finding as the load- bearing result. The case completes the replication-arc thread alongside Cases 132 and 147.

CASE 173 TUTORING HYBRID HUMAN-AI SYSTEMS K-12 EDUCATION 2024

Hybrid Human-AI Tutoring — Augmentation, Not Delegation

Three quasi-experimental studies of hybrid human-AI tutoring deployments reported improvements in student learning relative to comparison conditions; the AI is positioned as augmentation, not delegation; the human tutor retains authorization to override and re-direct

THE FULL CASE, IN FIVE BEATS

Background	Thomas et al. LAK 2024 best paper: three quasi-experimental studies of hybrid human-AI tutoring
The Intervention	Headline: learning outcomes improved relative to comparison conditions in each of the three studies
How It Worked	Design positioning: AI as augmentation, human tutor retains override authorization, measured outcome is student learning
The Evidence	Educational analog of Case 118 (TREWS clinician-AI teaming); contrast with Case 60 (Epic Sepsis delegation collapse)
What Transferred	Open: longitudinal persistence; transfer to lower-resource tutoring where human-tutor availability is the binding constraint

CONNECTIVITY

INDUCED	6.4
LENS	D5/PT6
CLO	CLO-2, CLO-5
COURSES	LEN 2, LEN 5, LEN 7

KEY REFERENCES

- Thomas, D. R. et al. (2024), “Improving Student Learning with Hybrid Human-AI Tutoring: A Three-Study Quasi-Experimental Investigation,” LAK ‘24, doi:10.1145/3636555.3636896.
- Case 118 (TREWS) reference set — Henry et al. (2022), Nature Medicine — clinician-AI teaming analog.
- Case 60 (Epic Sepsis) reference set — Wong et al. (2021), JAMA Internal Medicine — delegation-collapse analog.

> LENS LESSON Hybrid human-AI tutoring is the educational analog of the TREWS clinician-AI teaming pattern. AI is positioned as augmentation; human tutor retains override authorization; student-learning outcome is the measure. Three quasi- experimental studies converge on positive learning effects. The case pairs with Cases 118 / 101 / 118 / 119 in the human-AI teaming thread and grounds the delegation-with- revocation CLO at the educational deployment.

CASE 174 K-12 SCIENCE COLLABORATIVE LEARNING DESIGN-BASED RESEARCH 2009

Zhang/Scardamalia — Knowledge Building Across Three Cohorts

Three-year design study in a single Grade 4 classroom (Institute of Child Study, Toronto) using Knowledge Forum across an optics curriculum; documented progression from fixed small-group to opportunistic collaboration across cohorts, with associated gains in the depth and distribution of scientific explanations across the class community

THE FULL CASE, IN FIVE BEATS

Background	Single Grade 4 classroom at Institute of Child Study, Toronto; one teacher across three years; optics curriculum; Knowledge Forum platform
The Intervention	Design-based research across three successive cohorts; iterated redesign of how collaboration was organized as the unit of inquiry
How It Worked	Progression: fixed small-group collaboration → opportunistic collaboration around emerging questions; cohort is the iteration unit
The Evidence	Outcomes: improved depth and improved distribution of scientific explanations across the class community as the structure progressed
What Transferred	Hedges binding: single teacher, single school, design study not causal; transferability to non-ICS contexts and non-KB-expert teachers is open

CONNECTIVITY

INDUCED	2.2
LENS	D2/PT4
CLO	CLO-2, CLO-3
COURSES	LEN 2, LEN 3, LEN 7

KEY REFERENCES

- Zhang, J., Scardamalia, M., Reeve, R., & Messina, R. (2009), “Designs for Collective Cognitive Responsibility in Knowledge-Building Communities,” Journal of the Learning Sciences 18(1):7–44, doi:10.1080/10508400802581676.
- Scardamalia, M., & Bereiter, C. (2006), “Knowledge Building: Theory, pedagogy, and technology,” in K. Sawyer (ed.), Cambridge Handbook of the Learning Sciences — programmatic backdrop.
- Bereiter, C., & Scardamalia, M. (2014), “Knowledge Building and Knowledge Creation: One Concept, Two Hills to Climb,” in S. C. Tan, H. J. So, & J. Yeo (eds.), Knowledge Creation in Education — extension into the post-2009 program.

> LENS LESSON Zhang and colleagues’ three-year design study is the classroom-scale longitudinal record of how a collaborative structure can be iteratively redesigned across cohorts in a direction that improves the depth and distribution of scientific explanation. The hedges are binding — single teacher, single school, design study not causal — and the case’s pedagogical value depends on the hedges being preserved. The cohort is the iteration unit.

CASE 175 K-12 EDUCATION AI-SUPPORTED INSTRUCTION EDUCATIONAL EQUITY 2025

Chen et al. — Rural China AI Devices and the Equity-Direction Finding

Quasi-experimental study across 12 schools (4 urban, 8 rural) and 268 teachers, September to November 2024; rural experimental classes gained 17.93% on mathematics and 13.46% on history, while urban experimental classes gained 10.96% on mathematics and 9.55% on history — the rural gain exceeded the urban gain across both subjects

THE FULL CASE, IN FIVE BEATS

Background	Chen, Wu, Chen, Zhou (2025) Frontiers in Psychology; 12 schools (4 urban, 8 rural), 268 teachers, Sep–Nov 2024
The Intervention	Rural experimental classes +17.93% math, +13.46% history; urban experimental classes +10.96% math, +9.55% history
How It Worked	Equity-direction finding: rural gain exceeds urban gain across both subjects — the load-bearing teaching point
The Evidence	Load-bearing hedges in prose: 3-month horizon, 12 schools, non-randomized assignment, self-report and observation bias acknowledged
What Transferred	First-author-from-deployment-country structure; pair with Cases 186 (Gándara), 187 (LiveHint bias), 186 (Doer Effect non-WEIRD)

CONNECTIVITY

INDUCED	8.3
LENS	D2/PT4
CLO	CLO-3, fairness beyond omission
COURSES	LEN 3, LEN 7, LEN 8

KEY REFERENCES

- Chen, R., Wu, Y., Chen, Z., & Zhou, P. (2025), “Advancing educational equity in rural China: the impact of AI devices on teaching quality and learning outcomes for sustainable development,” Frontiers in Psychology 16:1588047, doi:10.3389/fpsyg.2025.1588047.
- Paired Case 186 (Gándara et al., AERA Open) — community-college predictive equity at the higher-education scale.
- Paired Case 189 (LiveHint AI bias across foundation models, AIED 2025) — bias-surfacing in AI-supported instruction.

> LENS LESSON Chen and colleagues’ 12-school three-month deployment of AI- supported instructional devices across rural and urban Chinese classrooms reported an equity-direction finding — rural gain exceeded urban gain across both mathematics and history. The load-bearing hedges are explicit and carried in the prose: three-month horizon, twelve schools, non-randomized assignment, self-report and observation bias acknowledged by the authors. The case anchors the CLO on fairness beyond omission with a rare published equity-narrowing result.

CASE 176 GOVERNMENT HIRING ALGORITHMS AUDIT AND DISCLOSURE 2023 – PRESENT

NYC Local Law 144 — Bias Audits as Governance Artifact

New York City Local Law 144 of 2021, implemented through Department of Consumer and Worker Protection rules effective July 5 2023, requires employers using "automated employment decision tools" (AEDTs) for hiring or promotion decisions in NYC to conduct annual independent bias audits, publish audit summaries, and provide candidate notice; the first national municipal regulation of algorithmic hiring tools at this scope

THE FULL CASE, IN FIVE BEATS

Background	NYC Local Law 144 of 2021; Department of Consumer and Worker Protection implementing rules effective July 5 2023; first US municipal AEDT regulation at this scope
The Intervention	Three requirements: annual independent bias audit, publication of audit summary, candidate notice and alternative-selection request process
How It Worked	Audit metrics: selection rate and impact ratio by sex, race/ethnicity, intersectional categories; computed by independent auditor not associated with the AEDT
The Evidence	Yam & Skirpan 2024 "bias audits without bias data": employers often lack protected-attribute data; Wright & Brown 2024 audit-quality study finds wide variability
What Transferred	Pair with Case 122 (OU Analyse), Case 186 (Gándara), Case 81 (Amazon hiring AI); whether audits reduce actual disparate impact remains under-evidenced

CONNECTIVITY

INDUCED	5.4
LENS	D4/PT5
CLO	CLO-3, CLO-4
COURSES	LEN 3, LEN 5, LEN 8

KEY REFERENCES

- New York City Department of Consumer and Worker Protection, Rules Implementing Local Law 144 of 2021 (Automated Employment Decision Tools), effective July 5, 2023.
- Yam, K. A., & Skirpan, M. W. (2024), “Bias Audits Without Bias Data: An Investigation of Auditor Practice Under NYC Local Law 144,” in Proceedings of FAccT 2024, doi:10.1145/3630106.3658955.
- Wright, L., & Brown, I. (2024), “Auditing the auditors: an empirical study of the first cohort of Local Law 144 bias audits,” AI and Ethics, doi:10.1007/s43681-024-00461-2

> LENS LESSON NYC Local Law 144 is the bias-audit-as-governance-artifact intervention at municipal-regulatory scale. The audit-and- notice regime is the first national municipal regulation of algorithmic hiring tools at this scope; the Yam/Skirpan and Wright/Brown critiques name the data-availability and audit-quality limitations the regulatory theory does not provide for. The intervention is real; whether it reduces actual disparate impact at the hiring level is under- evidenced and is the open evaluation question the case carries.

CASE 177 HEALTHCARE CLINICAL DECISION SUPPORT MACHINE LEARNING 2022 – 2024

COMPOSER Sepsis Prediction — The Third Clinical-AI Sepsis Case

Boussina et al. NPJ Digital Medicine 2024 prospective multi-site implementation of the COMPOSER (COnformal Multidimensional Prediction Of SEpsis Risk) deep-learning model at UC San Diego Health; reported 1.9 percentage-point absolute decrease in in-hospital sepsis mortality and 5.0 percentage-point absolute decrease in 28-day sepsis-related readmission during the deployment period, evaluated against the pre-deployment baseline period within the same health system

THE FULL CASE, IN FIVE BEATS

Background	Boussina et al. NPJ Digital Medicine 2024; COMPOSER deep-learning sepsis-risk model at UC San Diego Health, two-site prospective implementation
The Intervention	Conformal-prediction framework: calibrated uncertainty intervals enable model abstention when input is out-of-distribution
How It Worked	Deployment outcomes: 1.9 pp absolute decrease in in-hospital sepsis mortality; 5.0 pp absolute decrease in 28-day sepsis-related readmission vs pre-deployment baseline
The Evidence	Hedges binding: prospective not RCT; mortality reduction multifactorial — concurrent process improvements at UCSD cannot be cleanly separated from COMPOSER's contribution
What Transferred	Completes AI-delegation typology in sepsis: Case 118 (TREWS), Case 60 (Epic Sepsis Score), COMPOSER (Case 177)

CONNECTIVITY

INDUCED 3.1
LENS D3+D5/PT6
CLO CLO-3, CLO-5
COURSES LEN 3, LEN 5, LEN 9

KEY REFERENCES

- Boussina, A., Shashikumar, S. P., Malhotra, A., Owens, R. L., El-Kareh, R., Longhurst, C. A., Quintero, K., et al. (2024), “Impact of a deep learning sepsis prediction model on quality of care and survival,” NPJ Digital Medicine 7:14, doi:10.1038/s41746-023-00986-6.
- Shashikumar, S. P., Wardi, G., Malhotra, A., & Nemati, S. (2021), “Artificial intelligence sepsis prediction algorithm learns to say ‘I don’t know,’” NPJ Digital Medicine 4:134 — the methodological-foundation paper for the conformal-prediction abstention structure.
- Wong, A., Otles, E., Donnelly, J. P., Krumm, A., McCullough, J., DeTroyer-Cooley, O., et al. (2021), “External Validation of a Widely Implemented Proprietary Sepsis Prediction Model in Hospitalized Patients,” JAMA Internal Medicine 181(8):1065–1070 — the load-bearing external-validation paper on Epic Sepsis (Case 60).

LENS LESSON COMPOSER is the third clinical-AI sepsis case in the delegation typology — prospective-positive with conformal- prediction abstention. Th [illegible] al sepsis mortality by 1.9 percentage points and 28-day readmission by 5.0 percentage points; the load-bearing hedges are that the evaluation is prospective not RCT and that the mortality reduction is multifactorial. The abstention structure is the methodological contribution the case anchors.

CASE 178 AUTONOMOUS SYSTEMS 2016 – PRESENT

Tesla Autopilot — Recurring Fatalities

Dozens of fatalities documented in NHTSA's Standing General Order data; first U.S. cases of Level-2 automation contributing to fatal injury

THE FULL CASE, IN FIVE BEATS

The Shift	Level-2 driving automation reaches untrained consumers with no qualification gate for monitoring
What Is Emerging	Fatal crashes accumulate as reliable operation erodes the vigilance the system silently requires
The Capability Question	Liability rests with drivers whose inattention the engagement design itself helped to produce
Early Evidence	NHTSA finds driver-engagement inadequate; automation-complacency evidence points the same direction
Open Problems	Whether any engagement architecture can make consumer Level-2 monitoring sustainable stays open

CONNECTIVITY

COURSES LEN 7, LEN 2, LEN 6

KEY REFERENCES

- NTSB, Highway Accident Report HAR-17/02 (Williston, FL, 2016) — the quoted disengagement finding.
- NTSB, Highway Accident Report HAR-20/01 (Mountain View, CA, 2018) — Autopilot crash analysis.
- NHTSA Standing General Order 2021-01 reports — documented Autopilot fatal crashes.

LENS LESSON Tesla Autopilot at consumer scale is the largest live test of Level-2 monitoring as a sustainable role. The early evidence is that it is not. The case is the strongest currently available test of whether consumer-side training and engagement design can produce sustained attention to automation. The recurring fatality pattern suggests the answer.

CASE 179 AUTONOMOUS SYSTEMS 2023

Cruise Robotaxi — Pedestrian Drag

GM Cruise robotaxi struck a pedestrian and then dragged her ~20 feet; California suspended Cruise's permit; the program was substantially shut down

THE FULL CASE, IN FIVE BEATS

The Shift	Driverless robotaxi oversight rests on operator-controlled disclosure to regulators who cannot inspect directly
What Is Emerging	Pedestrian struck by another car was hit then dragged by a Cruise vehicle
The Capability Question	Whether commercial autonomy programs disclose fully under pressure becomes the institutional question
Early Evidence	Permit suspended; NHTSA opened investigation; GM shut down commercial operations entirely
Open Problems	What auditable pre-committed disclosure commitment commercial autonomy should require before operating

CONNECTIVITY

COURSES LEN 10, LEN 7

KEY REFERENCES

- California Public Utilities Commission decision suspending Cruise permits (2023) — the omitted-facts finding (paraphrased).
- California DMV order of suspension (2023) — the partial video disclosure.
- Quinn Emanuel report on Cruise incident response (2024) — the disclosure failures.

> LENS LESSON Cruise is the foundational governance-failure case in commercial autonomy. The incident itself was a single pedestrian injury; the institutional response converted it into a company-ending event. The capability that was missing was the institutional commitment to full disclosure of operational incidents to regulators.

CASE 180 HEALTHCARE TECHNOLOGY 2018 – PRESENT

Radiology AI Miscalibration

Recurring documented cases of FDA-cleared radiology AI tools performing worse in deployment than in validation, often along demographic lines; the canonical v1 anchor for clinical-AI deployment without surveillance, cross-referenced by the Epic Sepsis (Case 60) and pulse-oximetry (Case 120) deployment-evidence cases

THE FULL CASE, IN FIVE BEATS

The Shift	Machine-learning diagnostics enter clinical workflow with validation that may not survive deployment
What Is Emerging	Cleared radiology tools repeatedly perform worse in deployment, concentrated in under-represented patient groups
The Capability Question	How a regulator certifies safety without checking behavior across populations and labels it meets
Early Evidence	Degradation reported across imaging, sepsis, dermatology; 510(k) requires no demographic stratification
Open Problems	Capability gap sits in regulatory architecture; demographic post-market surveillance machinery unbuilt

CONNECTIVITY

INDUCED	7.2
LENS	D3+D5/PT5
CLO	CLO-3, CLO-5
COURSES	LEN 4, LEN 7, LEN 9

KEY REFERENCES

- Larrazabal, A. J., Nieto, N., Peterson, V., Milone, D. H., & Ferrante, E. (2020), “Gender imbalance in medical imaging datasets produces biased classifiers for computer-aided diagnosis,” PNAS 117(23):12592–12594 — sensitivity drops for under-represented groups on NIH ChestX-ray14 and CheXpert.
- Seyyed-Kalantari, L., Zhang, H., McDermott, M. B. A., Chen, I. Y., & Ghassemi, M. (2021), “Underdiagnosis bias of artificial intelligence algorithms applied to chest radiographs in under-served patient populations,” Nature Medicine 27:2176–2182 — disparities across Black, Hispanic, female, and lower-socioeconomic subgroups; persistence across model architectures.
- Obermeyer, Z., Powers, B., Vogeli, C., & Mullainathan, S. (2019), “Dissecting racial bias in an algorithm used to manage the health of populations,” Science 366(6464):447–453 — proxy-label bias in care-management algorithm covering 200 million Americans.

> LENS LESSON Radiology AI is the canonical contemporary case for the gap between regulatory clearance and clinical deployment performance in medical AI. The clearance dataset and the deployment population diverge. The institutional architecture to surface the divergence — demographic post-market surveillance — has not yet been built.

CASE 181 HEALTHCARE TECHNOLOGY 2023 – PRESENT

ChatGPT in Healthcare — Hallucination Cases

Documented cases of clinicians using LLMs that produce hallucinated citations, fabricated dosages, and fictitious clinical guidelines

THE FULL CASE, IN FIVE BEATS

The Shift	Clinicians adopted ChatGPT immediately at point of care without guidelines or institutional gate
What Is Emerging	Fabricated citations, hallucinated dosages, and fictitious guidelines recur across the documented literature
The Capability Question	Whether clinicians can develop verification routines the model's fluent confidence actively discourages
Early Evidence	Early reports show LLM output accepted less critically than colleague recommendations would be
Open Problems	Routine clinical verification practice analogous to bibliographic discipline remains to be defined

CONNECTIVITY

INDUCED	5.2
LENS	D5/PT6
CLO	CLO-3, CLO-5
COURSES	LEN 10, LEN 7, LEN 2

KEY REFERENCES

- JAMA editorials on LLM integration into clinical practice (2023–2024) — the hallucination/verification problem (synthesized).
- Sallam (2023), "ChatGPT Utility in Healthcare Education, Research, and Practice" — documented benefits and risks.
- Case reports of LLM hallucinated citations and dosages in clinical and academic use (2023–2024).

> LENS LESSON LLM use in clinical practice is the live frontier case for human-AI teaming when the AI's output is fluent across both accurate and hallucinated content. The capability that does not yet exist is the routine verification practice — a clinical analog to the bibliographic discipline of academic writing.

CASE 182 GOVERNMENT TECHNOLOGY 2011 – PRESENT

Predictive Policing — PredPol

Predictive-policing tools deployed by hundreds of U.S. police departments; several cities have abandoned them after equity analyses

THE FULL CASE, IN FIVE BEATS

The Shift	Police adopted statistical prediction tools lending discretionary judgment a veneer of objectivity
What Is Emerging	Training on arrest records, models learn enforcement patterns and create self-confirming feedback loops
The Capability Question	Whether a model trained on institutional behavior can predict the underlying phenomenon
Early Evidence	Dirty data documented; Santa Cruz, New Orleans, Los Angeles suspended deployments after equity reviews
Open Problems	Pre-deployment construct-validity audit remains absent in most jurisdictions adopting these systems

CONNECTIVITY

COURSES LEN 7, LEN 9

KEY REFERENCES

- Lum, K. & Isaac, W. (2016), "To Predict and Serve?," Significance — the enforcement-vs-crime feedback loop (paraphrased).
- Richardson, Schultz & Crawford (2019), "Dirty Data, Bad Predictions" — biased records feeding predictive systems.
- Brantingham et al. (2018) — predictive-policing field experiments.

> LENS LESSON Predictive policing is the canonical case for the difference between training data and ground truth at law-enforcement scale. The institution being predicted (where crime occurs) is not the institution producing the training data (where police make arrests). The capability gap is at the construct.

CASE 183 HEALTHCARE TECHNOLOGY 2020 – PRESENT

AlphaFold — Protein Structure Prediction

Substantially solved a 50-year protein-structure prediction problem; 200M+ structures publicly released; foundational positive AI case

THE FULL CASE, IN FIVE BEATS

The Shift	Deep learning offered computational solution to a fifty-year experimental bottleneck in biology
What Is Emerging	AlphaFold released 200 million predicted structures folded into research workflows worldwide
The Capability Question	Which conditions allow an AI capability to be safely and widely useful
Early Evidence	Success rested on agreed benchmark, clean data, verifiable output, and open release
Open Problems	Whether fields lacking those preconditions can deliberately build them remains the frontier question

CONNECTIVITY

COURSES LEN 1, LEN 7, LEN 9

KEY REFERENCES

- Jumper et al. (2021), "Highly accurate protein structure prediction with AlphaFold," Nature — the method and accuracy.
- Varadi et al. (2022), AlphaFold Protein Structure Database, Nucleic Acids Research — the 200M+ structures and open release.
- Moult, J. (CASP organizer) commentary on AlphaFold2 (2020) — the benchmark and the achievement (paraphrased).

> LENS LESSON AlphaFold is the strongest positive AI case in the dataset. The technical achievement is real. The conditions that made the benefit possible — agreed benchmark, training data, verifiable output, open release — are the capability infrastructure around the model, not the model itself. The case is the strongest available evidence for what supports beneficial AI deployment.

CASE 184 TECHNOLOGY 2021 – PRESENT

AI-Augmented Coding Tools

Tens of millions of developers using GitHub Copilot, Cursor, and peers; productivity gains documented; security and correctness implications still being characterized

THE FULL CASE, IN FIVE BEATS

The Shift	AI coding assistants became infrastructure for tens of millions of developers without controls
What Is Emerging	Productivity gains documented alongside unsettled security findings on AI-generated code quality
The Capability Question	Whether developers using these tools are becoming more capable or more dependent
Early Evidence	Jagged frontier shows performance degrading just outside competence, where users over-trust output
Open Problems	Longitudinal study and training practice that keeps human skill growing remain unbuilt

CONNECTIVITY

INDUCED	5.2
LENS	D5/PT6
CLO	CLO-5
COURSES	LEN 10, LEN 2, LEN 8

KEY REFERENCES

- Peng et al. (2023), "The Impact of AI on Developer Productivity" — short-term productivity gains.
- Pearce et al. (2022), "Asleep at the Keyboard? Assessing the Security of GitHub Copilot's Code Contributions."
- Sandoval et al. (2023), "Lost at C," USENIX Security — AI assistance did not significantly increase critical security-bug rates.

> LENS LESSON AI-augmented coding is the live foundational case for human-AI teaming in professional knowledge work. The short-term gains are real. The long-term capability question — does the tool make the operator more capable, or more dependent — is the question the discipline must learn to ask and answer.

CASE 185 NEUROSCIENCE/CONNECTOMICS HUMAN-AI TEAMING WORKFORCE TRAINING 2017 – PRESENT

CIRCUIT / MICrONS — The Human Correction Layer at Petabyte Scale

Under IARPA's MICrONS program, automated electron-microscopy segmentation produces brain maps too large and too error-prone to deploy without human correction; CIRCUIT trains undergraduate cohorts as the proofreading workforce that is the recovery mechanism for automation failure at petabyte scale

THE FULL CASE, IN FIVE BEATS

The Shift	Automated EM segmentation at petabyte scale produces brain maps too large and too error-prone to deploy without verification
What Is Emerging	CIRCUIT trains undergraduate cohorts as the proofreading workforce; APL's BossDB infrastructure stores and serves the data
The Capability Question	Human capability as the recovery mechanism for automation failure at the petabyte boundary — the design decision that makes the output usable
Early Evidence	Evidence-tier flag: connectomics method is peer-reviewed; CIRCUIT training-outcome evidence is institutional/program documentation
Open Problems	Frontier sub-competency candidate: design of the human correction layer for generative/automated systems at scale

CONNECTIVITY

INDUCED 3.4
LENS D5/PT6
CLO CLO-3, CLO-5
COURSES LEN 2, LEN 5, LEN 7

KEY REFERENCES

- MICrONS program literature (IARPA) — connectomics method and automated segmentation evidence base.
- APL BossDB documentation — petabyte-scale connectomics data infrastructure.
- CIRCUIT program documentation (JHU Hub, 2017 – present) — institutional/program description; the training-outcome evidence is at this tier and the evidence-tier flag is binding.

LENS LESSON CIRCUIT / MICrONS is the frontier instance of designing the human correction layer for automated systems at scale. The connectomics method is peer-reviewed; the program training outcomes are institutional documentation, and the gap is rendered as the evidence-tier flag under the title. The forward-looking question the case names — how to design the human correction layer for generative and automated systems at scale — recurs across domains and is not well-named in the existing curriculum.

CASE 186 HIGHER EDUCATION ALGO FAIRNESS POLICY ANALYSIS 2024

Gándara — Detecting and Mitigating Algorithmic Bias in College Student-Success Prediction

Gándara, Anahideh, Ison, and Picchiarini (AERA Open, 2024) audited predictive models of college student success and showed that models which look acceptable on overall accuracy are systematically less accurate for Black and Hispanic students and overestimate success for white and Asian students — small-tier frontier evidence that the choice of construct and the stratification used in evaluation, not only model-bias mitigation, determine whether an equity-oriented prediction is fair to the groups the equity commitment is meant to protect

THE FULL CASE, IN FIVE BEATS

The Shift	Predictive student-success modeling is routine: models score students on predicted graduation / retention / benefit / need; scores feed downstream support and outreach decisions
What Is Emerging	Gándara et al. (AERA Open, 2024): models less accurate for Black and Hispanic students; overestimate success for white and Asian students — overall accuracy masks the disparity
The Capability Question	Stratified evaluation by income, race/ethnicity, first-generation status reveals disparity that overall-accuracy summary metrics hide
Early Evidence	Decision context and mitigation matter: train end users to contextualize a flagged prediction; bias-mitigation reduces but does not eliminate the cross-group gap
Open Problems	Cross-references v2 race-construct trio (Cases 119 eGFR, 106 pulse oximetry, 107 Hoffman) — construct definition is the upstream design decision in each

CONNECTIVITY

INDUCED 8.2
LENS D3/PT5
CLO CLO-3, CLO-4
COURSES LEN 3, LEN 6, LEN 9

KEY REFERENCES

- Gándara, Anahideh, Ison, & Picchiarini (2024), “Inside the Black Box: Detecting and Mitigating Algorithmic Bias across Racialized Groups in College Student-Success Prediction,” AERA Open 10, doi:10.1177/23328584241258741 — primary case source on cross-group accuracy disparity, stratified evaluation, and bias mitigation in student-success prediction.
- Barocas, Hardt, & Narayanan, Fairness and Machine Learning (fairmlbook.org) — methodological backdrop on construct definition and stratified evaluation.
- Friedler, Scheidegger, & Venkatasubramanian (2021), “The (im)possibility of fairness: different value systems require different mechanisms for fair decision

LENS LESSON Gándara's student-success-prediction audit is the small-tier frontier instance of fairness-as-construct-definition. Which construct the model maximizes (predicted graduation vs. benefit vs. need), which stratification is used in validation, and the decision context the prediction is consumed in each determine the fairness properties of the deployed system; the audit found models systematically less accurate for Black and Hispanic students. Cross-references the v2 race-construct trio at the construct-definition layer.

CASE 187 LEARNING ANALYTICS ALGO FAIRNESS ED-TECH RESEARCH 2021 – 2024

Yu / Lee / Kizilcec — Protected Attributes in Learning-Analytics Models

Yu, Lee, and Kizilcec, publishing in the LAK/EDM literature, examined whether and how protected attributes (race/ethnicity, gender, socioeconomic status) should be included in learning-analytics predictive models, and showed that whether including or excluding the attribute produces fairer outcomes depends on the construct, the model class, the downstream intervention, and the population — small-tier frontier evidence that the include-or-exclude question is the wrong framing

THE FULL CASE, IN FIVE BEATS

The Shift	Long-running learning-analytics question: include or exclude protected attributes (race/ethnicity, gender, SES) as features?
What Is Emerging	Fairness-through-unawareness intuitive but incomplete: omitted attributes reconstructable from correlated features (zip code, prior achievement)
The Capability Question	Yu, Lee, Kizilcec (LAK/EDM): include-or-exclude effect depends on construct, model class, downstream intervention, population — no general answer
Early Evidence	Right framing is governance and audit: explicit decision recorded with reasoning, stratified evaluation, audit cadence that catches the consequences
Open Problems	Cross-references Case 186 (Gándara), the v2 race-construct trio (105 eGFR, 106 pulse oximetry, 107 Hoffman) — five-case equity-construct frontier set

CONNECTIVITY

INDUCED	8.4
LENS	D3/PT5
CLO	CLO-3, CLO-4
COURSES	LEN 3, LEN 6, LEN 9

KEY REFERENCES

- Yu, R., Lee, H., & Kizilcec, R. F. (2021), “Should College Dropout Prediction Models Include Protected Attributes?” in Proceedings of the Eighth ACM Conference on Learning @ Scale (L@S ‘21), doi:10.1145/3430895.3460139 — primary paper on the include-or-exclude empirical analysis.
- Kizilcec & Lee, “Algorithmic Fairness in Education,” in Holmes & Porayska-Pomsta (eds.), Ethics in Artificial Intelligence in Education — broader synthesis of the fairness-in-learning-analytics frontier.
- Dwork, Hardt, Pitassi, Reingold, & Zemel (2012), “Fairness through awareness,” Proceedings of ITCS — foundational paper on the inadequacy of fairness-through-unawareness.

> LENS LESSON Yu, Lee, and Kizilcec’s protected-attributes work is the frontier instance of surfacing bias through governance, not just technique. Whether inclu[illegible]s on the construct, the model class, the intervention, and the population. The governance architecture around the model is the carrier of the answer in any specific deployment.

CASE 188 LEARNING ANALYTICS HIGHER EDUCATION AFRICA 2022

Learning Analytics on the African Continent — A Scoping Review as the Present-State Map

A 2022 scoping review found only 15 learning-analytics studies on the entire African continent, concentrated in Morocco, South Africa, Nigeria, and Ethiopia; the structural finding — limited LMS access, limited institutional resourcing, limited African-scholar visibility at SoLAR — is itself the evidence the field requires before construct-travel claims can be made

THE FULL CASE, IN FIVE BEATS

The Shift	Prinsloo et al. (2022) scoping review of learning analytics on African continent — 15 studies total
What Is Emerging	Publication concentrated in Morocco, South Africa, Nigeria, Ethiopia; adjacent SA studies extend but do not change magnitude
The Capability Question	Structural barriers: limited LMS access, limited institutional resourcing, limited African-scholar visibility at SoLAR
Early Evidence	Construct-travel problem stated as research-base evidence; pairs with African data-privacy case for the governance side
Open Problems	Frontier case; practice-synthesis-tier flag preserved; future validation ongoing as the literature matures

CONNECTIVITY

INDUCED	8.4
LENS	D4/PT4
CLO	CLO-3, CLO-4
COURSES	LEN 4, LEN 7, LEN 8

KEY REFERENCES

- Prinsloo, P., Khalil, M., & Slade, S. (2022), “Learning Analytics on the African Continent: An Emerging Research Focus and Practice,” Journal of Learning Analytics; ResearchGate publication 361096718.
- Lemmens, J.-C., & Henn, M. (2015), South African Association for Institutional Research (SAAIR) proceedings — adjacent SA higher-education learning-analytics work.
- SciELO (2020), “Development of a contextualised learning-analytics framework for South African higher education.”

> LENS LESSON The scoping review is the present-state map of learning-analytics research on the African continent: 15 studies, geographically concentrated, with the structural barriers (LMS access, institutional resourcing, African-scholar SoLAR visibility) underneath the count. Practice-synthesis-tier — a snapshot of an early-stage literature; future validation ongoing as the field matures.

CASE 189 AI IN EDUCATION INTELLIGENT TUTORING ALGO FAIRNESS 2025

LiveHint AI — Evaluating an AI Tutor for Bias Across Foundation Models

Repeated probing of LiveHint AI (an LLM-based tutor under development at Carnegie Learning) with identity-marked student queries surfaced response differences in tone, detail, and pedagogical appropriateness across identities; choice of foundation model materially affected the level of differentiation

THE FULL CASE, IN FIVE BEATS

The Shift	LiveHint AI (Carnegie Learning) probed with identity-marked student queries across tone, detail, pedagogical appropriateness
What Is Emerging	Choice of foundation model materially affects differentiation level; vendor-selection decision is itself fairness-relevant
The Capability Question	Methods-development paper (LiveHint in development), not deployment-bias audit; grounds demographic-stratification at foundation-model layer
Early Evidence	Structurally new variable beyond race-construct trio (Cases 119/105/106/155): the foundation-model layer above the deployed system
Open Problems	Open: lab probing vs. deployed-conversation match; vendor selection as routine fairness deliverable; pair with Case 173

CONNECTIVITY

INDUCED	8.2
LENS	D3/PT6
CLO	CLO-3, CLO-5
COURSES	LEN 3, LEN 5, LEN 7

KEY REFERENCES

- AIED 2025, “Evaluating an AI Tutor for Bias Across Different Foundation Models,” Springer/ACM proceedings, doi:10.1007/978-3-031-98465-5_43; preprint at renzheyu.com/papers/AIED2025_Tutor.pdf.
- Bommasani, R. et al. (2021), “On the Opportunities and Risks of Foundation Models,” Stanford CRFM — the foundation-model framing the case builds on.
- Race-construct trio reference set: Hoffman et al. (2016), Sjoding et al. (2020) pulse oximetry, Inker et al. (2021) eGFR-without-race — paired with Cases 119, 120, 63.

> LENS LESSON LiveHint AI is the methods-development case at the foundation- model layer of LLM-tutoring fairness evaluation. The matched- pair identity-probing protocol surfaces tone, detail, and pedagogical-appropriateness differences across identities; the foundation-model choice materially affects the differentiation level. The case extends the race-construct trio into the LLM- tutoring layer with a structurally new variable — the foundation model — that did not exist at the deployment-audit layer.

CASE 190 MULTIMODAL LEARNING ANALYTICS CLASSROOM DEPLOYMENT DESIGN-BASED RESEARCH 2023

Multimodal Learning Analytics In-the-Wild — A First-Person Lessons-Learned Account

First-person practitioner reflection on multiple in-the-wild multimodal learning analytics (MMLA) deployments — eye-tracking, audio capture, spatial positioning in classroom contexts; documents what worked, what failed, what the team would have done differently

THE FULL CASE, IN FIVE BEATS

The Shift	Martinez-Maldonado et al. 2023 arXiv: first-person practitioner reflection on multiple MMLA in-the-wild deployments
What Is Emerging	Content: what worked, what failed, what the team would have done differently — operational knowledge not in results papers
The Capability Question	Offered as published-first-person Practice Flywheel exemplar, not as deployment-results case; pair with front-matter 'unpacking is the method'
Early Evidence	Grounds sustaining-tacit-capability anchor — practitioner knowledge walks out the door if not narrated
Open Problems	Preprint-tier flag binding: arXiv consolidated synthesis; future validation ongoing on peer-review and on genre adoption across LE

CONNECTIVITY

INDUCED	2.2
LENS	D2/PT4
CLO	CLO-2, CLO-3
COURSES	LEN 2, LEN 5, LEN 7

KEY REFERENCES

- Martinez-Maldonado, R. et al. (2023), “Lessons Learnt from a Multimodal Learning Analytics Deployment In-the-Wild,” arXiv:2303.09099 — preprint, sections published in adjacent LAK and IEEE TLT outlets.
- Worsley, M., & Blikstein, P. (2018), “A multimodal multisensor framework for examining how students engage in design,” Journal of Learning Analytics — broader MMLA literature backdrop.
- Schon, D. (1983), The Reflective Practitioner — the genre’s theoretical underpinning, referenced across the reflective-practice case tier.

> LENS LESSON Martinez-Maldonado et al. is the LE-specific published- first-person Practice Flywheel exemplar at MMLA in-the-wild deployment scale. The case is offered not as a deployment- results case but as a genre exemplar — the reflective- practice account at the field’s preprint tier. Preprint-tier flag binding; future validation ongoing on peer-review pipeline and on whether the genre takes hold across the LE community.

CASE 191 HEALTHCARE MEDICAL IMAGING MACHINE LEARNING 2020

DeepMind Mammography — High-Profile Nature Paper, Replicability Pushback

McKinney et al. 2020 Nature paper reported a deep-learning mammography screening system outperforming radiologists on retrospective UK and U.S. screening datasets, with reductions in false-positives (5.7 percentage points in the U.S. set, 1.2 in the UK set) and false-negatives (9.4 and 2.7 percentage points respectively); Haibe-Kains et al. October 2020 Nature comment critiqued the paper for failing to release code and trained models, arguing that reproducibility could not be assessed and that screening-comparison claims required deployment-grade evidence

THE FULL CASE, IN FIVE BEATS

The Shift	McKinney et al. Nature Jan 1 2020: deep-learning mammography reduces false-positives 5.7 pp (US) / 1.2 pp (UK), false-negatives 9.4 / 2.7 pp vs radiologists
What Is Emerging	Press framing: "AI outperforms radiologists"; paper's careful claims do not carry the framing's deployment implications
The Capability Question	Haibe-Kains et al. Nature Oct 14 2020 comment: code not released, models not released, methodology not reproducible from publication
Early Evidence	Comment does not allege error; argues reproducibility not established; large fraction of methodology not independently verifiable
Open Problems	Pair with Case 180 (Radiology AI miscalibration), Case 60 (Epic Sepsis), Case 120 (pulse oximetry population heterogeneity)

CONNECTIVITY

INDUCED 7.2
LENS D3+D5/PT6
CLO CLO-3, CLO-5
COURSES LEN 3, LEN 5, LEN 9

KEY REFERENCES

- McKinney, S. M., Sieniek, M., Godbole, V., Godwin, J., Antropova, N., Ashrafian, H., Back, T., et al. (2020), "International evaluation of an AI system for breast cancer screening," Nature 577:89–94, doi:10.1038/s41586-019-1799-6.
- Haibe-Kains, B., Adam, G. A., Hosny, A., Khodakarami, F., MAQC Society Board of Directors, Waldron, L., Wang, B., et al. (2020), "Transparency and reproducibility in artificial intelligence," Nature 586:E14–E16, doi:10.1038/s41586-020-2766-y.
- McKinney et al. (2020), reply to Haibe-Kains et al., Nature 586:E17–E18 — the developers' response on the reproducibility-infrastructure question.

> LENS LESSON DeepMind Mammography is the verification-as-deployment-event case at the high-profile-publication scale. The McKinney paper's retrospective [illegible] a [illegible] t [illegible] he Haibe-Kains comment named the reproducibility infrastructure as the condition for the arc, and subsequent deployment evidence has refined the screening-comparison framing the original paper offered.

CASE 192 EDUCATION AT SCALE HEALTHCARE DEFENSE ONGOING

The Discipline We Build Next

Open: the case that has not yet been written. The discipline that this casebook documents is the one practitioners now in training must build.

THE FULL CASE, IN FIVE BEATS

The Shift	The closing case belongs to practitioners now in training across the program's domains
What Is Emerging	A teachable discipline of engineerable capability is emerging from training, interfaces, evidence, institutions
The Capability Question	Whether the discipline can be built fast enough to close the documented gaps
Early Evidence	Capability engineering sits where INPO did in 1979 or CRM in 1981
Open Problems	What practitioners build next; a failure prevented, success engineered, institution built

CONNECTIVITY

COURSES LEN 1, LEN 10, LEN 8, LEN 3

KEY REFERENCES

- LDT / LENS program statement, Johns Hopkins School of Education — the program and its commitments.
- Goodell & Kolodner (2022), Learning Engineering Toolkit — the discipline's methods.
- The preceding cases of this volume — the cumulative record of cost and of possibility.

> LENS LESSON Every case in this book is closed. This one is open. The discipline LENS represents is the discipline that closes cases that are now open. The institutions of capability engineering that exist in 2026 are insufficient to the volume of capability work the contemporary world requires. The graduates of this program are the people who will build the institutions that are insufficient now.

APPENDIX · CLOS AND COURSE COVERAGE

The competencies, the CLOs, and the courses.

COMPETENCY → CONCENTRATION LEARNING OUTCOME

The five capabilities the cases turn on are named, in the curriculum, as five concentration learning outcomes (CLOs):

Systems Analysis	CLO-1 · Systems Thinking and Analysis
Iterative Development	CLO-2 · Learning Engineering Design and Implementation
Test and Evaluation	CLO-3 · Data, Measurement, and Evaluation
Navigating Sociotechnical Constraints	CLO-4 · Context and Domain Fluency
Machine Teaming and Adaptation	CLO-5 · Human-AI Teaming and Adaptive Systems

THE LEN COURSES · CASES PER COURSE

Each case names the LEN courses it serves as a worked example for, and most cases serve several. The counts below are therefore not a partition of the 202 cases — they sum to well more than 202, because a single case typically appears under three or four courses.

LEN 1	Principles of learning engineering	36
LEN 2	Human-AI teaming	54
LEN 3	Systems integration	49
LEN 4	Evidence and measurement	67
LEN 5	Capability analysis	86
LEN 6	Applied problem-solving	18
LEN 7	Bias and governance	114
LEN 8	Knowledge transfer	98

LEN 9	Computational methods	30
LEN 10	Capstone studio	19

WHAT THE DISTRIBUTION SHOWS

The distribution is itself diagnostic. The strongly covered topics are the ones the existing case literature most readily supports: bias and governance (LEN 7) and knowledge transfer (LEN 8) lead, with capability analysis (LEN 5) and evidence and measurement (LEN 4) close behind. Across 202 cases the corpus is saturated with examples of what goes wrong when ethics, governance, evidence, and institutional learning are not engineered — and, in the success cases, what it takes to engineer them.

The thinner coverage is equally informative. Computational methods (LEN 9) has the failure modes of algorithmic systems well represented but the computational tools for capability instrumentation under-represented. The capstone (LEN 10) maps to fewer cases than the number of studio projects each cohort undertakes. Systems integration (LEN 3) and applied problem-solving (LEN 6) required focused tagging to reach defensible coverage, surfacing that operational systems-engineering content is under-represented in the published case literature relative to its curricular centrality, and that stakeholder-analysis and problem-framing have not historically been packaged as case material by other disciplines. These are pedagogical gaps the LENS studio sequence is organized to close — not by adding more failure cases, but by having students produce the missing artifacts in real operational settings and entering them into the dataset for the next cohort.

WHAT LENS OFFERS

The program behind the casebook.

LENS — **Learning Engineering for Next-Generation Systems** — is a specialization within the Johns Hopkins University School of Education's **Learning Design & Technology** program. It operationalizes the casebook's central claim — that capability is a designable, measurable, engineerable property of a complex system — for high-consequence work in healthcare, defense, and education. It is housed in a School of Education because the binding constraint is rarely the engineering, which is mature; it is the learning sciences, educational measurement, implementation science, and equity-and-governance machinery that decide whether a capability intervention actually scales.

THE FIVE PILLARS

LENS stands on five institutional pillars, load-bearing together rather than singly. **Mission Literacy** — reading what an operational setting actually requires and translating it into design and measurement decisions a serious reviewer would accept. **JHU Ecosystem** — the School of Education's centers, doctoral programs, and faculty in learning sciences, measurement, and design-based research, alongside university practice partners (the Armstrong Institute for Patient Safety and Quality; the Bloomberg School's implementation-science tradition; Whiting School engineering and cognitive science; the Berman Institute's governance and ethics work). **Intersectional Expertise** — the distinguishing pillar: holding the engineering, the learning, the measurement, the equity, and the operational context in one conversation. **Capability Focus** — keeping every artifact, measurement, and iteration accountable to what the system can actually do. **Flywheel Iteration** — the Practice Flywheel (**Identify → Activate → Prototype → Analyze → Transition**) that turns each cohort, case, and pilot into the starting condition of the next.

THE RECORD AT JOHNS HOPKINS

The placement is not a default. The School of Education has studied the learner, designed the assessment, and engineered the intervention since before learning engineering had its current name. The Center for Talented Youth (1979) holds one of the longest continuous evidence bases in U.S. education on the identification and accelerated instruction of advanced learners; the Center for Research and Reform in Education has, since 2004, run large-scale evaluations against rigorous evidence standards (ESSA tiers, What Works Clearinghouse) and publishes the Best Evidence Encyclopedia. The Learning Design & Technology program inherits that environment: for two decades its M.Ed. and EdD tracks have graduated designers and evaluators of learning systems whose pedagogical core is design-based research. LENS formalizes that orientation as a specialization rather than departing from it.

THE PILOTS

The claim that capability is engineerable rests on a documented record of learning-engineering pilots run at the Johns Hopkins Applied Physics Laboratory before LENS enrolled its first cohort. **CIRCUIT** was built as the workforce solution to the IARPA MICrONS connectomics program; the cohorts it trained contributed materially to the MICrONS reconstructions now published in Nature, with student development a deliberate byproduct of the mission result. The model was generalized into the **MERIT** framework and the **COMPASS** mentor-intervention layer, documented in the peer-reviewed engineering-education literature. The **PISTA** incubator (DTRA, Summer 2025) compressed the same cycle into ten weeks: nine student fellows shipped a working CBRNE decision-support prototype to agency leadership through three full redesigns. No single pilot settles the proposition; the cumulative record shows the program is itself the institutional flywheel its case studies argue every high-consequence domain requires.

ABOUT THE EDITORS

The editors.

The casebook's central argument — that capability engineering is the applied form of the educational sciences, taught from the School of Education and informed by operational practice — is embodied in the two editors. Each has spent a career working in one half of that pairing and a substantial portion of it in the other. Each contributes to the five pillars on which LENS rests: Mission Literacy · JHU Ecosystem · Intersectional Expertise · Capability Focus · Flywheel Iteration.

William Gray-Roncal, Ph.D.

APPLIED RESEARCH · LEARNING ENGINEERING · STEM WORKFORCE

William Gray-Roncal, Ph.D., is Principal Research Engineer at the Johns Hopkins University Applied Physics Laboratory, where he applies systems engineering methods to challenges spanning defense, space, biomedical, and learning systems. His research — sponsored by DARPA, IARPA, NIH, and DoD — bridges computational neuroscience, artificial intelligence, and large-scale data infrastructure. He developed the College Prep and CIRCUIT workforce-development models, through which he has mentored nearly 500 students, and his current work develops quantitative measures of human cognitive performance in operational contexts. He leads the development of the LENS (Learning Engineering for Next-Generation Systems) specialization within the Johns Hopkins Master of Education in Learning Design and Technology, the program directed by James Diamond at the School of Education. Gray-Roncal holds a Ph.D. in Computer Science from Johns Hopkins, an M.S. in Electrical Engineering from USC, and a B.S. in Electrical Engineering from Vanderbilt.

James Diamond, Ph.D.

LEARNING SCIENCES · EDUCATIONAL DESIGN · EVALUATION

James Diamond, Ph.D., is Assistant Professor in the Johns Hopkins University School of Education and Program Director of the Master of Education in Learning Design and Technology, where he applies design-based research and learning-sciences methods to the design and evaluation of technology-mediated learning environments. His research — supported by the National Science Foundation, the Institute of Education Sciences, the Bill & Melinda Gates Foundation, the MacArthur Foundation's HASTAC Digital Media and Learning Initiative, the Robin Hood Learning + Technology Fund, the Wellcome Trust, and the UK Economic and Social Research Council — bridges digital games and simulations for learning, digital badges and micro-credentialing, classroom technology integration, and disciplinary literacy in history, social studies, civics, and STEM. Before joining Johns Hopkins, he was Senior Research Associate at the Education Development Center's Center for Children and Technology in New York City, where he led projects including Zoom In and the Who Built America? Teacher Mastery Badge System, contributed to Mission US and Possible Worlds, and taught at New York University as an adjunct. As Program Director, he oversees the LDT program within which the LENS (Learning Engineering for Next-Generation Systems) specialization is offered. Diamond holds a Ph.D. in Educational Communication and Technology from New York University, an Ed.M. in Educational Technology from Boston University, and a B.A. in History from Boston University.

These biographies reference careers documented in publications, program reports, faculty pages, and the curricular materials of the LDT program and the LENS specialization.